

The Swords of Armageddon

U.S. Nuclear Weapons Development since 1945

VOLUME II

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POSTWAR U.S. FISSION WEAPONS DEVELOPMENT

Overview

At the end of World War II, the United States possessed the best war materiel on the planet. The Boeing B-29 Superfortress (**Figure II-1**) was the most advanced piece of movable technology in the world: it featured state-of-the-art radar and bombing systems, an entirely electronic computerized remotely-controlled defensive fire control system, the highest performance supercharged aero-engines then in existence, and heated and pressurized crew cabins within which crewmen could fight off enemy aircraft and bomb targets without being burdened by heavy and cumbersome flight suits or exposed to the low-pressure, subzero atmosphere outside the aircraft.

The B-29's wing, engines, crew compartment pressurization system, and computerized electronic gunfire control system gave it unprecedented range, altitude, speed, defensive, and payload capabilities that were unsurpassed until the B-36s and later B-47s appeared. By August 1945, the aircraft was a combat-proven weapon, with high-speed long-range bombing capability at both low and high altitudes.

When mated with nuclear fission, then the greatest source of explosive power at mankind's disposal, the B-29 and the atomic bomb were a formidable weapon-delivery system pair. If available in sufficient numbers, and located within striking range of targets in enemy territory, the B-29 and its atomic bombs could deter potential future aggression against the U.S. and its interests and allies.

However, it was also likely that the U.S. monopoly in nuclear weaponry would not long endure: by the end of the war, the Soviets were already hard at work on both their own atomic bomb and an exact replica of the B-29, the former derived largely from espionage and the latter copied mainly from three U.S. Army B-29s forced down in Soviet airspace after bombing attacks on Japan.¹ As usual, American

¹TARGET AMERICA: THE SOVIET UNION AND THE STRATEGIC ARMS RACE, 1945-1964, Steven J. Zaloga, Presidio Press, Novato, California, 1993, pp. 69-74. For an excellent account of the postwar Soviet bomber program based on the B-29, see "The Billion-Dollar Bomber," William Green, AIR ENTHUSIAST, Vol. 1 Nos. 2, 3, 4, and 5 (July, August, September, October 1971), pp. 104-107, 160-163, 214-218, and 263-268. By September 1945, the Soviets also had exact descriptions of not only every component of the wartime U.S. atomic bombs, but also many details of the engineering and design of the large industrial facilities required to produce the fissionable materials for the bombs.

engineers and military officials assumed that the "primitive Russians" would never be able to reverse-engineer and mass-produce such a complex piece of military equipment.

As much as early U.S. atomic weapons represented the "state of the art" of applied fission physics by 1945, they were nonetheless very imperfect devices, in terms of their military usefulness and their as-yet-unrealized potential efficiency in the use of scarce and expensive fissionable materials. There were many improvements -- some obvious, some not so obvious -- that could be made in the design, components, and operation of both the implosion (FAT MAN) and gun-assembly (LITTLE BOY) type bombs. By 1945, the U.S. nuclear weapons industry was still in its infancy, like many other blooming postwar technologies, including radar, rocketry, and jet propulsion.

As former LASL director Norris Bradbury stated in 1965,

The task which we did at Los Alamos during the war had to be the most simple-minded, the most direct, the least-likely direction to run into difficulties or obstacles that we could imagine, that would lead to an atomic bomb.

And we were aware of lots of other ways of going about it, we were aware of lots of other potential needs of the military, both in size and bang and efficiency, size of device...

So we knew these tasks had to be done.²

Later, Bradbury recalled that the wartime weapons were almost offensively crude: "We had only scratched the surface of atomic bombs; we had, to put it bluntly, lousy bombs ... which were totally wrongly matched to the production empire."³

In 1944, even before the first bombs had been produced, U.S. Navy captain W. S. "Deke" Parsons ruminated on likely postwar goals of the U.S. nuclear weapons program:

It appears that even after an "endless" peace is achieved, nuclear weapon research and development should still be based on

²Address by Norris Bradbury to the Lawrence Livermore Laboratory on November 3, 1965. Lecture No. 3, "Los Alamos," LLNL Family Lecture Series.

³Rhodes, DARK SUN, p. 212.

time scales. The aims of this research and development might be set down as follows:

(a) To produce in the shortest time an immediately usable nuclear weapon. If this has been accomplished, then "(a)" would be to keep the U.S. arsenal supplied with nuclear weapons which are modified as accumulated knowledge permits design for increased power and efficiency.

(b) Looking ahead, say two years, and dovetailing predictable possibilities in nuclear weapons development with developments in carriers for such weapons, maintain a middle-term project whose success renders "(a)" weapons obsolete.

(c) Long-term research to discover ways by which sudden nuclear reactions can release much greater amounts of energy.⁴

By the early summer of 1945, even before the TRINITY test, improvements in the implosion bomb core and its high explosive (HE) charge were predicted:

For your information, the principal developmental improvements now contemplated in the field of implosion are the use of a pre-assembled core and the development of flatter (HE) lenses, using lower-velocity explosives. It is clear that this program may be affected by the results of Trinity, or by further insight into the problems of implosion.⁵

Possible postwar directions and objectives for LASL were outlined in a talk by Norris Bradbury at the beginning of October 1945. Bradbury emphasized the importance of continuing nuclear weapons development and testing:

The project cannot neglect the stockpiling or development of atomic weapons in this interim period. Strongly as we suspect that these weapons will never be used; as much as

⁴Memorandum dated 5 October 1944 for Dr. Richard C. Tolman, Chairman, Committee on Post-War Policy, via Dr. J. R. Oppenheimer, from W. S. Parsons.

⁵Memorandum dated 22 June 1945 to Messrs. Greison, Krets, Neddermeyer, Peierls, and Tuck from J. R. Oppenheimer.

we dislike the implications contained in this procedure, we have an obligation to the nation never to permit it to be in the position of saying it has something which it has not got.

The world now knows that we have a weapon. How many or how good it does not know. To weaken the nation's bargaining power in the next few months during the administration's attempt to bring about international cooperation would be suicidal.

....

To bring peace by threatening war is possible; to bring peace by requesting and promising cooperation seems more dignified. But the request and the promise, and surely the threat, are both fortified by weaponeering now; and the results of weaponeering may be that it may never have to be done again.

Bradbury also described LASL's short-term goals:

What shall our general project program be as far as atomic bombs and atomic weapons are concerned?

We will stockpile the current FAT MAN up to a number of 15. We will develop internal modifications, possibly in the method of fusing, almost certainly in the method of detonating. We will develop a levitated model. We will set up a more careful program of gadget testing so that we will know the degree of reliability of each component.

We will set up surveillance tests which at least must have the possibility of extending over a considerable time. We will set up Sandia Field as a field test site. It may not last there for more than a year, but we will learn how the ideal field test site for weapons should be set up, and it can either stay there, be moved, or become, let us hope, unnecessary in the course of time.

We will initiate the engineering of a new weapon whose aims should be -- although again we hope it will never need use --

increased reliability, ease of assembly, safety, and permanence; in short, a better weapon. ...

We will propose subsequent TRINITYs. (Author's note: nuclear tests.) The TRINITY bomb was a bomb and not a weapon, if you will permit the distinction.⁶

By late 1946, Bradbury, by then the second director of LASL, noted that a better understanding of the largely-empirical science that developed the wartime weapons was required in order to further improve later models:

Believing that the present weapon was primarily the fruit of fundamental research extending over a period of years, it was considered that such research should play a definite role in the life of the laboratory. Such research has been conducted in all the technical fields which bore upon the development of the weapon. This includes not only nuclear physics, but high explosives, equations of state, radiation, hydrodynamics, and phenomena of solids. Since the development of many portions of the weapons had proceeded on an almost entirely empirical basis, attempts are now to be made to increase the understanding of the processes involved (in the detonation of the bombs).⁷

In this letter, Bradbury also noted that while LASL "has made progress in the design of nuclear components for weapons, it has not made satisfactory progress in the ordnance engineering of weapons."

Some possibilities for improved weapons were ideas that had been put forth at early stages of the Manhattan Project but which had been "shelved" temporarily to permit the quickest-possible development of the wartime bombs in whose designs the number of unknowns were kept to a minimum. These enhancements were put aside until time became available at the end of the war, when the government would decide whether or not to continue the nuclear weapons program:

⁶Notes on Talk Given by Comdr. N. E. Bradbury at Coordinating Council, October 1, 1945. Emphasis in original.

⁷Letter dated November 14, 1946 to U.S. Atomic Energy Commission from N. E. Bradbury, Director, LASL.

The wartime work had bypassed, for the time being, very promising principles. Los Alamos Scientific Laboratory had, in 1945-1947, opened new paths toward more efficient, more versatile weapons which needed exploration.⁸

Continuance of the weapons program was not a foregone conclusion, even in spite of the country's enormous capital investment in industrial plants that produced materials for the Army's Manhattan Engineer District (MED): the Army's original wartime contract with the University of California to operate the Los Alamos lab extended only for the duration of the war plus six months. Army contracts with civilian companies that operated the uranium enrichment and plutonium production plants at Oak Ridge, Tennessee and Hanford, Washington were for the duration of the war plus five years.⁹

While General Leslie Groves had built special and expensive facilities that could turn out bombs in large numbers after World War II ended,¹⁰ there was little immediate postwar impetus outside the Pentagon to continue the U.S. nuclear weapons program, or even to maintain the physical plants and technical staffs required to keep the program functioning.

The continued operation of the Hanford and Oak Ridge reactors, and the laboratories in Chicago, Los Alamos, Ohio, and elsewhere was also hindered by unnecessary and obsessive secrecy which kept not only congressional funders in the dark, but also many members of the highest echelons of the armed forces and the President himself.

Although it was economically, militarily, and politically unlikely that the weapons-making facilities would be shut down or abandoned at war's end, continuance of the U.S.'s atomic weapons program would be contingent upon postwar presidential and congressional decisions and funding.

In addition, the nation's atomic strength after the end of the war would not be gauged just by the number of weapons in the nuclear stockpile (which was terribly small until 1950),

⁸"Background Information on Nevada Nuclear Tests," Office of Test Information, Nevada Test Organization, Nevada Test Site, Mercury, Nevada, May 1, 1957, p. 5.

⁹Comment by Dr. Stanley Goldberg at Los Alamos, New Mexico, during Smithsonian Institution seminar, October 1991. Dr. Goldberg is currently writing a biography of General Leslie Groves.

¹⁰Rhodes, DARK SUN, p. 224.

but also by the explosive power of each weapon and the weapon's adaptability to easy delivery to a designated target.

Preservation of the U.S. margin of superiority in atomic bombs required a dynamic effort to improve weapons in all three respects, namely, the technical excellence of the weapons, adequate numbers of warheads, and the deliverability of the warheads.¹¹ This effort toward improvement was the primary goal of the U.S. nuclear weapons program immediately following the end of World War II.

In early 1946, Major General Groves, commander of the MED, which oversaw the production of the first U.S. atomic bombs, noted several aspects of the change in military thinking required by the advent of nuclear weapons, and of their current limitations and of the importance of better continental U.S. air defenses to protect against a nuclear-armed foe:

The atomic bomb has been employed three times, once in test in New Mexico and twice against Japan. Each time the bomb has demonstrated its great power and its characteristic suddenness and total destruction. It is a devastating weapon of surprise such as no military organization has ever before possessed. With it, warfare can be carried to the heart of a nation where it can utterly destroy that nation's capacity to produce and even to live.

Three well-placed bombs can knock out any large (1,000,000 population) modern city. One hundred bombers striking simultaneously could reasonably be expected to cripple any nation in one blow, delivering utter destruction to 20 or 30 of the critical cities or industrial areas of the nation. There can be no doubt that such power demands a revolution in military thinking.

....

The delivery (of the bomb) by airplane would be by special air units, located at bases primarily in the United States but using overseas bases for staging and final

¹¹Memorandum for Record, June 6, 1949, by Capt. James S. Russell, USN, Deputy Director of Military Application, USAEC, Subject: Visit by Mr. Froman with Certain Members and Staff of the Joint Congressional Committee on Atomic Energy.

arming. The mission of these special air units would be the delivery of atomic bombs anywhere in the world. Scientific and technical personnel of the Manhattan Engineer District would assist these units in the assembly and technical supervision of the bombs.

This will be necessary for this is not just another bomb. It is expensive; not adaptable to mass production methods; requires special handling and storage; shipping and assembly are highly technical problems; and delivery on the target requires highly skilled and specially trained personnel.

....

Defenses against the atomic bomb will probably always be inadequate. The only active defense foreseeable now is to stop the carrying vehicle. A thoroughly alerted defense with the most modern equipment and under the most favorable conditions would be able to stop a high percentage of the attacking vehicles.

Nevertheless, the experience of World War II indicates that sufficient numbers would get through to wreak great destruction. Continued research of the highest quality and urgency must be carried on in the active defense field.

....

... Our anti-aircraft defenses against aircraft carrying atomic bombs must be improved.¹²

The present technique of employment of the atomic bomb involves the use of the B-29 for delivery on the target. It is to be expected that any enemy will attempt with all his resources to destroy the carrying plane. It is essential that we make realistic estimates of losses enroute to a target area in order that the final delivery of a sufficient number of bombs may be insured.

¹²Memorandum to Brigadier General C. C. Alexander dated 8 January 1946 from Major General L. R. Groves, U. S. Army, pp. 1, 2, 3, 5.

The effective employment of the atomic bomb requires highly trained personnel using the best available equipment and adequately supported by the necessary military forces. Our armed forces must be trained to render this support if we are to succeed in the any future war which involves the use of atomic bombs.¹³

Concurrent with changes in military strategy would be changes in atomic weapons. During World War II, significant improvement in fission weapons, leading to megaton yield bombs, was foreseen quite apart from the development of thermonuclear weapons:

... By various methods that seem quite possible of development within six months after the first bomb is perfected, it should be possible to increase the efficiency up to perhaps [deleted] in which case the same amount of (fissionable) material would yield something like 24,000 tons TNT equivalent.

Further developments along this same line hold a possibility of producing a single bomb with such amounts of materials and such efficiencies as to run this figure up to several hundred thousand tons TNT equivalent, or even perhaps a million tons TNT equivalent. ... All these possibilities reside only in perfecting the efficiency of the use of elements "25" (U-235) and "49" (Pu-239). You will thus see that a considerable "super" bomb is in the offing quite apart from the use of other nuclear reactions.

... the purpose of this memorandum is to show that even without this development (of a thermonuclear bomb) the potentialities of a bomb of tremendously devastating effect are not far off if the efficiency of the fission bomb can be increased, as it undoubtedly can, by straightforward research

¹³Memorandum for The Chief of Staff, U.S. Army dated 12 February 1946 from Major General L. R. Groves, USA, Subject: New Weapons Development.

and development along the lines now in progress.¹⁴

We are convinced that weapons quantitatively and qualitatively far more effective than now available will result from further work on those problems. This conviction is motivated not alone by analogy with past developments, but by specific projects to improve and multiply the existing weapons, and by the quite favorable technical prospects of the realization of the superbomb.¹⁵

Even before the successful TRINITY test of the FAT MAN in July 1945, LASL realized that the relatively-inefficient LITTLE BOY was likely to be limited by tactical circumstances to "certain specific missions" and that the majority of postwar weapons would be the more-advanced implosion type, whose efficiency was still rather low.¹⁶

The 6.2 kilograms of plutonium in the FAT MAN had a potential energy of 124 KT, but only 21 KT was converted to blast, heat, light, and radiation in the TRINITY test, for an efficiency of about 17%. (One early post-shot estimate of TRINITY "gadget" nuclear efficiency was about 20%.)¹⁷

¹⁴Letter dated October 20, 1944 from James B. Conant to Vannevar Bush, Subject: Possibilities of a Super Bomb. Vannevar Bush-James B. Conant Files, Records of the Office of Scientific Research & Development, S-1, Record Group 227, folder 3, National Archives of the United States. The numbers "25" and "49" derive from the last digits of the atomic numbers and the atomic weights of $_{92}\text{U}^{235}$ and $_{94}\text{Pu}^{239}$.

¹⁵Letter dated 17 August 1945 to the Secretary of War from J. Robert Oppenheimer, reproduced in THE MANHATTAN PROJECT: A DOCUMENTARY INTRODUCTION TO THE ATOMIC AGE, edited by Michael B. Stoff, et. al., Temple University Press, Philadelphia, Pennsylvania, 1991, p. 254.

¹⁶Letter dated May 7, 1945 to General Leslie R. Groves, USA, from J. Robert Oppenheimer, Director, LASL.

¹⁷Transcript of telephone message to General L. R. Groves from Col. Stafford Warren dated 2:05 PM, 17 July 1945; Serber, THE LOS ALAMOS PRIMER (1992 edition), p. 60. In a memorandum for the Secretary of War from Maj. Gen. L. R. Groves, USA, dated 18 July 1945, appendix P in A WORLD DESTROYED, Martin J. Sherwin, Vintage Books, New York, 1987, pp. 308-3314, Groves states that the TRINITY test device used about 13.5 lbs. of

The FAT MAN core had other drawbacks. It was so large that it generated heat sufficient to decay adjacent weapon components; in addition, the final nearly-solid "Christy pit" used in the bomb was barely subcritical within its natural uranium tamper and was partially prevented from premature ignition by a removable cadmium wire.¹⁸

In a letter dated July 30, 1945 to Army Chief of Staff General George C. Marshall, General Groves reported that the FAT MAN, if perfectly efficient, could theoretically yield 100 KT.¹⁹ With sufficient concentrated postwar research, wartime bomb efficiency might conceivably be doubled or tripled.

LASL was not happy even with the as-yet-unused FAT MAN; however, improvements would have to wait until the end of the war:

There are a number of aspects known to all of us which make the FAT MAN an unsatisfactory gadget. Some of these are fundamental and have to do with the emergency release of the gadget. Some of these have to do with the clumsiness of its design and engineering. It would be my understanding of our duty that we should attempt to rectify these defects, subject always to the condition that this should not be allowed to interfere with the production of weapons so long as the war continues.

Pu.

¹⁸Rhodes, DARK SUN, p. 119.

¹⁹"Eclipsed by Hiroshima and Nagasaki: Early Thinking About Tactical Nuclear Weapons," Dr. Barton J. Bernstein, INTERNATIONAL SECURITY, Vol. 15 No. 4, Spring 1991, p. 160. This figure of 100,000 tons of TNT equivalent also appears in a 14 May 1945 draft Presidential press release on the atomic bombing of Japan, which had not yet occurred. Cochran, Arkin, and Hoenig, in NUCLEAR WEAPONS DATABOOK, Vol. I, p. 32, claim that the first two MK III FAT MAN shots in New Mexico and Japan had efficiencies on the order of 17%. Early pre-TRINITY estimates of FAT MAN efficiency, even as late as April 1945, were as low as 5%. (Memorandum to J. R. Oppenheimer dated April 25, 1945 from J. O. Hirschfelder, subject: Strategic Possibilities Arising if a Thunderstorm Is Induced by Gadget Explosion.)

I list below certain technical improvements which should be made in the gadget. This list is not exhaustive, and it may turn out that certain improvements are not practicable:

A. From the point of view of performance, the most important change is probably in the tamper active material design. We believe that [deleted] Correspondingly, this change in design should make possible a reduction of material used in the gadget, providing tactical demands favored a less powerful unit.

The actual engineering of this change is a relatively trivial matter, but a series of studies would have to be undertaken by us, going far beyond those now in our program, to ascertain with certainty whether the anticipated improvements would really result. It is improbable that any improvement in initiator design, or any other redistribution of active material, could lead as rapidly to as great an improvement in performance.

B. The lenses now being planned for the FAT MAN are clumsy and imperfect. As of February 1945, the Laboratory (sic) had several leads on promising lens developments which were abandoned in order to make way for the "crash" program. The development of superior lenses should make possible either of two alternative developments: (1) further reduction in power of the gadget; or (2) perhaps more profitable, a reduction in size and weight of the gadget to make more practicable the delivery from carrier-based aircraft.

C. Present explosive assembly techniques involved in making the gadget are irksome and unstandardized and add enormously to the "special" character of the weapon. We would like to develop an explosive system in which there was considerably more preassembly and in which there would be no exposed explosive shipped from (Los Alamos).

D. The electrical system now used to fire the detonators is clumsy and marginal and

we are sure that a more reliable and more compact unit can and should be developed.

E. We are not too well satisfied with the fundamental fuzing elements of the gadget and have already asked that the Radiation Laboratory undertake an improved design.

F. It would be our intention to incorporate as many of these changes as possible in a revised model gadget and to review the engineering of the gadget and its structural elements, and redesign them from scratch.²⁰

Many of these design changes and improvements would later be incorporated into postwar U.S. nuclear weapons. In fact, so many of the ideas for postwar improvements in U.S. fission weapons were discussed at least in passing at Los Alamos during World War II that by 1953, with each new U.S. development or advance in atomic weaponry, a Central Intelligence Agency (CIA) study was prepared to determine what information related to the idea had been made available to Klaus Fuchs, a Soviet spy at LASL during the war.²¹ Presumably most of what Fuchs learned, wrote, or heard at Los Alamos was turned over to the Soviets before his arrest in England in early 1950.

²⁰Letter dated May 7, 1945 to General Leslie R. Groves, USA, from J. Robert Oppenheimer, Director, LASL. The point under discussion at "A" is probably the principle of core levitation described later in this chapter. The "Radiation Laboratory" was the wartime radar lab at the Massachusetts Institute of Technology in Cambridge, which supplied the fuzing radars for both the LITTLE BOY and the FAT MAN (this MIT facility should not be confused with either of the University of California Radiation Laboratories at Berkeley and, after the war, also at Livermore). In February 1948, Oppenheimer referred to the FAT MAN as "the haywire contraption that was slapped together in 1945." (Draft Minutes of the Eighth Meeting of the General Advisory Committee to the Atomic Energy Commission, p. 27, quoted in U.S. NAVAL BOMBARDMENT MISSILES, 1940-1958: A STUDY OF THE WEAPONS INNOVATION PROCESS, PhD thesis by Berend Derk Bruins, Columbia University, New York, 1981, pp. 155, 156.)

²¹Letter dated July 14, 1953 to Allen W. Dulles, Director, Central Intelligence Agency, from W. Sterling Cole, Chairman, JCAE; letter dated 23 July 1953 to W. Sterling Cole, Chairman, JCAE, from Allen W. Dulles, Director, Central Intelligence Agency.

A little more than a year after the end of the war, Los Alamos was already looking forward to new weapons. In the fall of 1946, Dr. Edward Teller proposed five types of new bombs, including implosion designs using uranium-235.²² A few months later, in January 1947, Norris Bradbury proposed nine new schemes which might either improve the efficiency of implosion systems or make possible more economical use of uranium-235,²³ including improved implosion bomb designs and modifications to the uranium gun-type weapons to make them more efficient.²⁴

In April 1947, the commissioners of the AEC wrote to President Harry Truman to inform him of the precarious and weakened state of both the U.S. nuclear weapons stockpile and the country's nuclear warfighting capabilities:

After three months of authority over the American Atomic Energy (sic) enterprises, with access to sources of information and opportunity gradually to fit facts together, the Atomic Energy Commission must report to the President certain weaknesses in the situation from the standpoint of national defense and security:

1. The present supply of atomic bombs is very small. The actual number for which all necessary parts are available is _____. None of these bombs is assembled. The highly technical operation of assembly hitherto has been effected by civilian teams no longer organized as such. Training of military personnel to effect assembly is not yet complete.
2. The atomic core of the most important type of bomb currently being produced is new and has never been tested by explosion.

²²"Proposed Outline of Laboratory Program," memorandum by Edward Teller dated October 1, 1946.

²³Hewlett and Duncan, p. 58.

²⁴OPPENHEIMER: THE YEARS OF RISK, James W. Kunetka, Prentice-Hall, Inc., Englewood Cliffs, New Jersey, 1983, p. 120; Hewlett and Duncan, p. 58.

3. Production of critical parts of the bomb, including ingredients of the atomic core, has been and is badly out of balance. For instance, plutonium production is far behind uranium-235 production. To increase plutonium production adequately requires a bold decision and a major undertaking.
4. One of the other vital parts of the bomb, the nuclear (initiator) deteriorates rapidly with age, and we are not equipped to produce enough of these (initiators).
5. The present atomic (reactors) at Hanford (so-called piles) on which we are wholly dependent for making plutonium have developed structural changes which limit their useful lives to about 18 months more. This general condition has been under study for over a year. A major rejuvenation operation may extend their useful lives; but this rejuvenation operation will involve a staggered shutdown of at least four months.
6. These atomic (reactors) actually utilize only a very small part of the uranium fuel that is put in(to) them; hence, practically all of the uranium is lost because we do not yet have a recovery process to recapture it. It is still, as of today, waste.
7. Quantities of uranium as great as our whole above-ground stockpile have been poured into underground storage tanks as part of this "waste;" no method of recovering this is known today; and no serious attack on the problem of finding a way has yet been made.
8. Our present usable stockpile of uranium is sufficient only to feed our present refining and production plants, at full capacity, for about two years.

9. Nearly all our supply of raw uranium comes from the Belgian Congo; the demand in 1947 will exceed the probable supply; and supply from this source will become less year by year.
10. Use of thorium as an alternative source material is undeveloped, though presumably possible; and processes of extracting uranium from large possible sources at home and abroad, other than the sources now being exhausted, are also undeveloped.
11. We found that the Joint Chiefs of Staff had not been informed of the foregoing facts and we have taken steps to see that they are briefed.

The Commission has been moving to find solutions to these and other serious problems which are of great importance to the Nation. (sic) The Commission asked its General Advisory Committee to study concurrently the same problems, and it has this week received from the General Advisory Committee certain important recommendations.

Whatever corrective measures are adopted are certain to call at the earliest possible (date) for extraordinary efforts in terms of money, materials, equipment, and human energy. In several critical fields there is need for a full head of steam.

This situation is one which it is our duty to report directly to the President of the United States.²⁵

²⁵"Report to the President of the United States from the Atomic Energy Commission, January 1 - April 1, 1947," April 3, 1947, signed by AEC Commissioners Robert F. Bacher, Sumner T. Pike, Lewis L. Strauss, and William W. Waymack; and Carroll L. Wilson, General Manager, USAEC, and David E. Lilienthal, Chairman, USAEC.

The omitted number of bombs was somewhere between 9 and 13. In the early days of the AEC, while the stockpile was very small, this number was so sensitive that it was seldom

By mid-1947, two weapon types, little more than slightly-improved versions of the wartime bombs, were being produced in extremely limited quantity. Units going into stockpile

written and was only given orally to intended recipients. As in this example, only a blank space appeared in which to orally enter the figure when the document was presented. These numbers were very closely held: in August 1950, the Joint Committee on Atomic Energy was unable to obtain a top-secret report on military aspects of the hydrogen bomb because the report contained stockpile numbers. (Memo to the Files dated August 3, 1950 from Bill Borden.) Ten months later, when the JCAE requested a copy of an AEC chart, the chart was denied on the grounds that it "contained information verging on stockpile data." (Memorandum for the Files dated June 28, 1951 from Hal Bergman.)

In August 1951, AEC commissioner Lewis Strauss told William Borden of the JCAE that the JCAE should have access to stockpile data because "the Russians already know this data, largely through (the espionage of Klaus) Fuchs." Strauss added that the compelling reason for first withholding this data from the JCAE was that the stockpile was very small. Strauss added that the current (August 1951) stockpile size was "not astronomical but would seem so from the way things looked in 1947." By this time, the U.S. nuclear weapons stockpile numbered somewhere around 438; the 1947 number was only 13. (Memorandum for the File dated August 13, 1951 from Bill Borden.) In a September 27, 1948 LIFE magazine article entitled "A Progress Report on Atomic Energy - A Dialog with David Lilienthal," AEC Chairman Lilienthal stated (on p. 127):

There is one piece of vital information which is known only to the commissioners and few others. It is not disclosed to the Joint Congressional Committee on Atomic Energy -- at the Joint Committee's request. It wasn't known even to the top military people when the commission took over. ... This information is information about how many atomic weapons we have, the number and (yield), and rate of production.

In the report to the President, the initiator is called a "fuse" and the Hanford reactors are called "furnaces." The "structural changes" are a reference to swelling of graphite in the reactors as a result of the intensely radioactive environment in which they operated. The "atomic core of the most important type of bomb currently being produced" is probably a reference to a composite plutonium-uranium unlevitated "Christy type" solid core; by December 1947, this type of core constituted the majority of the stockpile.

were essentially laboratory-built, and the limited production chain had only one source for each major component and some of the sources were rapidly wearing out from World War II operation. Three major modifications of one MARK weapon, the MK III, were in early development and would not reach production for some time.²⁶

These weapons were far from combat-ready:

We had lots of capsules -- nuclear cores -- but we didn't have weapons, we had lots of pieces. The idea was, if there was a threat, you would start putting them together. The fuzing systems weren't there; the initiators had to be changed; the detonators had to be stored in dessicated boxes and you put them in(to the bombs) when you needed them .. And it went on and on this way. We didn't have any weapons, we had piles of pieces.²⁷

The Search for Simplicity

One of the highest-priority projects at the end of the war was to reduce bomb size, so that more types of aircraft could carry them, and to simplify bomb assembly and handling. The wartime FAT MAN implosion bomb was almost 11 feet long, five feet in diameter, and weighed about 10,000 lbs.

The large diameter was dictated mainly by two thick layers of high explosive (HE) charges which imploded the pusher-tamper-plutonium-initiator assembly.²⁸ This double layer of explosives was composed of an outer layer of "fast" burning Composition B (a suspension of RDX in TNT) "lens" charges with "slow" burning Baratol (a mixture of TNT and barium nitrate) centers, and more "fast" burning Composition B charges on the inner layer.²⁹ This double layer of HE charges

²⁶Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 2.

²⁷Rhodes, DARK SUN, p. 284.

²⁸See, for example, the FAT MAN core diagram in Lansing Lamont's DAY OF TRINITY (Atheneum Books, New York, 1965), p. 176, and the photo of the FAT MAN core mockup in Kunetka's CITY OF FIRE. The Lamont diagram is in error, because it shows voids, or empty spaces, among the high-explosive "lenses" around the core. The photo in the Kunetka book shows the proper configuration of these "lenses" (without voids).

²⁹Hoddeson, et. al., p. 299.

spawned and shaped an inwardly-directed spherically-symmetric implosive shock wave to compress the aluminum-uranium-plutonium weapon "pit."³⁰

These layers of dense explosives could be quite thick and heavy: the MK 6 and MK 18 weapons, for example, had cores surrounded by a 17" (43.97 cm)-thick blanket of HE charges; devices detonated during the TUMBLER-SNAPPER series of nuclear tests at the Nevada Proving Grounds in the spring of 1952 used arrays of HE around their cores between 5" (13.17 cm) and 17" thick.³¹ Approximately half the weight, about two and a half tons, of the FAT MAN (MK III prototype) was high explosives used to detonate the nuclear charge.³²

Bomb diameter could be reduced by either the development of more-powerful high explosives, requiring less material for the same degree of "pit" compression, or by developing new techniques to get more energy into the core from less explosives (most early postwar 60-inch diameter atomic weapons and thermonuclear weapon triggers used a double layer

³⁰J. ROBERT OPPENHEIMER, SHATTERER OF WORLDS, Peter Goodchild, Houghton Mifflin Company, Boston, Massachusetts, 1981, pp. 114-115; THE MAKING OF THE ATOMIC BOMB (hereafter "TMAB"), Richard Rhodes, Simon and & Schuster, New York, 1986, pp. 575-577.

³¹INITIAL GAMMA DATA FROM NUCLEAR WEAPONS TESTS 1948 THROUGH 1962, Nuclear Defense Laboratory, Edgewood Arsenal, Maryland, 1965.

³²NUCLEAR WEAPONS DATABOOK, Vol. I, Cochran, et. al., p. 31; Kunetka, CITY OF FIRE, p. 164; memorandum for the Secretary of War from Maj. Gen. L. R. Groves, USA, dated 18 July 1945, appendix P in A WORLD DESTROYED, Martin J. Sherwin, Vintage Books, New York, 1987, pp. 308-3314. Necah Stewart Furman, in SANDIA NATIONAL LABORATORIES: THE POSTWAR DECADE (University of New Mexico Press, Albuquerque, New Mexico, 1990, p. 74), states that the FAT MAN explosive assembly tested in July 1945 during Project TRINITY contained 5,300 lbs. of HE. (The Furman book is an exhaustive and excellent account of Sandia's first ten years, including details of many of its projects and personnel.) Rowe, on p. 107, notes that each FAT MAN contained "over two tons of conventional explosives in the sphere;" he also states on p. 183 that eight PUMPKINS, FAT MAN dummies loaded with HE, "represented 20 tons of explosives." Other sources give figures as low as 4,000 lbs. for the FAT MAN HE weight. Baratol has a specific gravity of 2.61 g/cc; Composition B has a specific gravity of 1.68 g/cc. If a 17.31" thick HE sphere with an outer radius of 26.3651" is composed of these two materials on a 1:1 basis, then total weight would be slightly over 5,708 lbs.

of high explosives, even after new core suspension techniques had evolved).

Reductions in the weight, density, and volume of high explosives could be another path to lighter-weight weapons. Nuclear efficiencies that might be lowered by a reduction in HE volume could be restored to acceptable or even high levels by improved use of the fissionable cores, by either mechanical means or by new tamper-reflector materials.

Postwar progress in the development of new HE systems was rapid. For example, by mid-1951, a large amount of work had been done on the high explosives used in U.S. atomic weapons, including designing and testing of smaller HE systems to reduce weapon size, increasing the energy content of the HE in order to increase implosion force to raise nuclear efficiency, and designing smaller detonators to reduce weapon diameter.³³

Early U.S. atomic bombs contained very thick and heavy natural uranium tampers around their cores. The tamper, which inertially-confined the fission explosion to allow its force to grow to an appreciable extent before the core explosively disassembled, was quite massive: a 9" diameter spherical uranium tamper with a 3.6" diameter hollow center weighed about 255 lbs. Obviously, lighter, more-efficient tampers could make a significant contribution to overall weapon weight reduction.

The weight of the bomb could also be reduced substantially by substituting a lighter, internally-braced aircraft-style ballistic casing structure for the heavy armored castings used to protect early bombs from anti-aircraft fire while being carried to the target and after release over the target. This weight reduction could be as much as 1,000 to 3,000 lbs. without any change in the internal components.

The wartime FAT MAN was also awkward and difficult to service and arm: the useful life of the lead-acid batteries in it which stored power for the detonators was only a few days; recharging or replacing them involved disassembling most of the bomb. Weapon arming and safing procedures were also complicated and lengthy, requiring nearly complete assembly or disassembly of the bomb.

Weapon performance would also be improved by an increase in both yield reproducibility and nuclear efficiency. The

³³Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 19.

reliability and efficiency of the nuclear explosives in the bomb core depended upon many variables:

- the performance of the chemical high explosives
- how rapidly the subcritical mass could be assembled into a supercritical mass
- the density, inertia, composition, and mass of the tamper encasing the fissile core
- the performance and efficiency of the internal or external initiator or neutron source
- the amount of kinetic energy transferred to the core, and the resultant degree of core compression
- the extent of corrosion of the core materials caused by electrochemical reactions between dissimilar metals, accelerated by the heat of the slowly-decaying plutonium or uranium core (a relatively large piece of plutonium is warm to the touch because of the energy released in alpha decay; larger pieces will produce enough heat to boil water).³⁴

All of these factors needed careful and thorough study and evaluation before any improvement in yield for a given amount of fissile material could be realized.

³⁴Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report on Atomic Weapon Tests at Eniwetok 1951, Annex 9.2, Sandia Corporation Proving Ground Group, Part V -- Activities of the Nuclear Division, WT-102, R. D. Krohn, Supervisor, Task Unit 3.1.4.5, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, August 1951, p. 21. One of the features of the Los Alamos Scientific Laboratory technical research program for 1949 was "attention to problems of deterioration of plutonium coatings;" plutonium hemispheres were usually plated with nickel which sometimes blistered or pitted. (Letter dated 7 September 1948 from N. E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Office of Santa Fe Directed Operations, U.S. Atomic Energy Commission, p. 12.) Alpha decay of plutonium produces about 0.027227 calories per minute ("Explosive Properties of Various Types of Plutonium," paper presented by Richard L. Garwin at the NATO Advanced Research Workshop on Managing the Plutonium Surplus: Applications and Options, Chatham House, London, January 24, 1994, p. 4.), or about two watts per kilogram. ("Nuclear Weapons Supply and Demand," Michael M. May, AMERICAN SCIENTIST, Volume 82, November-December 1994, p. 530.)

In addition, bomb casing ballistics needed much improvement: the FAT MAN dropped at Bikini atoll in July 1946 for the "Able" shot of Operation CROSSROADS missed its target (the battleship NEVADA) by 2,130 feet.³⁵ The "California Parachute" high-drag tail fin structure was quite primitive and clearly left much to be desired.

New ballistic casings and fins, or drogued (parachute-retarded) delivery would be necessary for future weapons, in order to achieve better accuracy, and to slow the falling bomb to allow the delivery aircraft time to escape the blast wave of the nuclear explosion.

During the early days of the immediate postwar U.S. nuclear weapons program, ideas were proposed and developed that promised more versatile and efficient weapons; these concepts had been bypassed during World War II and required postwar testing before they could be introduced with any measure of certainty into new weapons.³⁶

Improved weapons concepts and systems, ideas that had been around the laboratory but had not been incorporated because of time pressure, were brought out, dusted off, and put into practice. New designs promised to double the efficiency of the U.S. atomic weapons stockpile, effectively doubling either the size or the total yield of the stockpile at one stroke.³⁷

By 1948, development work promised greatly improved weapons if certain projected changes in their design and content were

³⁵TECHNICAL REPORT OF OPERATION CROSSROADS, W. A. Shurcliff, Defense Atomic Support Agency, November 1946, Section 19.009 (A). A contemporary report, HISTORY OF AAF TECHNOLOGICAL PARTICIPATION IN OPERATION CROSSROADS, 1946 (Army Air Forces Air Materiel Command, August 1946) states (p. 16) that the miss distance was 1,500 to 2,000 feet west of the assigned target, the battleship USS NEVADA. Shurcliff, however, states that the actual slant distance was 2,130 feet (710 yards). According to Col. Paul W. Tibbets in FLIGHT OF THE ENOLA GAY (Buckeye Aviation Book Company, Reynoldsburg, Ohio, 1989, pp. 250, 251), the bombardier of the drop plane miscalculated ballistic data and ignored corrected ballistic computations offered by bombardiers who had dropped the two wartime FAT MAN weapons less than a year earlier.

³⁶A HISTORY OF PROJECT REDWING: THE ATOMIC WEAPONS TESTS IN THE PACIFIC, 1956. Headquarters, Joint Task Force Seven, Washington, D.C., 1 December 1956, p. 7.

³⁷NUCLEAR HOSTAGES, Bernard J. O'Keefe, Houghton Mifflin Company, Boston, Massachusetts, 1983, pp. 134-135.

made. Some of the very straightforward wartime options in weapon design, dealing primarily with core composition and mechanical assembly, were implemented in the new generation of fission weapons tested during Operation SANDSTONE at Eniwetok atoll in the spring of 1948:

Senator Hickenlooper: Were not the weapons tested at Eniwetok conceived during Manhattan District days and laid aside for testing later?

Dr. Bacher: If by "conceived" you mean that there was a gleam in the eye, then I would agree with you.

Senator Hickenlooper: I meant that some of the mechanical ideas were proposed under the Manhattan District.

Dr. Bacher: One of the principles incorporated in the Eniwetok tests had been planned during the war. But one of the major developments -- I would say the major development -- we would not have dared to do at that time. I would say that what was tested at Eniwetok was designed on the basis of work carried out at Los Alamos during 1947. The Eniwetok tests led to the design of weapons making considerably better use of fissionable material than any we knew about before. One point was completely new, and even contrary to some of the ideas we had during the war.³⁸

The SANDSTONE tests gave highly satisfying results that led to essentially immediate plans for changes in the kinds of weapons for the military stockpile.³⁹

³⁸BULLETIN OF THE ATOMIC SCIENTISTS, 1949, pp. 241, 242. "Dr. Bacher" was Dr. Robert F. Bacher of the Los Alamos Scientific Laboratory; "Senator Hickenlooper" was Senator Bourke B. Hickenlooper of Iowa, then chairman of the Joint (Congressional) Committee on Atomic Energy (JCAE). These statements were made during testimony before the JCAE in early July 1949. Dr. Bacher was probably referring to composite cores and levitated pits; the point contrary to wartime ideas was uranium implosion, which Los Alamos believed impractical during World War II.

³⁹Interview with J. Carson Mark in LOS ALAMOS SCIENCE, Volume 4 No. 7, Winter/Spring 1983, p. 34.

One of the first and most important results of SANDSTONE was the abolition of the so-called "doctrine of scarcity" which had dictated U.S. Air Force strategic war planning, and indirectly, the entire foreign policy of the United States defining when, where, and under what circumstances the government would be prepared to wage nuclear war.⁴⁰

This doctrine assumed that because of a shortage of raw materials and processing capability, and because relatively large quantities of fissionable material were required at great cost for each weapon, the U.S. would continue to have for quite some time -- possibly for as long as the next 10 to 20 years -- only a very limited supply of atomic bombs:⁴¹

The limited amounts of fissionable, weapons-grade material likely to be available and the great costs associated with facilities and bomb production, which itself was a meticulous and time-consuming process, indicated that the number of bombs on hand for any future conflict would be small. For some time to come, the bombs were expected to remain experimental "special weapons" of great cost and complexity, difficult to build and transport, and of limited availability.

... Cost and limited numbers meant that the weapons would have to be used sparingly and then only against the most important targets.⁴²

Early in 1946, Manhattan District chief Major General Groves informed the Army Chief of Staff that "the atomic bomb which will be available to the armed forces of the United States in January 1948, 1949, and 1950 will be of the same general type as the bombs dropped on Japan in August of 1945."

⁴⁰DEVELOPMENT OF THE SNARK GUIDED MISSILE, Robert L. Perry, U.S. Air Force Aeronautical Systems Division, 1956, pp. 62, 63.

⁴¹"Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, pp. 84, 169.

⁴²"The Atomic Bomb -- Early Air Force Thinking and the Strategic Air Force, August 1945-March 1946," John T. Greenwood, AEROSPACE HISTORIAN, Vol. 34 No. 3, Fall/September 1987, p. 161.

Groves noted while U.S. atomic weapons had "terrific destructive power," they would be available in very limited quantities and should therefore be used against only targets of major military importance. These targets included major industrial production centers, ports of embarkation, major supply installations, enemy beachheads, large concentrations of troops, important military headquarters, and enemy atomic bomb storage and production centers.⁴³

Groves did not inform the Army Chief of Staff that if a war between the Soviet Union and the U.S. broke out within the next few years, the U.S. did not have an adequate stockpile of nuclear bombs, and trained aircrews and bomb assembly teams, to even begin attacking such an ambitious list of potential targets.

The results of Operation SANDSTONE rendered the "doctrine of scarcity" obsolete. Changes in weapon design resulting from SANDSTONE, as well as an expansion of the United State's atomic bomb production complex (including the establishment of the new Burlington, Iowa bomb fabrication plant, where the first MK 4 was assembled during 1949) caused the U.S. stockpile of nuclear bombs to triple between June 1948 and June 1949, from 56 to 169 weapons.

Another consequential early postwar consideration in U.S. atomic bomb design was the efficient use of available fissionable materials. The general postwar objective of LASL was to maintain a continuous and significant overall U.S. superiority in its ability to wage atomic warfare, through research and development leading to the most effective use of current production of fissionable materials.

This objective implied constant change in the distribution in the national stockpile of weapon designs of various intrinsic efficiencies and deliverability; improvement in the efficiency of each design having a given deliverability; development of weapons of acceptable efficiencies and wide deliverability; and increasing the absolute number of weapons available from fixed quantities of fissionable and other scarce materials without unduly decreasing weapon efficiency or deliverability.⁴⁴

By November 1949, work had begun on the so-called "fractional crit" bombs which used only a fraction of the amount of

⁴³Memorandum for the Chief of Staff, U.S. Army, dated 12 February 1946 from Major General L. R. Groves, USA, Subject: New Weapons Development.

⁴⁴"Feasibility of Frequent Bomb Tests," ADWD-63, LASL paper by Darol Froman, October 10, 1949, pp. 2, 3.

fissionable materials in earlier designs. Dr. Theodore Taylor, an important LASL bomb designer, says of this period:

(Test) results that we had on hand at the time I arrived at Los Alamos in November 1949 were from (Operations) TRINITY, SANDSTONE, and Bikini (CROSSROADS). But there was a lot of work to do on new designs so I spent a lot of time on that when I first started.

There were already some things (that had been) started to improve the performance beyond what had been (obtained) at SANDSTONE which basically involved not adding anything, but rearranging what was inside the implosion systems; rearranging in them such a way as to get more energy to go into compressing the uranium-235 or plutonium, or mixtures involved.

The entire emphasis at that time when I got there was on what some people called the "full-crit" assembly, in which one put in close to a critical mass (of fissionable material) as it sat there. And then the function of the implosion system was to make it substantially supercritical. We wanted the biggest yield we could get out of what we had. So that was the context.⁴⁵

At the end of World War II, there was no shortage of ideas or ambition within the armed services, especially the Army Air Forces, for future applications of nuclear energy. Almost every conceivable type of delivery vehicle and delivery method was envisioned; even nuclear-powered missiles and aircraft might eventually be feasible:

Development of a super-velocity anti-aircraft artillery guided missile, using atomic power for the warhead, and if practicable, atomic power as the propelling force, for employment as a defense against V-2 type missiles and for guided missiles employing atomic power as the explosive.

⁴⁵ "An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 2.

Development of an anti-aircraft artillery guided missile similar to that described above, except that the required velocity may be lower, for employment as a defense against massed formations of hostile bombers (engaged in) saturation bombing attacks.

Development of a relatively short-range rocket or guided missile or artillery projectile using atomic power as the explosive for employment as an offensive weapon to blast out hostile ground defenses of all types in preparation for a breakthrough similar to the operations at Anzio, Cassino, and St. Lo, or as a preliminary to a landing on a hostile shore.

Development of a seacoast defense guided missile, using atomic power in the warhead, for employment against hostile landings in force, and, if proposed tests prove such employment feasible, against hostile fleets at sea.

Long-range research and development to develop, if possible, the use of atomic power to replace explosives as propellants for all types of missiles.⁴⁶

The Army Air Forces (sic) has the following general program in view with regard to further development of the atomic bomb and possible adaption of the atomic unit in the bomb to other methods of delivery. This program will be subject to continuous modification and change, but at present it is believed developments along the lines indicated are most practicable and feasible.

Missiles and projectiles of interest to the Air Forces fall generally into the following categories:

- a. Bombs -- either controlled or uncontrolled in fall but depending on the airplane and gravity for their delivery.

⁴⁶Memorandum dated 11 October 1945 for Brigadier General W. A. Borden, New Developments Division, War Department General Staff, from Lt. Col. R. A. Meredith, Adjutant General Division, Assistant Ground Adjutant General, Headquarters Army Ground Forces, Subject: Research Projects in Atomic Field.

b. Airplane-carried missiles -- carried by aircraft to the vicinity of the target, then released and directed to the target by remote control.

c. Aircraft-type missiles -- crewless aircraft, remotely controlled, and flown to a target by remote control.⁴⁷

Development of atomic energy to use as the explosive in the warheads for all missiles and projectiles such as:

- a. Conventional bombs
- b. Large-caliber artillery projectiles
- c. Rocket-assisted artillery shells
- d. Gravity-powered guided missiles (both vertical and glide-bomb types)
- e. Pilotless aircraft to be used as controlled missiles
- f. Ground-launched missiles that will include the following:
 - (1) Ground-to-air -- antiaircraft or antimissile guided missiles
 - (2) Ground-to-water -- seacoast-type guided missiles for the destruction of capital ships
 - (3) Ground-to-ground

⁴⁷Memorandum for General Borden dated 16 October 1945 from Major General Lauris Norstad, USA, Subject: Air Forces Development Program for Atomic Missiles.

- (a) Short-range tactical missile
- (b) Medium-range missile such as the German V-101 (the Germans anticipated a range of 3,000 miles)
- (c) Strategic missiles of hemispheric range⁴⁸

Most of the desires on this early postwar nuclear weapons "wish list" were formulated by persons who had absolutely no knowledge of either the limited numbers of atomic weapons then available or the physical characteristics of those weapons.

Despite this ignorance, within a few years of the end of the war, some of the Army's desires had been fulfilled. In comparison to the status of the U.S. fission weapons program in 1947, which featured only two basic designs, by July 1950, although only one type of bomb (the MK 4) was being stockpiled, it was factory-produced and was being manufactured in considerable numbers.

Additionally, two other types of implosion bombs (the MKs 5 and 6) and one gun-type bomb (the MK 8) were being readied for production. Two other weapons programs (the MK 7 bomb and MK 9 artillery shell) were well along in development and a major modification program (of MK 4s to 6s) had been started.⁴⁹

⁴⁸Memorandum dated 18 October 1945 for the Chairman, War Department Atomic Energy Advisory Board, from Brig. Gen. William A. Borden, Director, New Developments Division, War Department Special Staff, Subject: War Department Research and Development Program for the Employment of Atomic Energy.

⁴⁹Report of the Manager, Santa Fe Operations, USAEC, July 1950

Also at this time, based on a figure of merit derived from many variables, current U.S. implosion-type composite-core atomic weapons were making four times more efficient use of the same types of fissionable materials (uranium-235 and plutonium) than the first two wartime bombs.

In other words, this meant that LASL's work up to then had made possible the generation of four times as much explosive power, with the same amounts of fissionable materials, as would have been possible if wartime design and manufacturing methods and principles had continued unchanged.

Looking at the figures in another light, they indicated that production of plutonium by the aging Hanford reactors and production of enriched uranium by the Oak Ridge gaseous diffusion plant would have to have been quadrupled to produce an equivalent total explosive power if bomb designs had not been improved since the end of the war.⁵⁰

At the end of 1950, a special ad hoc committee set up to investigate military objectives in the field of atomic energy made several recommendations for future weapon improvements, including both high-yield, high-efficiency bombs and low-yield tactical warheads:

The most urgent requirements for research and development lie in the field of fission weapons:

- a. In the improvement of the fission warheads themselves to increase the flexibility of their use and their economy of fissionable material;
- b. In the development of varied and flexible (delivery vehicles) and weapons systems and operations for the delivery of fission weapons;
- c. In the development of production techniques to provide the material support for the fission weapon program.

Our present stockpile of atomic weapons is designed for high-altitude delivery and consists almost entirely of airburst

to January 1954, 1 January 1954, p. 5.

⁵⁰Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 16.

implosion weapons. The magnitude of this stockpile is determined by the available implosion techniques and fission weapon designs, and by the amount of fissionable material available.

It is anticipated that this amount of material will increase at least [deleted] within the next five years. Of even more immediate importance is the possibility of making more effective use of fissionable material by redesign of the weapons themselves. This we believe to be possible. Of equally high importance is the improvement, in accuracy and reliability, of existing delivery systems.

...

A. Fission Warheads

Perhaps the most substantial, if not the most dramatic, development of the last few years has been the increasing mastery at Los Alamos of the implosion techniques for atomic weapons. This opens up a variety of applications of great importance. Thus, it should make possible the development of effective weapons using substantially smaller amounts of fissionable material [deleted]

It should also make possible the development of fission weapons with energy release on the order of 100 to perhaps 500 kilotons TNT equivalent. It has already made possible the development of smaller diameter weapons, and we are probably only at the beginning of this road.

Among the objectives suggested by past developments, and partially reflected in the Joint Chiefs of Staff requirements for 1952 and 1953, two urgent programs deserve special high priority emphasis. One of these is the reduction in the amount of fissionable material to be used in airburst bombs of current dimensions (60 inch and 45 inch diameters).

The second is the development of implosion weapons of small diameter. In fact, the Panel believes that Los Alamos should be encouraged to develop, within 12 months,

the smallest implosion weapon which both gives an energy release in excess of five kilotons and makes a substantial improvement, compared to the airburst gun-assembly, in the number of weapons which can be made from fissionable material, taking into account the relative value of uranium-235 and plutonium.

The first of these two urgent undertakings is in the first instance directed toward an immediate increase in weapons available for high-altitude air delivery. The second is more clearly directed, as more fully indicated below, toward opening up new weapon systems for the delivery of atomic weapons which may be of especial importance for striking at tactical and other military targets and for the support of ground and naval operations.

The report also suggested a number of specific improvements in weapon technology and design, including hollow cores, external initiators, and interchangeable nuclear components:

There are several more fundamental developments that may well go beyond the two specific programs emphasized above. One of these, possibly of short but more probably of intermediate range, is the development of hollow implosions. It has been recognized for many years that if this technique could be successfully realized, it would give the most efficient use of fissionable material, [deleted] in atomic warheads of military interest, and that with suitable initiation it could lead to atomic weapons of very great energy release.

A second development, probably of the short term, is that of an external initiator which should be designed to give a pulse of neutrons or of gamma rays at a time most favorable for the initiation of the nuclear explosion. At present, both types of initiators appear hopeful, the neutron initiator offering perhaps the greatest promise of simplicity and compactness.

These external initiators, while expected to give some increase in yield of many existing warheads, are probably indispensable if hollow implosions are to

realize their full benefits. They also of course are of interest in alleviating the stockpile problems of initiators based on radioactive materials of relatively short life time.

To the maximum extent practical, the nuclear components of atomic weapons should be interchangeable. Although this objective is desirable, it should clearly not be allowed to conflict with the development of an effective, very small implosion weapon, or with the use of hollow implosions, should they fulfill the promise of high efficiency.⁵¹

In August 1951, the AEC and Department of Defense (DOD) were funding at least 19 separate and specific models of atomic weapons including five different types of missile warheads. These weapons comprised penetrating bombs, a very-high yield bomb, an atomic artillery shell, and airburst bombs that could be carried either externally or internally by aircraft as small as jet fighters.

At the end of 1951, advances made in the development of "fractional crit" weapons by LASL during Operation RANGER, coupled with an ambitious AEC expansion in production of fissionable materials (150% in oralloy, and 50% in plutonium), would effectively double the national stockpile size within the near future.⁵²

By 1955, the AEC expected to be producing fissionable material and weapons annually at a rate in excess of total current stockpile size, and with a total yield in excess of all the conventional high explosives dropped by Allied aircraft over Europe during World War II. A figure of nine million tons, or nine megatons, of TNT-equivalent explosive power was possible⁵³ (these estimates would soon be dwarfed by the development of thermonuclear weapons, each of which could have yields in excess of 15 megatons).

⁵¹"Military Objectives in the Use of Atomic Energy," Department of Defense Research and Development Board Committee on Atomic Energy, 29 December 1950, pp. 3-5.

⁵²Minutes of the Twenty-Eighth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, December 12, 13, and 14, 1951, p. 4.

⁵³Memorandum to the File dated July 11, 1951 from Wayne P. Brobeck, subject: Meeting with Walter Williams, AEC, on Friday, July 6, p. 2.

The Army, Navy, and Air Force had also started to battle over the allocation of available fissionable materials and over tactical and strategic nuclear battlefield combat roles.⁵⁴

In the fall of 1951, key installations for U.S. nuclear weapon components included Oak Ridge, Tennessee; Mound, Ohio; Hanford, Washington; and the new Savannah River works in Aiken, South Carolina. These facilities all produced nuclear components, including plutonium, enriched and unenriched uranium, tritium, lithium, polonium, beryllium, and deuterium.

Non-nuclear weapon parts were being fabricated at or by Project "Eye," the Salt Wells Pilot Plant of the Naval Ordnance Test Station, at Inyokern, California which had opened during World War II;⁵⁵ Project "Orange," the Pantex ordnance plant, near Amarillo, Texas which was activated in October 1951; Project "Royal," the Bendix Aviation plant in Kansas City, Missouri that opened in February 1949; Motorola; Douglas Aircraft in El Segundo, California; Eclipse Pioneer; Frise Instruments; Kollsan Instruments; Ford Instrument; Project "Pepper," the U.S. Army Picatinny Arsenal in New Jersey where AEC activity had begun in 1948; and Amphenol Corporation.

Table II-1 lists early AEC facility code names and DOD nuclear weapons storage sites.

Starting in 1952, a major portion of nuclear assembly work would be done at Project "Apple," the Rocky Flats, Colorado plant that opened in April 1951 with the balance of assembly at Los Alamos; the major portion of non-nuclear bomb assembly work would be conducted at Project "Sugar," the Army's Burlington, Iowa ordnance plant which had been activated for the AEC late in 1947, and at Pantex and Sandia (the latter was in Albuquerque, New Mexico).⁵⁶ Fuse assembly and manufacture of spare parts would be divided between Project "Royal" and Sandia. Production of atomic weapons began at Burlington in 1950.⁵⁷

⁵⁴Minutes of JCAE Meeting, August 16, 1951, p. 1264.

⁵⁵Memorandum dated 25 October 1945 to N. E. Bradbury from G. B. Kistiakowsky, subject: Stockpiling of FM.

⁵⁶FIRST YOU TAKE A PICK & SHOVEL: THE STORY OF THE MASON COMPANIES, Ann Arnold Lemert, The John Bradford Press, Lexington, Kentucky, 1979, pp. 161 - 164, 166, 167. According to Lemert, Burlington was code-named "Sugar" because the powdered high explosives handled there resembled brown sugar.

Rocky Flats was expected to relieve much of the production and inspection workload of LASL. Early in 1952, "Apple" was to take over a major share of LASL's presently-assigned routine nuclear inspection and assembly responsibilities; by 1953, when other facilities were expected to be in operation at "Apple," LASL would stop its heavy nuclear rework and obsolete weapon retirement programs.

By late 1953 or early 1954, Rocky Flats was to conduct production engineering for new LASL weapons and start serving as the production control and stockpile surveillance center for the AEC's nuclear fabrication and storage empire.⁵⁸

In June 1952, the AEC's General Advisory Committee summed up the state of the U.S. fission weapons program at that time:

We have learned to make much better weapons with very much less (fissionable) material. We have learned to make practical weapons more than ten times as powerful as the first atomic bomb. We have learned to make weapons adaptable to a great variety of delivery systems, to fighter aircraft as well as to bombers, to rockets and missiles as well as to aircraft, and to battlefield situations as well as to the bombing of industrial targets.⁵⁹

By the end of 1952, the U.S. implosion weapons development program alone had resulted in no fewer than seven types of HE assemblies (MK 3, MK 4, MK 5, MK 6, MK 7, TX-12, TX-13); eight or nine pit designs (Type A, Type B/48LP-B, Type C/49LP-C, Type D/52LP-D, Type D/53LP-D, Type E/59P-E, Type E/60P-E, Type F/61P-F, and at least one "special" design); 14 production cores (Types 030, 050, 080, 090, 100, 110, 120, 130, 140, 150, 160, 170, 180, and 190); at least four special cores (including boosted designs) used during Operations RANGER, BUSTER, GREENHOUSE, and TUMBLER-SNAPPER; and two types of internal initiators ("Urchin" and "Tom").⁶⁰

⁵⁷Lemert, p. 167.

⁵⁸Memorandum dated 26 October 1951 to Col. K. E. Fields, Director of Military Application, USAEC, from Carroll L. Tyler, Manager, Santa Fe Operations, USAEC, Subject: Maximizing the Use of Atomic Energy for Defense.

⁵⁹Letter dated June 14, 1952 to The President, The White House, from J. R. Oppenheimer, Chairman, General Advisory Committee.

⁶⁰"Vade Mecum of Implosion Bombs," W-4-380, R. C. Anderson and B. A. Mohr, Los Alamos Scientific Laboratory, Fall 1952;

In January 1953, a construction program was underway which would add eight plutonium production reactors and ten gaseous diffusion U-235 production plants to the five reactors and two gaseous diffusion plants already operating in mid-1950. These plants and reactors supported an enormous expansion of the U.S. nuclear weapons stockpile.⁶¹

By July 1953, no fewer than 47 separate nuclear bomb and warhead developments were in varying stages of completion, from studies to stockpiling. These programs, compared to the statuses of similar programs three years earlier, are shown in **Table II-2**. As can be seen, the largest expansion was in the area of missile warheads. **Table II-3** shows that no fewer than 16 missile-warhead marriages, for all three U.S. military services, were underway or under consideration by December 1953. These included warheads for air-to-surface, surface-to-air, surface-to-surface, and surface-to-underwater delivery vehicles.

All weapons stockpiled before mid-1950 had been either retired or converted to improved types. Seven new weapon types were being delivered to nine continental U.S. storage locations; these weapons included a full range of implosion weapons, two gun-type bombs, and one rocket warhead installation. The delivery rate of weapons to stockpile was triple the 1950 rate.

Considerable developmental progress had been made on guided missile warheads, with three different types nearing production and stockpiling. As of July 1, 1953, 41 new weapon programs were in progress and six major weapon modification programs were underway.⁶²

letter dated 26 December 1951 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Norris Bradbury, Director, LASL, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 7; WT-102, p. 32; Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 267.

"Vade mecum" is Latin for "go with me;" it also means handbook or manual.

⁶¹"The Origins of Overkill: Nuclear Weapons and American Strategy, 1945-1960," David Alan Rosenberg, INTERNATIONAL SECURITY, Spring 1983, p. 23; "Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, p. 176.

⁶²Report of the Manager, Santa Fe Operations, U.S. Atomic

At the end of 1953, fission weapon developments since 1945 had produced:

- A gun-type weapon (the MK 8) having about one-third the weight and 60% greater yield than the MK I LITTLE BOY
- A gun-type weapon (the MK 9 artillery shell) having less than one-tenth the weight of the MK I gravity bomb but approximately the same efficiency and yield
- A range of stockpiled implosion bombs varying in diameter from 30 to 60 inches, in weight from 1,600 to 8,000 lbs., and in yield from a few kilotons up to 500 kilotons (not including the 22-inch diameter MK 12 then in development)
- A system of interchangeable cores and bombs with different HE assemblies so that yield and bomb size could be chosen relatively independently to match a specified target or delivery aircraft

The amount of fissionable material required to produce a yield equal to that of the wartime FAT MAN bomb was now only a quarter or a fifth of that required for the 1945 bomb; equivalent bombs in 1953 had a slightly higher yield and slightly smaller weight. Alternatively, the total amount of fissionable material expended in the Hiroshima and Nagasaki bombs could now be used to make several times as many equivalent-yield MK 7 bombs, each having a weight only one-sixth that of the 1945 weapons.

The military value of a wide choice of yields and bomb sizes was great enough to allow the U.S. to stockpile less than the maximum number of bombs possible from all available fissionable materials. A given amount of fissionable material could be used to make many times the number of weapons possible in 1945, with the new bombs having an average of more than twice the yield (on the order of 40 KT), and the average bomb weight carried to deliver a given blast force had been reduced by a factor of three.

The U.S. by 1953 was able to inflict the same damage, in terms of area, as in 1945 with fewer than 3/4 as many sorties carrying less than 1/4 the total bomb load and expending only a fraction of the fissile material. (A sortie was one aircraft carrying one weapon to a target.)

A very small (22" diameter) relatively-inefficient implosion bomb (the MK 12) had been developed for special applications. Half of the Nagasaki bomb yield, or 10 KT, was possible with only a fraction of the wartime fissionable material. The weight-to-yield ratio of the MK 12, in terms of total bomb

Energy Commission, July 1950 to January 1954, p. 13.

weight per kiloton delivered, was only about 100 lbs. per KT, as opposed to the approximately 500 lbs. per kiloton of the FAT MAN.

Only implosion weapons were being stockpiled; production requirements for gun-type weapons had been fulfilled before December 31, 1953. There were no major modification programs in progress, and 42 formal weapons programs were underway.⁶³

Also by the end of 1953, a very cheap, permanent replacement (an external fission initiator) had been developed to supersede the earlier short-lived internal initiator, production of which was costing the U.S. millions of dollars per year at this time and which would have posed a prodigious problem in the near future.⁶⁴

A year later, in accordance with the so-called "family of weapons" concept, the U.S. nuclear weapons stockpile had been significantly enlarged in both variety and versatility. **Table II-4** shows the progress in this area between fiscal years 1950 and 1956.

Implosion weapons, which comprised the majority of the stockpile, included free-fall bombs, demolition munitions, depth charges and missile warheads. Special implosion system designs ran from 54 inches in diameter down to a proposed 15 inch device (between 12 and 18 inches), with total warhead weights varying from 6,800 lbs. down to 200 lbs. Characteristics of delivery vehicles had a major bearing on the selection of each configuration.

At the time, with the exception of the half-megaton yield MK 18 fission bomb, there was a range of yields available for each implosion system, permitting considerable flexibility in the integration of atomic weapons with war plans. (The MK 18 core was specifically designed to be compatible only with the corresponding HE assembly.)

Gun-type weapons were being stockpiled for specific applications. Unlike implosion systems, gun-assembly systems were configured for optimum efficiency of use of fissionable material for the particular-size weapon. Optimization of design of gun-type weapons ruled out any interchangeability, although many non-nuclear components were either interchangeable or similar between weapons.⁶⁵

⁶³Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1950 to January 1954, p. 19.

⁶⁴Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, pp. 60, 62.

A primary objective of U.S. nuclear weapons development by this time was to produce a wide range of yields and a good selection of bomb and warhead sizes and weights in order to allow DOD to choose weapons appropriate to both a target and the available and applicable delivery vehicle.

Reductions in weapon sizes and weights allowed a wider spectrum of delivery vehicles and increased delivery vehicle range and performance. Nuclear weapons in the U.S. by the mid-1950s provided greatly increased nuclear efficiency and flexibility.⁶⁶

Table II-5 shows the growth in the U.S. nuclear weapons stockpile between 1945 and 1959. The U.S. nuclear weapons stockpile grew slowly after World War II for a number of reasons. First, there was little direction or guidance from the armed services for new or improved weapons, nor did Congress allocate funds during the immediate postwar period for weapons research and development of almost any type, either nuclear or conventional.

In addition, very few government officials in D.C. were even aware of the nation's minuscule stockpile size. This number was guarded jealously by General Groves and even President Truman did not become aware of it until late in 1946, nearly a year and a half after he had taken over from President Franklin Roosevelt.

Second, reactor problems (described below) at the Manhattan Engineer District's Hanford, Washington facility severely slowed the production of plutonium between 1946 and 1948.

Third, LASL was reluctant to stockpile a large number of what it considered to be obsolete or obsolescent bomb designs, when several much better, more efficient, and more practical designs were in the planning stage, or awaiting test.

Fourth, two fatal accidents at LASL's Pajarito critical assembly test site (see MK 4 history in Volume VI) effectively stopped new core stockpiling between May 1946 and 1947 until remote control equipment could be installed at the site.

Before 1951, all weapons were in the custody of the AEC; starting in 1951, weapons were held by both the AEC and the Department of Defense. **Figure II-2, Figure II-3, Figure**

⁶⁵Atomic Energy Commission, Program Status Report, December 31, 1954, Part III - Weapons Program, pp. 2, 4.

⁶⁶Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1950 to January 1954, p. 13.

II-4, and **Figure II-5** show graphically the numbers in several of the columns of Table II-5.

By contrast, the Soviet nuclear weapons stockpile grew very slowly: by 1950, there were only an estimated five bombs in the arsenal and this number had grown to as few as 20 to 30 by 1953.⁶⁷

The following sections summarize and highlight a few of the more noteworthy developments in some of the major components, or types of components, in postwar U.S. fission warheads and fusion warhead triggers. These components represent the most common elements among postwar weapons; as will be seen, there was much room for improvement in each of them.

Advances in High Explosives

The chemical conventional high explosive (HE) used in all nuclear weapons is one of the most important components. Because all subsequent fission and fusion explosions in a nuclear weapon are started by the detonation of several pounds of chemical compounds in the initial fission trigger, the performance of the "burn" determines the power of every other following stage: if the initial fissile material is not adequately or properly compressed, the fission explosion will not reach its predicted yield, and any following fusion stages in a multistage thermonuclear device will also fall short of their designed or certified yield, which could lead to ignition failure in subsequent stages.

Partly for this reason, the chemistry, formulation, and technology of manufacture of conventional chemical explosives continues to be one of the most closely-guarded and classified aspects of modern nuclear weapons design.

However, in spite of the absence of details in this area, certain explosives performance and design criteria can be deduced. For example, the explosives have to offer as much "push" as possible within a reasonable amount of material; they need to be uniformly cast and shaped, without bubbles or voids within the charges (structural imperfections interrupt and deform the shock wave generated within the charge); and they need to have a reasonably-long shelf life, without excessive "aging" via chemical deterioration of the components of the HE charge.

Achieving efficient use of high explosives in nuclear weapons places demands on explosive systems that are not encountered in any other ordnance practice: simultaneity of detonation at a large number of points, uniform propagation of detonation

⁶⁷Zaloga, pp. 77, 79.

waves in three dimensions, and maximum conversion of explosive energy into kinetic energy of metal parts at the center of the system.

In addition, the explosives must be insensitive to unwanted, unintentional shocks such as those encountered in aircraft or vehicle crashes or impacts in order to make the HE charges as safe as possible. The HE must also be heat-resistant, for the same reason.

Even the chemical composition is significant: the highly-hydrogenated and nitrogen-rich compounds used in most HE charges make them very good neutron absorbers, which is a drawback in weapons designed to maximize neutron flux or output. The thick HE layer around the core of the FAT MAN detonated over Nagasaki attenuated the fission neutron flux to such an extent that very few neutrons managed to escape. On the other hand, during tests of the MK 12 at Nevada in the spring of 1952, a high neutron flux was noted, as a result of the 5.18" thick HE layer around the cores.⁶⁸

The thermal neutron flux per kiloton of yield of the MK 5 weapon was twice as large as that of the MK 6;⁶⁹ the MK 5 used a 10.24" thick layer of HE as opposed to the 17.3" thick layer on the MK 6.

Desirable properties for a nuclear weapon high explosive include:

- Insensitivity to impact, fire, and fragment damage
- Initiability, i.e., ease of detonation

⁶⁸Operation REDWING, Technical Summary of Military Effects, Programs 1-9, WT-1344, Office of the Deputy Chief of Staff for Weapons Effects, Headquarters Field Command, Defense Atomic Support Agency, Sandia Base, Albuquerque, New Mexico, p. 140; "Studies Revise Dose Estimates of A-Bomb Survivors," Bertram M. Schwarzschild, PHYSICS TODAY, September 1981, reprinted in PHYSICS AND NUCLEAR ARMS TODAY, Readings from PHYSICS TODAY, David Hafemeister, ed., American Institute of Physics, New York, 1991, p. 12; OPERATION TUMBLER-SNAPPER, APRIL-JUNE 1952, EXTERNAL NEUTRON MEASUREMENTS, PROJECTS 17.1 and 17.2, WT-555(EX), Los Alamos Scientific Laboratory, University of California, Los Alamos, New Mexico, June 1952, pp. 59, 60.

⁶⁹Letter DSARA-4 927.3 dated 12 November 1959 to Director, Office of Special Weapons Development, U.S. Continental Army Command, Ft. Bliss, Texas, subject: Weapons Data for Use with TM 23-200.

- Mechanical properties
- Performance
- Compatibility and stability during service life
- Processability, i.e., ease of manufacture
- Cost-effectiveness⁷⁰

After the end of World War II, development and research effort to improve high explosive systems in U.S. nuclear weapons was directed at increasing the compression of the pit for a given total thickness of high explosives; simplifying and improving the reliability and timing accuracy of the detonation of the explosives; and developing simpler, safer, and more economical methods of fabricating high explosives.⁷¹

By late 1946, conspicuous advances had been made in high explosives; among them was the development of certain types of materials which had an explosive-like behavior when used in lenses, but which by themselves were largely inert. Studies of unusual implosion techniques also continued.⁷²

In the spring of 1947, representatives of the AEC and the U.S. armed forces met in Washington, D.C. to lighten the high explosives research workload of Los Alamos laboratory. Early in 1948, the AEC enlisted the support of the HE research and development facilities of the U.S. Navy's Naval Ordnance Laboratory in Silver Spring, Maryland, to assist LASL in a "program which is probably the weakest part of the work leading toward more efficient weapons, particularly smaller and lighter weapons."⁷³

⁷⁰"Formulating High-Explosive Materials," C. O. Pruneda and E. H. von Holtz, ENERGY & TECHNOLOGY REVIEW, UCRL-52000-88-1.2, January/February 1988, Lawrence Livermore National Laboratory, Livermore, California, p. 24.

⁷¹Letter dated 26 December 1951 from Norris E. Bradbury, Director, LASL, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 3.

⁷²Letter dated November 14, 1946 to U.S. Atomic Energy Commission from N. E. Bradbury.

⁷³Atomic Energy Commission, Armed Services Participation in Atomic Energy Commission Explosives Program, Report by the Division of Military Application, AEC 30, February 4, 1948.

One crucial consideration for good bomb pit compression is the number of high explosive charges surrounding it. The more points that can be detonated simultaneously, the closer the implosion approaches perfect spherical symmetry. Early postwar bombs used 32-, 60-, and 92-point HE ignition systems; by 1950, LASL had developed new types of explosive detonators that exploded instead of igniting, providing simultaneity of detonation not previously attainable.⁷⁴ During World War II, LASL was able to fire systems comprising several hundred detonators within several hundredths of a microsecond of each other.⁷⁵

The wartime 32-point system began as the solution to a problem of spherical geometry dating back to the time of the Greek philosopher Plato:

I learned a lot about the five regular polyhedra during this period. The largest number of points that can be spaced regularly on the surface of a sphere is 20, a number corresponding to the centers of the 20 triangular faces of an icosahedron.

The next largest number is 12, corresponding to the centers of the 12 pentagonal faces of an icosahedron. It's possible to interleave a dodecahedron with an icosahedron, as Plato showed, to get the nearly regular faces, alternately pentagons and hexagons, the shape of an object so familiar as a soccer ball.⁷⁶

The 92-point system may have had its origins in January 1945, when Dr. Hans Bethe studied methods of arranging detonation points on the outside of a sphere so that there would never be an intersection of more than three detonation waves

⁷⁴Memo to Weapons Panel Members, Subject: Action on Choice between 32 Point and 60 Point Detonation System (CROSSROADS Project), undated (but probably sometime in the spring of 1946); Affidavit of Dimitri A. Rotow in Civil Action No. 79-C-98, filed May 10, 1979, p. 6; Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 20.

⁷⁵Hoddeson, et. al., p. 173. At these detonation velocities, the lengths of wire which the signals have to traverse become an important factor: all signals should arrive simultaneously, regardless of the distance from the firing unit to the charges.

⁷⁶Hoddeson, et. al., pp. 171, 172.

(generated by the detonators) at any point within the sphere. This meant that the detonation points had to be positioned at the corners of triangles which had to be as equilateral as possible.⁷⁷

The new generation of sealed-pit weapons, starting with the W-25 air-to-air missile warhead, used fewer ignition points.⁷⁸ The fission primary of a late 1950s design, the MK 28, reportedly used a 40-point HE ignition system surrounding 44 lbs. of PBX 9404.⁷⁹

An important method of determining implosion efficiency and symmetry was (and is) hydrodynamic testing, in which a simulated weapon core, built to actual physical specifications but using neither fissile material nor initiator (depleted uranium, depletalloy, is used in place of either or alloy or plutonium), is surrounded by tamper, reflector, high explosives, and detonators, connected to a firing system.

One early method of hydrodynamic testing involved the use of a strong gamma ray emitter, such as lanthanum-140, placed at the center of the core. The test assembly was surrounded radially by ionization chambers connected to recording instruments.

When the high explosives detonated and imploded the core, the gamma radiation from the lanthanum "shone" through the surrounding material at a rate proportional to the density of the material. This escaping radiation induced currents in the ionization chambers; since the density of the core increased as it was imploded, the gamma ray flux and induced currents varied accordingly.

As a result of these hydrodynamic implosion tests, scientists can accurately determine both the rate of compression of an imploding core from the rate of change of material density,

⁷⁷Memorandum dated January 27, 1945 to G. B. Kistiakowsky from H. A. Bethe. The 92-point system may also have been a combined overlay of the 32- and 60-point systems.

⁷⁸Interview with Mr. William J. Brady, Las Vegas, Nevada, December 5, 1988, conducted by Philip L. Cantelon and Brian W. Martin, History Associates, Incorporated, pp. 21, 22, 63. This reduction in the number of ignition points emphasized the need for one-point safety tests of sealed-pit weapons: the fewer the number of detonators, the more likely a nuclear yield if a single detonator fired.

⁷⁹THE BOMBS OF PALOMARES, Tad Szulc, Viking Press, New York, New York, pp. 74, 98. These numbers are unconfirmed.

and the symmetry of the implosion, by comparing radiation-strength data from the radially-positioned ionization chambers.⁸⁰

A significant postwar development was the replacement of multiple, individually-fuzed and detonated high explosive lenses surrounding the warhead pit by a lensless HE sphere that was ignited simultaneously at several points on its outer surface. Plutonium core implosion by means of multipoint detonation over the surface of a uniform HE sphere had been considered at Los Alamos by early 1945.⁸¹

Unfortunately, a hemisphere of Composition B for the nonlensed design of this period would have weighed about a metric ton (2,200 lbs.); such a massive piece of HE required careful handling by special heavy equipment, and was obviously impractical.⁸² By February 1945, further work on the lensless HE system had been turned over to the California Institute of Technology.⁸³

The Los Alamos Scientific Laboratory program for 1949 included "a more detailed study of the possibility, virtues, and defects of multipoint initiation of a sphere of high explosive as opposed to lens initiation of the inner charge."⁸⁴

The Stoat shot of Operation NOUGAT at the Nevada Test Site on January 9, 1962 was the first in a series of developmental tests of a new multipoint detonation system. More multipoint detonation tests were conducted during Operation STORAX in 1962 and 1963 to develop systems that might have applications in new ballistic missiles.⁸⁵

⁸⁰"Laboratory Annual Report for 1947," LAMS-653, N. E. Bradbury, 3 December 1947, pp. 60, 61; Hoddeson, et. al., pp. 148-154.

⁸¹KLAUS FUCHS, ATOM SPY, Robert Chadwell Williams, Harvard University Press, Cambridge, Massachusetts, 1987, p. 190; Hoddeson, et. al., pp. 300, 301; Rhodes, DARK SUN, p. 154.

⁸²Hoddeson, et. al., p. 167.

⁸³Hoddeson, et. al., p. 312.

⁸⁴Letter dated 7 September 1948 from Norris E. Bradbury, Director, University of California Los Alamos Scientific Laboratory, Los Alamos, New Mexico, to Carroll L. Tyler, Manager, Office of Santa Fe Directed Operations, U.S. Atomic Energy Commission, Los Alamos, New Mexico, Subject: Los Alamos Laboratory Program for 1949, p. 3.

Current U.S. nuclear weapon primaries are imploded by hollow hemispheres of pressed and machined PBX.⁸⁶

The shape of the charge is also important: since only about a third of the total explosive power goes into compressing the fissile core, the generated shock wave must be shaped and directed as perfectly as possible. In an HE lens system, each separate HE charge contributes a portion of the entire spherically-symmetric implosion wave.⁸⁷

Development of New Postwar Explosives

A major step toward reducing bomb diameters and weights was accomplished by the development of new procedures for producing HE bomb components. Early methods of shaping explosives involved construction of molds in which the lenses were cast from molten HE. Whenever a slight change in shape or dimensions was required, a new mold had to be made, slowing and complicating testing of new HE shapes.

Just after the war ended, LASL was already planning the development of flatter, lighter HE lenses made from sintered explosives, which were powdered and slurried or molten HE molded into desired shapes by high-pressure machinery.⁸⁸ During World War III, some work had been done on producing slow-burning components of HE "lenses" by mechanical tamping of loose granular TNT to form a uniform but low-density explosive.

Mechanical and vibratory tamping received some study during the early stages of the "lens" program primarily to produce "lenses" for the implosion testing program where "lenses" penetrable by diagnostic x-rays were a necessity. This study was overtaken by adoption of cast Baratol "lenses" for the first FAT MAN model, but served as a starting point for postwar work on bonded and sintered "lenses." The latter

⁸⁵Memorandum for Chairman Seaborg, et. al. dated December 17, 1962 from Maj. Gen. A. W. Betts, Director of Military Application, USAEC, subject: STORAX III; Atomic Energy Commission, STORAX III, Note by the Secretary dated January 8, 1963.

⁸⁶Explanation of PBX-9502 fabrication techniques included in a display at the Bradbury Science Museum at the Los Alamos National Laboratory, Los Alamos, New Mexico, 1987.

⁸⁷McPhee, pp. 82, 218.

⁸⁸Teletype dated September 27, 1945 to Washington Liaison Office from N. E. Bradbury.

method in particular yielded a uniform and reasonably strong explosive charge.⁸⁹

By mid-1950, new methods of machining desired HE shapes directly out of pre-cast blocks or chunks of explosives, using saws, lathes, and automatic milling machines, allowed quicker, cheaper, and simpler development of new explosive geometries. By this time, techniques for casting and machining explosives at LASL were probably the most advanced in the world.⁹⁰

In the fall of 1950, the General Advisory Committee to the AEC thought that research in new explosives might make a significant contribution toward increased energy content; however, an improvement of anything like a factor of two over the power of Composition B seemed unlikely. Some gain could be expected, but as much as even 50% over the power of RDX would be optimistic. The biggest problem with high-energy content explosives was their high susceptibility to shock-induced detonation and their chemical instability; both characteristics posed very serious stockpiling and handling difficulties.⁹¹

At this time, the AEC believed its high explosives research and development program to be sufficiently advanced that it would not be necessary to bring other agencies more heavily into the program. The Navy's Bureau of Ordnance and the Army's Ordnance Department were already assisting the AEC with HE manufacturing.⁹²

⁸⁹Volume XI, "Explosives and Detonation Waves, Part IV: The Making of Explosive Charges," LA-1052, G. B. Kistiakowsky, ed., Los Alamos Scientific Laboratory, January 15, 1949, pp. 7, 8.

⁹⁰Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 19. During World War II, the use of large-scale machining methods was considered as a revolutionary advance in explosives manufacturing. (LA-1052, p. 9.)

⁹¹Minutes, Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, and November 1 1950, Los Alamos New Mexico, p. 24.

⁹²Attachment dated September 12, 1950 to letter of same date to William L. Borden, Executive Director, JCAE, from Brig. Gen. James McCormack Jr., USAF, Director of Military Application, USAEC, p. 8.

The Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953 declared the lab's near-term goals for HE research and development:

The program of basic research on the mechanism of detonation, its initiation and propagation in various types of explosives and pseudo-explosives, and the factors influencing the rate of detonation in an explosive will be continued.

...

(For high explosive systems), developmental and research effort will be spent on the high explosive components of implosion weapons with the objective of increasing the extent of compression for a given total thickness of high explosive; of increasing the reliability, simplicity, and accuracy of the timing of the initiation of the explosive components; and of developing appropriate fabrication techniques for high explosive materials.⁹³

A further measure toward reducing bomb diameter and weight was the creation of new "slow-burning" explosives by adding inert substances to existing HE mixtures. Slower explosives were desirable because the slower the detonation wave propagation rate in the inner components (or centers) of explosive lenses around the weapon pit, the flatter, or shallower, the lens could be. Since less explosives were required, and because new explosives offered more power for the same weight, bomb diameter and total weapon weight were both reduced.⁹⁴

These effects were dramatic: the 60" diameter MK 4 weapon used HE layers 17" thick; the 43" diameter MK 5 used a 10"

⁹³Letter dated 26 December 1951 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Norris Bradbury, Director, LASL, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, pp. 2, 3.

⁹⁴Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 19. The Los Alamos Technical Program for Calendar Year 1951 and Fiscal Year 1952, unveiled in November 1950, included plans for development and research effort to "(increase) the extent of (pit) compression for a given total thickness of high explosives ..."

layer of HE; the 30" diameter MK 7 and 22" diameter MK 12 both contained just over 5" thick HE spheres. The experimental, small-diameter prototype air defense warheads tested by LASL during Operation TEAPOT in 1955 -- the BEE, HORNET, WASP, and MOTH -- all used thin HE layers around their pits.⁹⁵

The wartime FAT MAN (MK III prototype) weapon used charges of Composition B, a fast-burning, high-powered explosive superior to TNT (trinitrotoluene) in its burning rate and calorie-per-pound output of energy. Composition B was a mixture of RDX, TNT, and beeswax.⁹⁶

This material was superseded in postwar weapons by mixtures containing HBX (desensitized Torpex) and RDX (Cyclonite), or Octol, and HMX. Unfortunately, a common difficulty with these high energy-content explosives was their high sensitivity to shocks and their chemical instability, very serious drawbacks to stockpiling and handling requirements.⁹⁷

RDX, also known as Cyclonite and Research Department Explosive, or Rapid Detonation Explosive, or sym-cyclotrimethylenetrinitramine, was developed independently early in the 20th century and was widely used during World War II by both Germany and England.⁹⁸ RDX was made by combining ammonia, formaldehyde, and nitric acid. Much more powerful than TNT, RDX was by war's end a major component in many conventional explosives.

By the end of 1951, plans were being made to increase the proportion of RDX in the RDX-TNT mixture in Composition B, which would give greater core compression and a corresponding increase in yield for a given amount of fissionable material.⁹⁹ This increase resulted in the formulation of Cyclotol 75/25.

⁹⁵OPERATION TEAPOT, TECHNICAL SUMMARY OF MILITARY EFFECTS, PROGRAMS 1-9, WT-1153 (EX), p. 90.

⁹⁶Hoddeson, et. al., pp. 164, 454.

⁹⁷Minutes, Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, and November 1 1950, Los Alamos New Mexico, p. 24.

⁹⁸The "Research Department" may have been a reference to the explosives research department at Woolwich in eastern London. (Hoddeson, et. al., p. 165.)

⁹⁹Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 15.

Early drawbacks to the use of RDX were its sensitivity and cost. The former problem was overcome by the addition of beeswax, plasticizing oils, and TNT to the other ingredients. The latter problem arose mainly from the fact that in order to get 100 lbs. of RDX, 1,300 lbs. of ingredients had to be combined and later separated. A new method of synthesization developed by the end of the war finally required only 641 lbs. of materials to get 100 lbs. of RDX.¹⁰⁰

Three different formulations of RDX were used during World War II; these were designated Compositions A, B, and C, and they differed from one another in the proportion of TNT added to the RDX. Composition A was 91% RDX and 9% beeswax; Composition B was 59% RDX, 40% TNT, and 1% beeswax; and Composition C was 88% RDX and 12% plasticizer.¹⁰¹

The prime component of Torpex was RDX; Torpex, or TPX, was formulated from RDX, TNT, and aluminum powder. On a pound for pound basis, Torpex was half again as powerful as TNT: because it was more dense than TNT, a given volume of Torpex was 70% more powerful than an equal volume of TNT. However, Torpex was more sensitive to shocks than TNT.¹⁰²

HBX arose from a Navy requirement to develop a desensitized variant of RDX that could be used in warm tropical environments without exuding ("sweating") the wax used to make it less sensitive to shock detonation. A newly discovered microcrystalline wax that was available in quantity solved the problem and was also a satisfactory desensitizer. HBX, a mixture of TNT, RDX, aluminum powder, and desensitizer was much less sensitive to shocks than Torpex, and only 2% less powerful than Torpex.¹⁰³

HMX (High Melting-Point Explosive) was first discovered as an impurity in RDX. When HMX proved to be both more energetic and safer to use than RDX, methods of synthesizing it were investigated. Several processes were developed in the U.S. and Europe; HMX is now an important component of many

¹⁰⁰U.S. NAVY BUREAU OF ORDNANCE IN WORLD WAR II, Lt. Cdr. Buford Rowland and Lt. William B. Boyd, Bureau of Ordnance, Department of the Navy, Washington, D.C., 1953, pp. 204, 205.

¹⁰¹AMMUNITION GENERAL, Department of the Army Technical Manual TM 9-1900/Department of the Air Force Technical Order TO 11A-1-20, Departments of the Army and the Air Force, June 1956, pp. 57, 58.

¹⁰²Rowland and Boyd, p. 205; Hoddeson, et. al., p. 454.

¹⁰³Rowland and Boyd, pp. 206, 207.

conventional and nuclear weapon systems. For the past thirty years, HMX has been the standard against which other HEs are measured.

Both HMX and RDX are used in rocket propellants, as well as in nuclear warheads. RDX is more widely used because it is three times cheaper than HMX to produce.¹⁰⁴

Charges manufactured from TNT-based compounds have a tendency to crack when roughly handled; they are also very shock-sensitive and susceptible to melting and burning at relatively low temperatures. Nuclear weapon primaries are often exposed to temperatures as high as 200° C (392° F). A new explosive was required that had better heat resistance than melt-cast explosives such as Composition B.¹⁰⁵

At the end of 1953, LASL began a search for an improved chemical explosive that might offer a 30% to 40% improvement over existing HEs. Such a mixture would be of great importance for small weapons, both because of the lower weight and the smaller amount required for high core compression.¹⁰⁶ The AEC was advised of the importance of the rapid development of new explosives:

This worry leads us to a suggestion which we strongly urge on the Commission, namely, to initiate a vigorous program of research in chemical explosives suitable for the implosion of atomic weapons. It has long been felt by some experts in the field of chemical explosives that great improvement in explosion yield per unit volume could be achieved by explosives research and development. The pressure of other programs, however, has caused this field to be largely neglected.

We feel that we should no longer leave this largely unexplored. The gains to be

¹⁰⁴"Synthesis of Energetic Materials," Clifford Coon, ENERGY & TECHNOLOGY REVIEW, UCRL-52000-88-1.2, January/February 1988, Lawrence Livermore National Laboratory, Livermore, California, p. 18.

¹⁰⁵EXPLOSIVES, PROPELLANTS, & PYROTECHNICS, A. Bailey & S. G. Murray, Brassey's Ltd., London, England, 1989, p. 154.

¹⁰⁶Minutes of the Thirty-Seventh Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, November 4-6, 1953; letter dated November 7, 1953 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

achieved from success in this direction are enormous both in the reduction in size of large fission weapons and even more importantly in the possibility of making smaller fission weapons of simple design and great economy of fissionable material. It is well known that both the Russians and the British are very expert in the field of chemical explosives. It is conceivable to us that they may have made significant advances in this field.

The explosives in present use in the United States were developed for the more usual military purposes. Many of the requirements which are put on such explosives can perhaps be relaxed for nuclear weapons in order to achieve a greater energy release per unit volume. With this in mind and with regard to the great gains to be achieved for the weapons program from such a development, we recommend that the Commission proceed toward the exploration of these possibilities with all speed.¹⁰⁷

By the summer of 1956, the University of California Radiation Laboratory at Livermore, California was working on explosives which were 70% more powerful than Composition B but which were also very prone to detonate when impacted.¹⁰⁸

During the mid-1950s and early 1960s, new composite explosives called DATB (diaminotrinitrobenzene) and TATB (triaminotrinitrobenzene) were developed and used in nuclear warheads.¹⁰⁹ These are relatively high-powered, shock-insensitive compositions; TATB is used in weapons under the name PBX (Plastic Bonded Explosive), such as PBX-9404 and the newer, more shock-insensitive PBX-9502. PBX has better heat and cracking resistance, improved strength, and dimensional stability. The explosive also is stronger and denser than its

¹⁰⁷Letter dated November 7, 1953 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, General Advisory Committee to the AEC.

¹⁰⁸Minutes of the Fiftieth Meeting of the General Advisory Committee to the USAEC, July 16-18, 1956, p. 33.

¹⁰⁹UCRL 5280, Weapon Development during June 1958, University of California Radiation Laboratory, Livermore, California, July 8, 1958, p. 14.

predecessors, resulting in more uniform detonation and a higher energy to weight ratio.¹¹⁰

PBX was first used in a full-scale nuclear detonation during the Blackfoot shot of Operation REDWING in June 1956. The biggest drawback to TATB or PBX is its cost: approximately \$30 per pound, as opposed to \$1 or \$2 per pound for TNT or Tritonal.¹¹¹ PBX did not go into widespread use until after 1960.¹¹²

HE Manufacturing Methods

High explosives used in U.S. nuclear weapons are manufactured in a variety of ways. Pure explosives are usually synthesized by sulfuric/nitric acid nitration of organic compounds. The product is separated from the mixed acids by filtration, then purified and dried.

TNT is one of the few pure explosives that can be fabricated directly by melting and casting into desired shapes, such as lenses. Most other materials must be diluted with either TNT to make them castable or with plastic to make them pressable before they can be formed into useful shapes. TNT melts at a relatively low temperature, around 176° F.

The procedure used for making castable TNT-containing formulations is as follows: TNT is melted, and the desired solid ingredients are stirred in. The molten mixture is then vacuum-cast into a mold. Cracking and variations in density and composition are minimized by careful control of the cooling rate. Wartime HE lenses cast at Los Alamos suffered from cracks and entrapped bubbles.¹¹³ Molds are generally cooled by water circulated around them.¹¹⁴

The FAT MAN HE lenses manufactured at Los Alamos contained "fast"-burning Composition B with "slow"-burning Baratol centers (Baratol was developed by the Navy). Composition B was poured as a hot slurry of wax, molten TNT, and a non-molten crystalline powder, RDX, that was 20% to 40% more powerful than TNT alone. Baratol was formed from a slurry of

¹¹⁰Lemert, p. 173.

¹¹¹"Dig Deep - Blast Hard," Colleen A. Nash, AIR FORCE Magazine, Vol. 74 No. 3, March 1991, p. 92.

¹¹²Lemert, p. 166.

¹¹³Rhodes, TMAB, pp. 577, 655-657.

¹¹⁴Lemert, p. 165.

barium nitrate, aluminum powder, TNT, stearoxyacetic acid, and nitrocellulose.¹¹⁵

If necessary, lenses were machined to precise shapes after molding; voids within the blocks (revealed by x-rays) were filled by injections of slurried explosive.

Plastic-bonded explosives (PBX) are pressed from "molding" powders, which may be produced in several ways. A typical preparation method is the water slurry technique: crystalline explosive and water are agitated in a covered container or kettle (called a Holston reactor) that contains a condenser and stirrer. A lacquer, which consists of a thermoplastic (together with a plasticizer, if required) dissolved in a suitable solvent, is added to the slurry.

The solvent is not a solvent for the HE but instead is used to wet the crystalline surfaces better than water. The solvent is immiscible with water and has a high vapor pressure; it is removed by distillation, which causes the plastic to precipitate out onto the explosive as a coating. Vapors resulting from the rapid stirring and heating of the slurry are removed by vacuum.

The plastic-explosive conglomerates into "beads" as the mixture is stirred and solvent removed. Finally, water is removed from the beads by filtering and drying. The product of the process is molding powder; good molding powders have a high bulk density and are free-flowing and dustless.

PBX molding powder can be pressed into usable shapes by either compression molding with steel molds or by hydrostatic (isostatic) pressing. In the latter method, the explosive is placed into rubber sacks and subjected to a very high fluid pressure. With either method, compression of the molding powder to reasonable densities (up to 97% of theoretical) is attained at pressures between 12,000 and 20,000 psi and molding temperatures between 25° and 150° C (77° to 302° F).

An important and necessary feature of molding is the use of vacuum; the molding powder is normally evacuated to a pressure of less than one millimeter of mercury before pressing.

Both pressed and cast explosives are usually remotely-machined to final shape. Many intricate forms have been cut successfully; as a rule, the machining of explosives is similar to the machining of conventional plastic, except that water instead of oil is used as a cutting-tool coolant when machining explosives. New explosives are machined by remote

¹¹⁵Rhodes, TMAB, p. 577.

control until their behavior under machining conditions has been carefully evaluated.¹¹⁶

An alternative method of making PBX is the paste mix method in which a TATB-based PBX with unique mechanical properties not easily obtained by the water-slurry method is treated with a special adhesive to enhance its adhesion to the thermoplastic. The TATB, thermoplastic, and lacquer are mixed in a Sigma-blade horizontal mixer; the mixer produces a very-low viscosity, paste-like material.

The solvent is removed by vacuum while the mixture is stirred and heated. When most of the solvent has been removed, the resultant molding powder is oven-dried, leaving a mixture with the consistency of moist sand. The powder is then isostatically-pressed and machined to shape like other PBXs.¹¹⁷

PBX charges have good mechanical properties, relatively inherent safety, and high thermal stability. Their disadvantages include toxic ingredients, long cure times, complex formulations, and special mixing and filling equipment.¹¹⁸

Since 1974, all TATB used in U.S. nuclear weapons has been produced by the Chemical Operations Division of the Aerojet Strategic Propulsion Corporation, a subsidiary of Aerojet General, in Sacramento, California (this division also makes solid propellants for ballistic missiles). TATB is a high-density, low-sensitivity, long-lived explosive; it can be stored up to 30 years without chemical change.

Final production of the high explosives used in U.S. nuclear weapons occurs at the Holston Army Ammunition Plant (originally the Holston Ordnance Works) in Kingsport, Tennessee.¹¹⁹ Since 1960, the Holston facility has been the exclusive source of all U.S. nuclear weapons explosives.

¹¹⁶LLNL EXPLOSIVES HANDBOOK: PROPERTIES OF CHEMICAL EXPLOSIVES AND SIMULANTS, UCRL-52997, B. M. Dobratz and P. C. Crawford, Lawrence Livermore National Laboratory, Livermore, California, January 31, 1985, pp. 2-1, 2-2; "Formulating High-Explosive Materials," p. 25.

¹¹⁷"Formulating High-Explosive Materials," pp. 25, 26.

¹¹⁸Bailey and Murray, p. 155.

¹¹⁹"Formulating High-Explosive Materials," p. 24; Rowland and Boyd, p. 205; "AMCCOM Facts: U.S. Army Armament, Munitions, and Chemical Command," AMCCOM PAM 5-1, 2 February 1988, p. 45.

Holston had produced up to 300 lbs. per minute of RDX during World War II.¹²⁰

In addition to its nuclear weapons work, the Holston plant makes a variety of explosives for the U.S. Army. The main charges of insensitive high explosives for weapons primaries are fabricated from PBX at Holston, after TATB is procured from Aerojet. The plasticized powder produced at Holston is then pressed and machined to final dimensions at the Department of Energy's Pantex assembly plant in Amarillo, Texas.¹²¹

At Pantex, the HE charges are inspected and sampled and then pressed into billets, which are in turn x-rayed to locate internal cavities or bubbles or foreign matter.

The billets go through a series of machining operations and inspections to produce a finished shaped charge. To assure quality, samples of a group (or lot) of charges are test-fired to ascertain that they behave as predicted.

When the charges are released for production, some subassembly preparation may be required. Certain attachments may be mated to the main charge before final assembly of the HE and pit. After the pit and HE have been mated, fuzing and firing components are installed.¹²²

PBX is procured in 10,000 to 30,000 lb. lots, with annual procurement ranging from 10,000 to 100,000 lbs. Since 1986, DOE procurement has averaged less than 5% of the annual total of 30 million lbs. of HE produced at the Holston plant.¹²³

Implementation of Insensitive High Explosives

One of the most significant advances in the science of high explosives for nuclear weapons is the recent development of insensitive high explosives, or IHE, as exemplified by the plastic-bonded explosive PBX-9502.

¹²⁰Hoddeson, et. al., p. 165.

¹²¹"Tennessee Fireworks," Nuclear Notebook, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 47 No. 6, July/August 1991, p. 47.

¹²²U.S. Department of Energy Amarillo Area Office Pantex Plant, June 1990 booklet, p. 24.

¹²³Transcript of questions & answers between Representative Fazio and W. Graham Claytor, Assistant Secretary for Defense Programs, DOE, undated (received May 1991).

IHE is 20 to 30 times less sensitive to shock detonation than older explosives used in U.S. nuclear weapons, so it is inherently much safer. Weapons containing IHE are less prone to accidental explosion if they are inadvertently dropped or subjected to rapid deceleration and sudden impacts. IHE can withstand the impact of a rifle bullet without detonating, which many older explosives used in American nuclear weapons could not.¹²⁴

However, because of its insensitivity, IHE is also somewhat less energetic and more difficult to ignite than earlier HEs. IHE contains, pound for pound, only about two-thirds the energy of earlier U.S. nuclear weapons explosives, and is therefore required in greater weight and volume to initiate a nuclear detonation. Hence the yield-to-weight ratio of a warhead decreases when IHE replaces conventional HE.¹²⁵

IHE varies in several important key respects from normal HE. It takes less than 0.001 ounces of detonator explosive to ignite normal HE: IHE requires slightly more than four ounces. IHE requires four and a half times as much shock pressure to ignite, and an impact velocity 12 to 13 times as great as normal HE to detonate.¹²⁶ IHE-equipped weapons are the only warheads that the Department of Energy routinely transports by air.¹²⁷

PBX-9502, for example, has both a slower detonation rate and a lower heat of detonation than PBX-9404, which it replaced (see **Table II-6**). Since PBX-9502 is less energetic, more of it must be used in a primary to obtain equivalent pit compression; this means that warheads containing PBX-9502 tend to be heavier than older warheads using more high-powered, less-insensitive explosives.¹²⁸ In addition, when older explosives are replaced in existing warheads by IHE, a nuclear test is usually required to recertify the weapon's yield.¹²⁹

¹²⁴Rosengren, RDA-TR-122100-001-Rev. 1, p. 103.

¹²⁵"Nuclear Weapons Safety," Report of the Panel on Nuclear Weapons Safety of The House Armed Services Committee, House of Representatives, One Hundred First Congress, Second Session, December 1990, pp. 14, 27 (hereafter referred to as "Drell-Foster-Townes Report").

¹²⁶Drell-Foster-Townes Report, p. 27.

¹²⁷Drell-Foster-Townes Report, p. 30.

¹²⁸"Safety Second," NUCLEAR NOTEBOOK, Bulletin of the Atomic Scientists, Vol. 44 No. 4, May 1988, p. 55.

DOE expects to use IHE in numerous future nuclear weapon systems. The thermal and shock insensitivity of IHE provides major improvements in weapon safety; however, its incorporation into new primary designs led to some problems, mainly temperature-related. The insensitive HE TATB poses design considerations that differ little from those with conventional HE except that TATB is much more difficult to initiate. Its initiation is sensitive to temperature and particle size. Moreover, IHE exhibits significant sensitivity to granular spaces within pieces of the explosive when weakly initiated.¹³⁰

By mid-1990, the U.S. had incorporated IHE in only 35% of its stockpiled nuclear weapons. Because IHE performs much differently from earlier explosives used in U.S. weapons, warheads retrofitted with IHE must be completely redesigned and tested.¹³¹ The last air-delivered warheads without IHE in them will not be retired until after the year 2000.¹³²

The major reason for this low percentage of weapons fitted with IHE is that technological and operational restrictions precluded incorporation of IHE in artillery-fired atomic projectiles and fleet ballistic missile warheads. The small diameters of 155MM and 8" cannon barrels posed very tight geometric constraints on the design of nuclear projectiles; if IHE were used, less nuclear material could be included, leading to lower yields. Submarine-launched ballistic missile re-entry vehicles are maximized for range; heavier warheads cost range and so do not use IHE.¹³³

The net result of more than 50 years of high explosives development for American nuclear weapons has been the

¹²⁹"Nuclear Weapons R & D and the Role of Nuclear Testing," ENERGY & TECHNOLOGY REVIEW, UCRL-52000-86-9, September 1986, Lawrence Livermore National Laboratory, Livermore, California, p. 9.

¹³⁰Rosengren, RDA-TR-122100-001-Rev. 1, p. 67.

¹³¹REPORT TO CONGRESS ON STOCKPILE RELIABILITY, WEAPON REMANUFACTURE, AND THE ROLE OF NUCLEAR TESTING, UCRL-53822, George H. Miller, Paul S. Brown, and Carol T. Alonso, Lawrence Livermore National Laboratory, October 12, 1987, p. 9. Another source claims that at the beginning of 1990, only 25% of the U.S. nuclear weapons stockpile was equipped with IHE. (Drell-Foster-Townes Report, p. 27.)

¹³²Officials Respond to Warhead Safety Report," ARMS CONTROL TODAY, Vol. 21 No. 3, April 1991, p. 26.

¹³³Drell-Foster-Townes Report, pp. 27, 28.

reduction of an explosive charge of several hundred or thousand pounds required for wartime weapons to a charge ranging from 15 to 40 pounds for most modern warheads. Along the way, explosives have been made safer, more reliable, and longer-lived without chemical degradation than their predecessors.

High Explosive Naming Conventions

Both the Los Alamos and Livermore laboratories maintain separate naming conventions for their high explosives. These names are assigned at distinct and separate points at each lab.

The Los Alamos National Laboratory uses a numeric code to designate PBX-materials that reach stages of either pilot or full-scale production. The code consists of the letters PBX (Plastic Bonded Explosive), a dash, four digits, a dash, and a two more digits; for example, PBX-9010-02. The first two digits are the percentage by weight of the major explosive ingredient in the formulation. The next two digits are an arbitrary serial number, assigned sequentially as the materials are developed. The two digits following the dash are an additional sequentially-assigned serial number that designate different modifications of a given explosive compound.

Thus PBX-9010-02 is an explosive compound that contains by weight 90% of the major explosive ingredient, is the tenth 90% material to be developed, and is the second major modification of that particular compound.

The last two digits, the modification level, are often deleted in reference to LANL explosives. Thus production PBX-9404 should, strictly speaking, be designated PBX-9404-03. The -03 identifies a product manufactured in Holston mixing equipment and containing HMX with a specific particle-size distribution.

LANL research explosives are identified by an X followed by a four-digit number.¹³⁴

At the Lawrence Livermore National Laboratory, a specific code designation is assigned to an explosive its development has reached a point where:

1. A set of reasonable manufacturing specifications can be written for a developed formulation.
2. The evaluation of the material's chemical, physical,

¹³⁴UCRL-52997, p. 18-1/18-2.

explosive properties and sensitivities is essentially complete.

3. The material has a definite application.

LLNL explosive designations consist of the letters LX (Livermore Explosive), followed by a dash, two digits, a dash, and a single digit; for example, LX-05-0. The first pair of digits is an arbitrarily-assigned sequential serial number. The final single digit identifies a subclass in the series; it indicates small changes in manufacturing specifications that invariably occur during the production of the explosive. For example, when the LX-04-0 explosive particle size was revised, new lots manufactured under the changed specifications were identified as LX-04-1.¹³⁵

Some Livermore-developed HEs used in postwar U.S. nuclear weapons referenced in this series of volumes are as follows:

- LX-04 A solid explosive characterized by excellent mechanical and compatibility properties, an energy release about 9% less than LX-09, and shock-sensitivity properties much superior to LX-09.
- LX-07 A modification of LX-04 with a higher energy release (5% less than LX-09) obtained at the expense of some degradation in mechanical properties, e.g., less elongation, and shock-sensitivity.
- LX-09 An explosive similar to the LANL explosive PBX-9404 but with significantly improved thermal stability and slightly poorer physical properties. LX-09 suffered from serious "aging" problems while in service with the U.S. Navy in W-68 POSEIDON missile warheads.
- LX-10 An explosive in the same energy class as LX-09 and PBX-9404 but which uses HMX and Vitron A (a binder also used in LX-04) and which has excellent thermal characteristics. It also exhibits high creep resistance but may be somewhat more shock-sensitive than other HEs.
- LX-14 An explosive similar to PBX-9404 in energy release but which uses HMX and Estane (a binder also used in PBX-9011) and that has excellent thermal characteristics. LX-14 exhibits higher creep resistance than LX-10 and shock-

¹³⁵UCRL-52997, p. 17-1.

sensitivity similar to LX-04. LX-14 is slightly less energetic than LX-10.

LX-17 A solid explosive characterized by dramatic insensitivity to mechanical stimuli (i.e., has a high resistance to shock-initiated detonation), outstanding mechanical and compatibility properties and an energy release about two-thirds that of PBX-9404. LX-17 contains TATB and Kel-F 800 (a binder).¹³⁶

A newly-developed "LX"-type explosive has been formulated for use in earth-penetrating weapons (EPWs). LX-17 is too brittle to withstand the relatively long-duration shock loading associated with an EPW impact; in addition, LX-17 undergoes some irreversible dimensional change during the -54° to +74° C (-65° to +165° F) temperature change in the stockpile-to-target sequence encountered by warheads using it. A combination of TATB with a new binder and manufacturing process resulted in an explosive that met all physical requirements without any loss of energy.¹³⁷

In addition, in 1977, the Lawrence Livermore Lab found that it was possible to mix HMX, which is relatively dangerous, with TATB, which is less energetic but much safer to handle.¹³⁸ The resultant material is named PBX-9503.

Even more recently, Los Alamos has developed a new class of HE. LAX-112, the first of the class to be synthesized, is less sensitive than existing types but must undergo tests to determine if it can be used in weapons.

Table II-6 lists some properties of explosives used in past and present U.S. nuclear weapons. Both RDX and HMX are used in solid rocket propellants. LX-10 is used in W-68 and W-79 SLBM and AFAP warheads; LX-17 is used in B 83, W-84, and W-87 warheads.¹³⁹ **Table II-7** lists some contemporary uses of various HEs.

¹³⁶UCRL-52997, p. 17-2.

¹³⁷"Designer Explosives," ENERGY & TECHNOLOGY REVIEW, UCRL-52000-88-7.8, July-August 1988, Lawrence Livermore National Laboratory, p. 14.

¹³⁸"The State of the Laboratory - 1978," NEWSLINE, Lawrence Livermore Laboratory, May 18, 1978, p. 8.

¹³⁹"Formulating High-Explosive Materials," p. 25.

Advances in Pit Design

The fissile core, or "pit" of a fission explosive, so-called because it usually resides in the center of a sphere, like the pit of an avocado, peach, cherry, or olive, also offered many options for improvement. Changes could be made in the structure, composition, and configuration of the core. A number of these modifications were foreseen at the time of the Manhattan Project as mentioned above; others were invented and implemented by new generations of nuclear weapons designers who came to the laboratories at Los Alamos and Livermore after World War II.

Levitation

One of the first improvements incorporated in the postwar family of fission weapons was the so-called "levitated core."¹⁴⁰ This was a fancy name for the idea of suspending the pit within the tamper-high explosives assembly.

¹⁴⁰The concept of "levitation" as applied to nuclear weapons first appeared in the open literature in John McPhee's *THE CURVE OF BINDING ENERGY*:

Taylor leaned back and looked up into the sky and seemed for the first time to be aware of the falling rain. He said there was something about the structure of implosion bombs that he had not gone into, and that he could not go into, which contributed greatly to their yield. He said it had proved out in a bomb he had designed, which had had a very descriptive name.

"What was the name?" I asked him.

He shook his head. After a moment, he said, "All I can say is this: They had known all along that the way to get more energy into the middle was to hit the core harder. When you hammer on a nail, what do you do? Do you put the hammer on the nail and push?" (p. 218)

A further description of "levitation" can be found in *BORN SECRET: THE H-BOMB, THE PROGRESSIVE CASE, AND NATIONAL SECURITY* by A. DeVolpi, et. al. (Pergamon Press, New York, 1981, p. 93). There is some justifiable doubt about whether Ted Taylor was the first, or only, U.S. nuclear bomb designer to develop this concept -- the idea of "levitation" went back to the days of the Manhattan Project, and Taylor did not join the staff at Los Alamos until November 1949, a year and a half

The heavy fissile material was usually suspended and positioned within the tamper by means of wires that would not significantly disturb the implosion shock wave.¹⁴¹ For manual in-flight arming, the entire fissile core and initiator, and a portion of the tamper and high explosive charges, could be inserted through a "trap door" in the high explosives assembly around the pit (see Volume VIII).

The fissile core may be solid, a hollow shell, or a solid fissile core levitated within a fissile shell. The portion of the pit accelerated into the innermost core may be either the

after the SANDSTONE tests. In a letter to the author dated 7 March 1986, Taylor stated:

What I had in mind, but was not at liberty to explain to McPhee, involved extensions of several ideas that had been conceived by others long before I came to Los Alamos. The overall result was new, not the conceptual building blocks. In particular, the concept of introducing a gap between explosively accelerated materials to increase peak pressures was extensively described in the open literature long before I met McPhee, and was well known before the start of the Manhattan Project.

Dr. Taylor's assertions notwithstanding, the concept of levitation, as applied to U.S. nuclear weapons, was still avowed to be a high state secret by DOE during the PROGRESSIVE case in 1979. Taylor later described levitation as meaning "standing back and hitting hard, as with a full swing of a hammer." ("An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 4.)

A telex from Alvin C. Graves, then director of nuclear testing at the Nevada Test Site in August 1951, referring to information that could be shared with Great Britain at that time, mentions "levitated systems."

¹⁴¹Los Alamos Laboratory Program for Calendar Year 1949, pp. 3, 10; letter dated December 9, 1949 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, p. 4.

reflector/tamper, or in a hollow core design, fissile material "pasted" to the reflector/tamper assembly.¹⁴²

Hans Bethe described the advantages of combining levitation with hollow composite cores:

Levitating is very useful for (reducing bomb diameter). And well, Los Alamos reduced the diameter step by step from '45 on. Here it was felt that the solid core clearly was very hard to compress. We knew that by '45. Originally we had wanted a hollow shell, but we didn't trust the symmetry (of implosion).

Then we decided, yes, we could after all have a hollow shell, and if we had a hollow shell, it was useful to have the very center solid. So that the hollow shell was pasted on the explosives, so to speak. And the core had to be levitated. So the question was, could you make thin enough wires (levitating the core within the tamper) which were strong enough, so that they could survive transportation by plane and (being dropped from an aircraft) and you don't need to have it impact on the ground.

But they had to be strong enough, and yet small enough, thin enough, so that they wouldn't disturb the spherical symmetry. Because the spherical symmetry is all important. Only then do you get the increase in density, which you bank on.

In addition to that, with the hollow construction, you can put a little more material in. You can put in more than a critical mass. Which again increases the yield. And then already in the last month of Los Alamos, we invented the composite, which is a great increase in the yield, (using) plutonium in the center and then uranium outside.

¹⁴² "The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 5.

Plutonium was much more expensive than uranium, the separated uranium. Three times maybe. Therefore, we would get a much better arsenal by having the mixtures. So in the many test series in Nevada and also occasionally in the Pacific, Los Alamos went to smaller and smaller diameter high explosives.¹⁴³

The technique of "levitation" followed directly from the need to get more energy into the pit for better compression to increase yield for a given critical mass. The purpose of the empty space ("air gap") between the pit and the tamper-HE assembly was to allow time and space for the relatively-massive tamper to gather momentum before striking the fissile core-initiator assembly.

This method of explosively accelerating a mass of material before it hits another mass was quite well-known during World War II; the effect it has is to multiply, by several orders of magnitude, the intensity of the compressive shock. This high-energy, high-speed impact results in better core compression. It was not employed in the first atomic implosion bombs at TRINITY site and over Nagasaki because LASL wanted to be conservative and keep design risks to a minimum to save time and improve the chances of success.¹⁴⁴

In a reorganization of the Los Alamos laboratory in the fall of 1945, the new M Division was established to study, among other things, "induced motion and compression of parts of the pit by HE" and "measurements of the (bomb) improvement effected by new explosives arrangements."¹⁴⁵

Levitated pits were under consideration as early as July 1945. A cable to Secretary of War Henry L. Stimson from George L. Harrison, a consultant to Stimson, dated 23 July 1945, stated:

First one of tested type should be ready at Pacific base about 6 August. Second one ready about 24 August. Additional ones ready at accelerating (production) rate

¹⁴³Interview of Hans Bethe by Richard Rhodes, 3 May 1993, reprinted in Rhodes, DARK SUN, pp. 188-189.

¹⁴⁴"Iraq's Bomb Program": A Smoking Gun Emerges." SCIENCE, Vol. 254, November 1, 1991, p. 644.

¹⁴⁵MANHATTAN DISTRICT HISTORY, PROJECT Y -- THE LOS ALAMOS PROJECT, Edith C. Truslow and Ralph Carlisle Smith, LAMS-2532, Vol. II, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, 1947, p. 75.

from possibly three in September to we hope seven or more in December. The increased rate above three per month entails changes in (bomb) design which (General Leslie R.) Groves believes thoroughly sound.¹⁴⁶

By late August, a test of a levitated pit was possible by the summer of 1946, along with an improved HE assembly:

Our operations should be conducted in such a way that over the coming year the number of units actually produced is within a few percent of those which could be produced on the basis of the material manufactured. The development of the levitated model should be completed by the summer of 1946, and the essential program leading to a new model in the explosives design should be completed by the autumn of 1946.¹⁴⁷

In an espionage report to his brother-in-law Julius Rosenberg, David Greenglass, then an Army machinist at Los Alamos, described levitation in September 1945 as an experiment "which was concerned with (reducing) the amount of plutonium to be used in the atomic bomb." This technique would also improve the efficiency of the use of fissionable materials:

I recall that in September 1945 when I was in New York City on furlough, I saw Julius Rosenberg, my brother-in-law. At that time, I gave him information relating to the atomic bomb. With that information, I now recall that I also gave him in writing a description of an experiment, then being conducted at Los Alamos, N.M., which was concerned with the reduction of the amount of plutonium to be used in the atomic bomb.

¹⁴⁶Rhodes (TMAB, p. 689) assumes that the cable hints of a composite core. Composite cores were under consideration at LASL at this time; however, it is equally likely that the cable refers to a levitated pit design. Since a levitated pit required less fissionable material for the same yield, more levitated weapons could be produced from a given quantity of plutonium.

¹⁴⁷Memorandum dated August 20, 1945 to All Division and Group Leaders from J. R. Oppenheimer, Subject: Near-Future Operating Principles for the Los Alamos Project.

This experiment, I informed Julius Rosenberg, consisted of one sphere of uranium inside a larger sphere of uranium with a large air gap between the two spheres and stilts to hold the inner sphere apart from the outer sphere. I informed Rosenberg that the air gap was used to increase the speed with which the outer sphere is imploded. I told him this would result in a greater explosion with the use of less plutonium.¹⁴⁸

By November 1945, Dr. J. Robert Oppenheimer envisioned a test of a levitated-pit weapon by the spring of 1946:

As you know, considerable progress has been made in the development of slow explosives, and some progress has been made on the levitation program. I believe that plans should now be initiated for a test of the gadget embodying these advances. ... It would seem that May 1st (1946) might appropriately be set as the target date for this test ...¹⁴⁹

Suspended or "levitated" bomb cores were developed by the summer of 1946 and proposed for inclusion in the Operation CROSSROADS nuclear tests:

The question of what weapon to employ in these (CROSSROADS) tests was the subject of much discussion. The recommendation of the (Los Alamos) Laboratory was to employ the Nagasaki type (implosion) weapon, in spite of the fact that a different type of weapon existed and was urgently in need of test.¹⁵⁰

¹⁴⁸Rhodes, DARK SUN, pp. 188-189, citing a signed statement to the Federal Bureau of Intelligence by David Greenglass, August 16, 1950. This statement was later introduced as evidence against Julius and Ethel Rosenberg during their espionage trial.

¹⁴⁹Letter dated November 21, 1945 from J. Robert Oppenheimer to Major General L. R. Groves.

¹⁵⁰MANHATTAN DISTRICT HISTORY, PROJECT Y, THE LOS ALAMOS PROJECT, Vol. II, p. 20. Hoddeson, et. al., p. 453 (footnote 103) states that "the first postwar step to improve the design of the implosion system was to adopt a conservative levitated design, which would in fact have worked perfectly well at TRINITY and given a significantly higher yield."

This recommendation was made in December 1945 or January 1946; since the Nagasaki FAT MAN implosion bomb had been detonated in combat over a city, the military wanted to learn the effects of the bomb over and under water so they could be compared with effects from its previous use in an airburst.

Also, had a new and untried weapon been used for the tests, and had it "fizzled," the laboratory would have been harshly criticized for not using a "proven" weapon in such an expensive military test operation with its attendant international publicity.¹⁵¹

By late 1946, the Los Alamos Scientific Laboratory envisioned the firing of at least one static airburst test to:

- a. test the principle of levitation,
- b. increase existing theory and knowledge of implosion type weapons, and
- c. determine the most efficient weapon design for the effective use of available fissionable material in building up the stockpile.¹⁵²

In November 1946, LASL director Norris Bradbury noted:

The possibility of a slightly different type of weapon designated as the "levitated" model was known at the close of the war. The potential improvement in performance of such a weapon, together with the possibility of employing uranium-235 more efficiently under these circumstances, indicated this to be one of the problems upon which effort should be expended and these efforts have been, in the main, successful.¹⁵³

Bomb models incorporating levitated pits were in dire need of proof-testing by early 1947, and by April, plans were being

¹⁵¹Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 27.

¹⁵²OPERATIONAL HISTORY OF ATOMIC ENERGY COMMISSION PROVING GROUND GROUP (Task Group 7.1), submitted 20 May 1948, USS ALBERMARLE (AV-5), Eniwetok Atoll, Marshall Islands, pp. 2-1, 2-2.

¹⁵³Letter dated November 14, 1946 to U.S. Atomic Energy Commission from N. E. Bradbury, Director, LASL.

made for the test-firing of three bombs; at least two would include levitated cores.¹⁵⁴

By May 1948, implosion calculations were being run on IBM computers at Los Alamos according to a program laid down at the beginning of the year to obtain data on a wide range of levitated configurations.¹⁵⁵

Levitated pits were tested in the devices exploded during Operation SANDSTONE in the spring of 1948. The objectives of SANDSTONE were to (1) prove the nuclear and initiator components of the stockpile; (2) increase existing theory and knowledge of implosion type atomic weapon design; (3) test the principle of levitation; (4) test the principle of composition (composite cores); and (5) determine the most efficient weapon design for the effective use of available fissionable material in building up the stockpile.¹⁵⁶

The SANDSTONE tests rendered obsolete virtually every component of existing stockpile weapons.¹⁵⁷ Weapons which first incorporated levitated pits included the MKs 4, 5, 6, and 7 aerial bombs and missile warheads developed soon after SANDSTONE. Levitated pits are now a standard assembly in all U.S. fission weapons and thermonuclear weapon primary stages.

Composition

¹⁵⁴Letter dated November 7, 1946, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Col. H. C. Gee, Santa Fe, New Mexico, Subject: Los Alamos Technical Program for 1947, p. 6; Draft Minutes of the General Advisory Committee, Second Meeting, February 2-3, 1947; Memorandum for the Commissioners, USAEC, Washington, D.C., April 21, 1947, from Carroll L. Wilson; letter dated April 25, 1947, from David Lilienthal, USAEC Chairman, to Lt. Gen. Lewis H. Brereton, Chairman, Military Liaison Committee to the AEC.

¹⁵⁵Los Alamos T (Theoretical) Division Monthly Progress Report, LAMS-743, 20 May 1948, p. 3.

¹⁵⁶OPERATIONAL HISTORY OF ATOMIC ENERGY COMMISSION PROVING GROUND GROUP (Task Group 7.1), 20 May 1948, Eniwetok Atoll, Marshall Islands, p. 2-1. A memorandum for Record dated July 3, 1952 from Lt. Col. R. P. Campbell, Jr., DMA/USAEC, Subject: Notes from the Commission's Briefing of July 1, 1952, states that SANDSTONE tested composite split levitated cores and U-235 implosion weapons.

¹⁵⁷Hewlett and Duncan, p. 165.

Another development which occurred soon after World War II was the use of composite cores, consisting of concentric thin shells (as opposed to solid hemispheres) of both uranium-235 (oralloy) and plutonium-239.¹⁵⁸

¹⁵⁸This peculiar arrangement of dual-element bomb cores is shown in an illustration in an article entitled "The Great Test Ban Debate," by Herbert York, which first appeared in SCIENTIFIC AMERICAN magazine in November 1972, and which was reprinted on page 302 of ARMS CONTROL: READINGS FROM SCIENTIFIC AMERICAN (W. H. Freeman & Company, San Francisco, California, 1973). Composite cores are also mentioned on pages 79 and 94 of NUCLEAR WEAPONS AND AMERICAN STRATEGY, 1945-1953, a doctoral dissertation by Mark Bernard Schneider (University of Southern California, Los Angeles, California, 1974). Howard Morland incorporated a composite core in the primary (fission) stage of his hypothetical 300-kiloton thermonuclear bomb design in THE PROGRESSIVE (November 1979); the accuracy of his conjecture was confirmed by Dr. George W. Rathjens of the Massachusetts Institute of Technology:

On November 4, 1978, George Rathjens, a government affiant in this case, told me that the two materials described in paragraph 30 above...were used in combination in a typical nuclear weapon. "How do you know?" I asked. "Because I've seen the designs," he replied. I asked if he could tell me the configuration and were they alloyed together. "Let's put it this way," he said, "they are not mixed at the molecular level." Thus, the two materials were not blended or alloyed. (Affidavit I of Howard Morland in Opposition to Motion for Preliminary Injunction, Civil Action 79-C-98, filed March 19, 1979, par. 3.)

Further credibility for the concept (and implementation date) of composite bomb cores is given by the following statement in AF ATOMIC ENERGY PROGRAM:

Later, in 1947 when the Atomic Energy Commission was struggling with the problem of production of fissionable material, the Los Alamos personnel made invaluable contributions to the fight against (material) scarcity by creating nuclear systems (bomb cores) that could employ U-235 as well as Pu-239, and which at the same time were more efficient than the core used in the FAT MAN 1561. (Vol. IV, p. 143).

The purpose of this arrangement is twofold: to reduce the amount of plutonium-239 in the core (which helped alleviate a critical shortage of plutonium in the U.S. immediately after World War II), and to improve the yield-to-weight ratio of the core itself. The faster-fissioning plutonium, when placed inside a shell of oralloy, makes more efficient use of the neutrons released by its own fissions.¹⁵⁹

One characteristic of the implosion of uranium is that it offers a lower yield, on a pound-per-pound basis, than a comparable amount of plutonium.¹⁶⁰ Because of this, a relatively large amount of oralloy can be assembled for a

Although the actual figures are deleted, page 684 of Vol. II of AF ATOMIC ENERGY PROGRAM shows "Solid Christy Pit (Nagasaki)" and "Composite Christy Pit" (Dr. Robert F. Christy was a premier bomb core designer in the Theoretical Division at Los Alamos during and after World War II). Development of a "Composite Christy Gadget" is described in "Operation SANDSTONE, LAJ-9B Activities."

An August 1951 telex from Alvin C. Graves, director of nuclear testing at the Nevada Test Site, referring to information that could be shared with Great Britain at that time, mentions "composite devices." In addition, an official Air Force history states:

In the implosion principle, a subcritical mass ... of fissionable material (U-235, plutonium, or a mixture of the two)..."(HISTORY OF TACTICAL AIR COMMAND, 1 July through 31 December 1951, Vol. VII, Special Weapons Activities, p. 34).

¹⁵⁹"An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 5; Zaloga, p. 89. A July 1951 CIA estimate of Soviet atomic weaponry notes that "as uranium-235 becomes available, it is reasonable to expect that the relatively more efficient composite weapons will be fabricated. (Klaus) Fuchs is known to have told the Soviets of the possibility of such weapons." ("Status of the Soviet Atomic Energy Program," CIA/SI 113-51, 28 July 1951, Central Intelligence Agency, p. 14.) Due to delays in building and operating their uranium isotope separation plants, the Soviets did not test a uranium bomb until September 1951; this was a composite uranium-plutonium implosion design, rather than a much simpler but more inefficient gun-assembly design. (Zaloga, pp. 41, 90.)

¹⁶⁰Kunetka, p. 117.

high-yield explosion, provided that the mass is not allowed to become prematurely critical. The highest-yield pure-fission weapon ever designed and tested by the United States, the 500 KT MK 18 "Super Oralloy Bomb," used uranium-235 as its explosive ingredient.¹⁶¹ Theoretically, uranium-fueled fission bombs could reach yields of 750 KT to a megaton.¹⁶²

Uranium production had always outpaced plutonium production in the U.S. By October 1945, weapons-grade U-235 was being produced at Oak Ridge at a rate of about 32 kilograms (70.15 lbs.) per month, while monthly plutonium production at Hanford was only about four to six kilograms per month.¹⁶³

Between October 1945 and June 1946, improvements in design and operating procedures at the electromagnetic uranium isotope separation equipment at Oak Ridge resulted in a 117% increase in output¹⁶⁴ to about 69 kilograms, 152 lbs., of U-235 per month.

This meant that in the fall of 1945, by maintaining current production rates of fissionable materials for use in the weapons then in existence (the LITTLE BOY, which used just over 64 kilograms of uranium, and the FAT MAN, which used 6.2 kilograms of plutonium), enough material would be available to fuel only about six LITTLE BOYS and only eight to 12 FAT MANS per year, if these bombs continued to use the wartime target-projectile and pit assemblies, respectively. By mid-

¹⁶¹McPhee, pp. 89-91; Schneider, pp. 180, 185-186, 188, 191; THE ADVISORS: OPPENHEIMER, TELLER, AND THE SUPERBOMB, Herbert York (W. H. Freeman and Company, San Francisco, California, 1976), pp. 83-85. The latter book was reprinted in 1989 by the Stanford University Press in Stanford, California; it is referenced hereafter as York, THE ADVISORS (1989) to differentiate it from the 1976 Freeman edition.

¹⁶²Zaloga, p. 90.

¹⁶³"Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, pp. 11, 12. The Report of Hugh H. Clegg and Robert J. Lamphere covering Interviews with Klaus Fuchs in London, England, between May 20 and June 2, 1950 (pp. 26, 32) claims that Oak Ridge was producing 100 kg of U-235 per month and Hanford was producing 20 kg of plutonium per month. These numbers may have represented goals, rather than actual monthly production levels.

¹⁶⁴Memorandum dated 8 July 1946 from Col. E. E. Kirkpatrick to Commanding General, Manhattan Project, Subject: Y-12 Operations, October 1945-June 1946.

1946, enough U-235 was being produced to fuel about 13 LITTLE BOYS per year, if the higher rate of U-235 generation could be maintained.

This disparity in production rates between uranium-235 and plutonium also meant that until a U-235/Pu composite core could be tested and certified for stockpiling, the large quantity of U-235 on hand was in effect a long-term reserve with no immediate practical military application,¹⁶⁵ exclusive of its use in inefficient LITTLE BOY bombs.

The total number of nuclear weapons stockpiled by the U.S. between 1945 and 1947 did not even approach these figures, while ambitious war plans called for the employment of hundreds of weapons. **Table II-8** lists some of the early postwar plans for nuclear combat, the numbers of weapons associated with each plan, and the actual size of the national atomic weapons stockpile when each plan was formulated.

By 1947, it was possible in the U.S. to produce, from one ton of uranium ore, eight times as much alloy at the K-25 plant (the gaseous diffusion uranium isotope separation facility at Oak Ridge, Tennessee) as plutonium at the Hanford, Washington reactors, where plutonium was produced by bombarding uranium-238 with neutrons.¹⁶⁶

In addition, on a per-gram basis, it cost between four and eight times as much to produce plutonium than U-235. Despite this fact, plutonium would remain a necessary material in low-yield implosion weapons.¹⁶⁷

By 1946, the three Hanford production reactors -- the wartime 250 megawatt B, D, and F piles -- had started to wear out, poisoned by fission products and in peril of being shut down completely by swelling of the graphite: one reactor (B pile) was unloaded in May and placed in standby status, and another (F pile) was cut back to only 80% of its rated power.

B pile, the world's first full-scale plutonium production reactor, was built in less than a year between 1943 and 1944, and had operated continuously between 1944 and 1946, producing the plutonium used in both the TRINITY test device and the Nagasaki FAT MAN bomb.

¹⁶⁵Rhodes, DARK SUN, p. 227.

¹⁶⁶AF ATOMIC ENERGY PROGRAM, Vol. II, p. 678.

¹⁶⁷Memorandum for the Files dated November 16, 1951 from William Borden, subject: Some Comments on the Ratio Problem.

None of the early operating problems at Hanford was more serious than this graphite distortion. By early 1946, expansion of the graphite reactor cores had led to severe bowing and binding of process tubes, the piping within the reactor which held the uranium fuel elements as they were irradiated into plutonium.¹⁶⁸

The swelling was caused by rearrangement of the crystal lattices within the graphite by intense neutron flux; this problem had been predicted by Eugene Wigner, a Hungarian theoretical physicist who worked for the Manhattan Engineer District, and it was called accordingly "Wigner's disease."¹⁶⁹

The shutdown and cutbacks at Hanford immediately affected bomb production. Los Alamos had planned to begin manufacturing 40 levitated composite cores (for MK 4 weapons) in August 1946 at a rate of two per week. LASL Associate Director Darol Froman had not checked with Hanford before making that plan. When he did so in June 1946, he discovered that the slowdown at Hanford would have a drastic effect. They would still accumulate their 40 cores, Froman believed, but it would take eight months:

In our conversation of June 13, 1946 I indicated that DP could fabricate two MK IV cores per week, beginning about August 19th. This represented production capacity at DP, assuming availability of material and trained personnel.

I was not fully aware of (plutonium) stocks and expected receipts at the time, but have since checked and find that this rate of production could be maintained only for about 13 weeks. Thereafter we would be limited by Hanford (plutonium) production to less than one MK 4 (core) per week, and would then require an additional five months to complete the required numbers of cores. Therefore, approximately eight months would be required to complete the total order regardless of the original production rate.

Mr. Taub advises me that the maximum expected production of (U-235) shells is sufficient for only one MK IV (core) per

¹⁶⁸THE HANFORD SITE: AN ANTHOLOGY OF EARLY HISTORIES, M. S. Gerber, Westinghouse Hanford Company, Richland, Washington, October 1993, p. 24.

¹⁶⁹Rhodes, DARK SUN, p. 277.

week. Since it would appear to be poor policy to maintain sufficient personnel to proceed at a rapid rate for a few months, I would suggest that the schedule be revised to a production of one MK IV (core) and one-half MK III (or equivalent MK IV) cores per week. This will suit Taub's production better and will permit completion of the total in about eight months.¹⁷⁰

Solutions to the problems at Hanford were under study by March of that year:

Potential trouble has started to develop at Hanford Engineer Works. Radiation effects on graphite are causing the graphite structure to expand at an alarming rate. Engineers from E. I. du Pont de Nemours & Company are at present investigating the problem to determine if any remedy is possible.

However, it is very probable that this phenomenon will limit the life of the Hanford piles. (Useful lifetime) estimates vary from a few months to a few years, and competent engineers will not attempt to make any prediction until further study. ... Du Pont has recommend shutting down one pile and operating the remaining two at different rates. The Du Pont recommendation will be followed.

In this way, one unit will approach the dangerous condition at a more rapid rate and one unit will be held in reserve so that there will always be an operable unit after the point is reached where the pile might become inoperable due to increased swelling of the graphite. It is desirable to always have one unit capable of operation in order to produce polonium (for initiators). ...

¹⁷⁰Memorandum dated June 15, 1946 to Darol K. Froman from M. Kolodney, subject: Fabrication of (Plutonium) Cores. DP Site was a chemical and metallurgical production plant established at LASL in the summer of 1945. "Mr. Taub" was J. M. Taub, supervisor of group CMR-6, Plutonium Metal Fabrication, at LASL. The U.S. did not stockpile even 40 bomb cores until well into 1948.

This difficulty at Hanford is typical of the many unknown features attendant to the production of plutonium by the transmutation method which is the only method known.¹⁷¹

Since these reactors also produced polonium by irradiating bismuth-209, and because polonium was a vital initiator ingredient which had to be replaced periodically because of its 138 day half-life, all three reactors could not be shut down completely for a prolonged period.¹⁷²

The distortion problem was traced to the helium gas generated within the piles by the neutron bombardment of the fuel elements. Helium was a highly-efficient heat remover, and higher temperatures were required within the graphite lattices to maintain their proper configuration. The addition of carbon dioxide, with its lower heat-transfer capacity, to the gas atmospheres of the Hanford reactors annealed and reduced the graphite expansion problem.¹⁷³

B pile at Hanford was shut down between May 1946 and April 1948; it was restarted when problems of graphite distortion were solved and when plans for Operation SANDSTONE led to the conclusion that less polonium would be required in future weapons and that it would not be necessary to hold a reactor in reserve to produce polonium for initiators.¹⁷⁴ (One of the findings of Operation SANDSTONE was that "aged" initiators with less "fresh" polonium in them were still adequate to start fission reactions.) B pile continued to produce

¹⁷¹Memorandum dated 1 March 1946 from Co. K. D. Nichols, USA, to Maj. Gen. L. R. Groves, USA, subject: Items to be Discussed with General Eisenhower at Your Next Meeting.

¹⁷²"Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, pp. 21, 22; Glasstone and Redman, WASH-1037 (Revised), p. 19.

¹⁷³Gerber, p. 24.

¹⁷⁴AF ATOMIC ENERGY PROGRAM, Vol. II, p. 678; Rhodes, TMAB, p. 765; "The Scale and Scope of Atomic Production: A Chronology of Leading Events," p. 1, attachment dated January 30, 1952 to letter dated May 8, 1952 from Brien McMahon, Chairman, Joint Committee on Atomic Energy, to Karl R. Bendetsen, Under-Secretary of the Army; "Production Reactors: An Outline Overview, 1944-1988," DOE/NP/00092T-H1, prepared for the U.S. Department of Energy by History Associates Incorporated, Rockville, Maryland, October 1992, pp. iii, 3, 9, 10.

plutonium for the AEC until it was shut down permanently in 1968.

Development of Uranium Implosion

Because so much more uranium than plutonium could be produced shortly after the end of the war, it became highly desirable to devise methods to implode or alloy, either by itself or in a composite pit. During World War II, or alloy had been used only in the LITTLE BOY gun-type assembly weapon dropped on Hiroshima, even though the MK II weapon had been conceived to use either or alloy or plutonium implosion (see Volume VI).

In July 1943 -- within the first year of the Los Alamos laboratory's establishment -- implosion of both plutonium and uranium was considered. With the acquisition of accurate means of calculation and reliable cross-section data, it became apparent that uranium implosion would be considerably less efficient than plutonium implosion.

This fact, added to the uncertainties of the whole implosion program at the lab at that time, made it seem desirable to plan for the use of uranium-235 by the gun method alone.¹⁷⁵

Although it was apparent by October 1943 that uranium implosion would drastically reduce the quantity of U-235 for a critical mass, an extremely important consideration at a time when uranium was in short supply,¹⁷⁶ this technique could not be pursued until more efficient methods of imploding uranium could be devised, perhaps after the war ended.

Another factor militating against a wartime test of a U-235-fueled implosion weapon was the extremely limited amount of enriched uranium on hand. Since the earliest bomb designs used heavy natural uranium tampers, to reduce the likelihood of instabilities and asymmetry in the spherical implosion wave, there was a chance that if U-235 were imploded within a U-238 tamper, and the test failed, then recovery of the precious U-235 might require tedious isotope separation if the metals of the core and tamper became mixed. On the other

¹⁷⁵A HISTORY OF THE UNITED STATE ATOMIC ENERGY COMMISSION, Vol. I, 1939-1946: THE NEW WORLD, Richard G. Hewlett and Oscar Anderson, Jr., WASH 1214, USAEC, Washington, D.C., 1962, 1972, pp. 251-252; MANHATTAN DISTRICT HISTORY, PROJECT Y, THE LOS ALAMOS PROJECT, Vol. I, p. 192; AF ATOMIC ENERGY PROGRAM, Vol. I, p. 75. The latter history states: "(The implosion bomb's) first design was discarded as too theoretical -- although it later proved highly efficient..." This may be a reference to a levitated composite design.

¹⁷⁶Hoddeson, et. al., p. 134.

hand, if plutonium were imploded within U-238, and that test failed, then the Pu could be easily chemically separated from the U-238.¹⁷⁷

When prospects for successful implosion appeared to be particularly unpromising during the fall of 1944 and winter of 1944-45, Los Alamos proposed, as an alternate to imploding either plutonium or uranium, to design and construct an "autocatalytic" bomb.

Autocatalysis had been considered as early as April 1943 when Robert Serber, in a series of lectures at the fledgling laboratory, suggested placing paraffin spheres coated with neutron-absorbent boron-10 in a mass of subcritical fissionable material; when the mass was made critical by adding more active material or by surrounding it with a tamper, the boron "bubbles" would be compressed and absorb fewer neutrons, increasing the nuclear reaction rate. Unfortunately, this scheme required a large amount of fissionable material and was very inefficient.¹⁷⁸

The most favorable feature of an autocatalytic system was that it would not require rapid assembly, thus eliminating the requirement for a large and cumbersome ordnance system. There were two main advantages believed to accrue from autocatalytic fission systems: small weapons sufficiently light and compact to open up many new avenues of delivery by both conventional means and by sabotage (e.g., covert smuggling of a weapon to its target) could be built; in addition, autocatalytic devices might be used to initiate thermonuclear reactions.¹⁷⁹

Because of their inherent inefficiency, autocatalytic systems, while pursued by LASL at least until the early 1950s, were never developed.¹⁸⁰

¹⁷⁷Hoddeson, et. al., p. 474.

¹⁷⁸"The Los Alamos Primer," E. V. Condon, Los Alamos Scientific Laboratory, 1943, pp. 23, 24; HISTORY OF MODERN PHYSICS, Vol. II, pp. 22, 167, 187-188; Smyth, p. 212.

¹⁷⁹Draft of The Long Range Military Objectives in Atomic Energy, Committee on Atomic Energy, Panel on Long-Range Objectives, p. 6, attachment to letter dated 18 August 1948 from J. Robert Oppenheimer, Chairman, Panel on Long Range Objectives, to Donald F. Carpenter, Chairman, Committee on Atomic Energy Committee, Research and Development Board, Department of Defense.

¹⁸⁰By October 1951, autocatalysis at LASL was "no more than a name." (Letter dated 9 October 1951 from Norris Bradbury to

Another early line of research at Los Alamos during World War II involved the use of uranium hydride for the gun and implosion weapons. This hydride seemed to have certain advantages over metallic uranium as a bomb material. One of the characteristics of uranium hydride is that the mean time between fissions in it is about 100 times that of metallic uranium. In addition, uranium hydride (uranium mixed with paraffin) was more readily available than uranium metal.

Since the uranium in the hydride was chemically bonded with hydrogen, a neutron moderator, in theory, less uranium hydride would be required for a given explosive yield than would be metallic uranium. Less fortunately, uranium hydride was also highly pyrophoric (this problem was later partially solved by coating the hydride with paraffin).¹⁸¹

By May 1943, LASL had noted the possibility of using uranium hydride as an explosive, and pointed out that the material had been discovered somewhat by accident. A positive feature of the hydride was that high-density compacts of it could be fabricated easily; because of this attribute, it was used in experimental reactors. The first uranium metallurgy study at Los Alamos was the preparation and powder metallurgy of the hydride, UH_3 .¹⁸²

When the implosion studies began, thought was given to imploding uranium hydride. UH_3 is about half as dense as metallic uranium, and the volume occupied by the hydrogen in it could be recovered under sufficient pressure. The feasibility of the uranium hydride bomb depended upon the hypothesis that the delaying effect of neutron slowing (moderation) by the hydrogen atoms, which caused the relatively long period between successive fission generations, was compensated for by a corresponding increase in the fission cross-section of the uranium with decreasing neutron energy.¹⁸³

Col. K. E. Fields.)

¹⁸¹GENIUS: THE LIFE AND SCIENCE OF RICHARD FEYNMAN, James Gleick, Pantheon Books, New York, 1992, p. 173; Hoddeson, et. al., pp. 181, 210.

¹⁸²THE HISTORY OF MODERN PHYSICS, 1800-1950, Vol. II, PROJECT Y: THE LOS ALAMOS STORY, Part II, Toward Trinity, by David Hawkins, Tomash Publishers, Los Angeles, California, 1983, pp. 23, 70, 138, 139. This is a further-declassified edition of LAMS-2532 and it contains some information deleted from the earlier edition released in 1961.

¹⁸³THE HISTORY OF MODERN PHYSICS, 1800-1950, Vol. II, PROJECT

If this hypothesis were true, the explosion rate would remain the same as in a metallic uranium bomb, although the critical mass would be decreased considerably, thus conserving a rare material in critically short supply.

A uranium hydride gun-type weapon would require the assembly of many small critical masses to achieve an appreciable yield. This presented severe design problems and the hydride gun was abandoned in February 1944.¹⁸⁴

Between May and July 1944, an integral experiment on the hydride implosion bomb was begun. A deuterium-deuterium collision neutron source was surrounded by a modified sphere simulating the hydride core as nearly as possible, and various tamper materials were investigated, as well as neutron distribution in the core and tamper.¹⁸⁵

Samples of uranium hydride prepared at Los Alamos were studied at a high pressure physics laboratory at Harvard University. Pressure density up to 10 kilobars (about 10,000 atmospheres), still very low pressure from the implosion standpoint, suggested that the hydride was not easily compressible.

Later studies showed that metallic uranium hydride would have an unexpectedly high critical mass -- about twice its previously-calculated value -- because it would also have a much lower density than earlier theoretical estimates had indicated.¹⁸⁶

In addition, more accurate critical mass and efficiency calculations completed in late 1943 indicated that the energy dependence that would make the hydride an efficient weapon did not occur. The calculations showed instead that metallic uranium would be a better fissile material.¹⁸⁷ Because of these factors, the hydride bomb program was continued at only a low priority after the spring of 1944.¹⁸⁸

Y: THE LOS ALAMOS STORY, Part II, p. 103; Minutes of the Meeting of the Governing Board, October 28, 1943, 7:45 PM.

¹⁸⁴Hoddeson, et. al., p. 181, claims that the uranium hydride gun had been abandoned at LASL by August 1944.

¹⁸⁵THE HISTORY OF MODERN PHYSICS, 1800-1950, Vol. II, PROJECT Y: THE LOS ALAMOS STORY, Part II, p. 104.

¹⁸⁶Hoddeson, et. al., pp. 206, 211.

¹⁸⁷Gleick, p. 173.

¹⁸⁸THE HISTORY OF MODERN PHYSICS, 1800-1950, Vol. II, PROJECT

By July 1944, LASL had high hopes for imploding fissionable materials, while recognizing that many problems still remained its realization:

There is little doubt that an implosion method can be made to work for both '25' (U-235) and '49.' (Pu-239) The question is how efficient can such bombs be made to be and how fast a moderately efficient bomb can be developed ... Eventually it may be possible to design an implosion bomb with such (high) speeds of assembly and such high compressions that very great efficiencies result (much greater than with the gun assembly). But at present, one must be content with lower compressions and low efficiencies.¹⁸⁹

One proposal put forth but not pursued at this time was for a low-efficiency composite implosion core with a yield of only a few hundred tons of TNT equivalent. This would be a "quick and dirty" approach to implosion.¹⁹⁰

With the reduction in hydride research, attention focused again on imploding metallic uranium. Towards the end of the war, in April 1945, LASL had created the lab's first metal critical assembly with U-235 metal hemispheres to study the amount of U-235 required for criticality. In addition, criticality studies had been conducted on 3.5 and 4.5 inch diameter U-235 spheres.¹⁹¹

Towards the end of the war, in May 1945, a group was assigned at Los Alamos to investigate the possibility of composite core implosion, to use or alloy with plutonium more efficiently than would be possible by using uranium alone in a gun assembly.¹⁹² At this time, the lab believed that the

Y: THE LOS ALAMOS STORY, Part II, pp. 18, 66, 69, 97, 118, 308; Hoddeson, et. al., p. 136. The UH₃ weapon was reportedly a "pet project" of Edward Teller; he continued research on it at his new weapons lab founded in Livermore, California in the fall of 1952.

¹⁸⁹Hoddeson, et. al., p. 241.

¹⁹⁰Hoddeson, et. al., p. 242; Hewlett and Anderson, pp. 251-252.

¹⁹¹Hoddeson, et. al., pp. 339, 347.

¹⁹²MANHATTAN DISTRICT HISTORY, PROJECT Y, THE LOS ALAMOS PROJECT, Vol. I, pp. 192, 198-199, 325. Composite weapons

majority of postwar nuclear weapons would be FAT MAN models fueled with either plutonium or uranium.¹⁹³

By the time of the first atomic test in July, LASL was estimating that composite core bombs could be made available within a relatively short time, that they could be used without testing, and that they would make better use of the limited U-235 available than the inefficient LITTLE BOY target-projectile assemblies:

Should like to be quite sure that the cost of going through with our present program is understood by you. If we go through with present immediate commitments on LB (LITTLE BOY) and FM (FAT MAN), then start on composite core FM work, estimate total number available by November first, including initial units, as about ten.

If we scratch LB but go through with first FM as planned, total number for this period would be about fifteen. If we scratch both initial units, total number for this period would be about sixteen. You have the dates for initial units. Our estimate for composite core FMs is that they will be available about a month after these dates. This may be squeezed somewhat but not much. If, on the other hand, we do not at present

were one of the "secrets" disclosed by Klaus Fuchs to the Soviets, possibly as early as September 1945. (Report of Hugh H. Clegg and Robert J. Lamphere covering Interviews with Klaus Fuchs in London, England, between May 20 and June 2, 1950, pp. 27, 34; Scientific Intelligence Report, Joint Atomic Energy Intelligence Committee, Status of the Soviet Atomic Energy Program, CIA/SI 118-51, Office of Scientific Intelligence, Central Intelligence Agency, 6 March 1952, p. 14.) A report dated October 18, 1944 to J. R. Oppenheimer from J. B. Conant referred to a possible weapon design using "both materials," presumably U-235 and Pu.

¹⁹³Letter dated May 7, 1945 to General Leslie R. Groves, USA, from J. Robert Oppenheimer, Director, LASL. A tentative draft dated 14 May 1945 of a proposed radio address to be delivered by President Truman after the successful use of the atomic bomb over Japan, written by William L. Laurence, press consultant to General Groves, stated that a "second type" of bomb (the implosion type) would, with future development, probably be the better of the two types (i.e., gun and implosion) and would be used for both U-235 and plutonium.

commit all our material, it may be possible to improve this date by ten days.¹⁹⁴

Groves responded to Oppenheimer's telex by stating that present requirements could not be changed; these included the dropping over Japan of one LITTLE BOY and possibly as many as three FAT MANs "in their best present condition."

Two weeks after the TRINITY test, Major General Leslie Groves anticipated that uranium-fueled FAT MAN bombs might be available as early as September 1945, although at a cost in efficiency compared to the plutonium-fueled FAT MAN:

In September, we should have three or four bombs. One of these will be made from 235 material and will have a smaller effectiveness, about two-thirds that of the test type, but by November, we should be able to bring this up to full power. ... By some time in November, we should have the effectiveness of the 235 implosion type bomb equal to that of the tested plutonium implosion type.

By mid-October we could increase the number of bombs slightly by changing our design now to one using both materials in the same bomb. I have not made this change because of the ever-present possibilities of difficulties in the new designs. We could, if it were wise, change our plans and develop the combination bomb.¹⁹⁵

¹⁹⁴Telegram dated 19 July 1945 from J. R. Oppenheimer to Maj. Gen. L. R. Groves; Hoddeson, et. al., p. 377. The minutes of an August 7, 1945 technical and scheduling conference at LASL on estimating bomb efficiencies from radiochemical sampling refer to discussion of "the usefulness of the method for composite 25-49 gadgets."

In addition, a memo dated August 9, 1945 from J. Hirshfelder and N. K. Sugarman to Samuel K. Allison discusses the difficulty of determining the efficiency of mixed plutonium-uranium "gadgets." The July 19, 1945 telex from Oppenheimer to Groves also reveals that the amount of uranium used in a LITTLE BOY (approximately 64 kilograms) was then estimated to be sufficient for five composite plutonium-oralloy cores.

¹⁹⁵Memorandum dated 30 July 1945 to the Chief of Staff from Major General Leslie R. Groves, USA. According to "A Few Words About This Picture," (Stanley Goldberg, INVENTION & TECHNOLOGY, Fall 1991, p. 51), enough U-235 to make a second

In September, a composite core with a uranium-to-plutonium ratio of 2:1 was recommended:

For the convenience of the Committee, it may be remarked that present atomic bombs, whose performance may be judged by the Nagasaki strike, use about 6.2 kilograms of plutonium or alternatively, in all probability, 3.2 kilograms of plutonium and 6.5 kilograms of enriched uranium U-235.¹⁹⁶

Such a core, it was believed, would make more efficient use of the limited U.S. plutonium stockpile:

In 1945, therefore, Los Alamos found itself confronting two important principles which pointed toward significant improvements: One was the fact that use of plutonium alone in a bomb is inefficient because this element, if employed in conjunction with U-235, permits manufacture of more weapons -- each having a respectable bang -- and because our project produces much more U-235 than plutonium.¹⁹⁷

By March 1946, the Army planned to continue production of both plutonium and uranium-235, and to use both materials in combination in a bomb then under development.¹⁹⁸ During March,

LITTLE BOY (possibly around 60 to 65 kilograms), had been accumulated at by Oak Ridge by January 1946.

¹⁹⁶Schneider, NUCLEAR WEAPONS AND AMERICAN STRATEGY, 1945-1953, p. 80. Composite cores was one of the U.S. nuclear weapons secrets passed along to the Soviets by Dr. Klaus Fuchs in the fall of 1945, then described as a "mixed" or "combination" fission bomb that used both plutonium and uranium. (KLAUS FUCHS, ATOM SPY, Robert Chadwell Williams, Harvard University Press, Cambridge, Massachusetts, 1987, pp. 81, 191; BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 40 No. 8, October 1984, p. 46.)

¹⁹⁷Memorandum for the Files dated November 8, 1949 from William L. Borden, re: Super Weapon Program, p. 1.

¹⁹⁸Memorandum dated 1 March 1946 from Co. K. D. Nichols, USA, to Maj. Gen. L. R. Groves, USA, subject: Items to be Discussed with General Eisenhower at Your Next Meeting.

Soviet spy Klaus Fuchs attended a symposium at LASL which discussed levitation and composite cores.¹⁹⁹

The basic design of a composite nuclear core was developed by the spring of 1946, even before the CROSSROADS tests that summer:²⁰⁰

Various ideas for modification of both these weapons (the LITTLE BOY and FAT MAN) were expressed in the laboratory during the war, but the necessity for completing a successful weapon resulted in concentration of all energy on the most sure-fire designs. Thus it was not until 1946 that a solid composite implosion weapon containing both U-235 and plutonium was constructed. This job was stimulated by a desire to keep the laboratory personnel together and working on something until policy could be formulated. The composite product was directed at greater economy of the more-scare plutonium with achievement of a larger bang than was thought could be obtained from a gun-type U-235 weapon.²⁰¹

By that time, one of the "outstanding and primary needs" at LASL was the development of an efficient use of U-235 in atomic weapons.²⁰² In March, Darol Froman of LASL had spoken to the Interim Council on the experimental data developed on the levitated implosion "gadget" and the elements of design of that device, including the composite "gadget" of plutonium and uranium.²⁰³

One of the most critical items facing the nuclear weaponeers at Los Alamos at the end of World War II was a full-scale test

¹⁹⁹Rhodes, DARK SUN, p. 246.

²⁰⁰MANHATTAN DISTRICT HISTORY, PROJECT Y, THE LOS ALAMOS PROJECT, Vol. II, p. 76; THE HISTORY OF MODERN PHYSICS, 1800-1950, Vol. II, p. 273.

²⁰¹Memorandum for the Files dated November 8, 1949, from Walter A. Hamilton, subject: Inquiry into the Aspects of a Superweapon Program, p. 3.

²⁰²Letter dated 2 December 1953 to Honorable W. Sterling Cole, Chairman, JCAE, from Norris Bradbury, Director, LASL.

²⁰³Notes on Interrogation of Klaus Fuchs from FBI serial file 65-58805-1246, dated June 12, 1950 (provided by Richard Rhodes), p. 5.

of a levitated composite pit weapon. If this test showed questionable results, then a solid non-levitated composite pit would be fired. In addition, a test of a pure uranium-235 pit, either levitated or not, was also highly desirable.²⁰⁴

Some thought was given to testing a composite core during Operation CROSSROADS; however, because the core required some additional development at that time,²⁰⁵ an opportunity to test it would not arise for almost two more years.

Another important factor entering into the future of the composite weapon core was the condition of the plutonium-producing reactors at the Manhattan District's Hanford Works on the banks of the Columbia River in Washington state. By the spring of 1946, prolonged irradiation of the graphite in the reactors was causing it to expand "at an alarming rate."

Estimates of the remaining useful life of the reactors ranged from a few months to a few years, so the E. I. du Pont de Nemours and Company, which ran the facility, recommended shutting down one pile and operating the other two reactors at a reduced power level, thus causing a large reduction in plutonium production. Under these operating conditions, one reactor would be held in reserve so that there would still be one operable unit even if the other two had to be eventually shut down.

In the meantime, running the two others at a slower rate would prolong their life, at the cost of reduced plutonium output. At this time, uranium and plutonium had been produced for only a handful of composite cores.²⁰⁶

A bottleneck in the testing of composite cores was the installation of remotely-operated core assembly equipment at LASL's Pajarito site in 1946. This equipment was put into use following the deaths of two physicists in August 1945 and May 1946 during manual critical assembly experiments and demonstrations. By the beginning of September 1946, parts had been fabricated for about five composite plutonium-oralloy cores.²⁰⁷

²⁰⁴Los Alamos Technical Program for 1947, p. 6.

²⁰⁵Memorandum dated 1 March 1946 from Co. K. D. Nichols, USA, to Maj. Gen. L. R. Groves, USA, subject: Items to be Discussed with General Eisenhower at Your Next Meeting.

²⁰⁶Memorandum dated 1 March 1946 for Major General L. R. Groves from Brig. Gen. K. D. Nichols, District Engineer, U.S. Army Corps of Engineers, subject: Items to be Discussed with General Eisenhower at Your Next Meeting.

²⁰⁷Memorandum for Dr. N. E. Bradbury dated 3 September 1946

By the fall of 1946, the principal nuclear testing goals of the Los Alamos laboratory were:

1. Completion of a test of a levitated composite weapon ...
2. Completion of a test of a solid composite ... in the event that (the first test) shows questionable results.
3. Completion of a test of a pure "25" (U-235) weapon of either the levitated or Christy type ...²⁰⁸

Experimental studies conducted under laboratory conditions at Los Alamos did not permit verification of theoretical calculations to the extent possible by actually detonating a test model of an atomic weapon. In fact, only by firing devices of experimental design would it be possible to determine precisely future research and development programs for nuclear weapons. Data from tests would enable Los Alamos both to develop more efficient weapon designs that would make the most effective use of available fissionable material, and to advance general theory of implosion type weapons.²⁰⁹

Composite levitated cores were being stockpiled by the beginning of 1947 for use in the MK III improved FAT MAN production model bomb,²¹⁰ although composite pits were not tested until the SANDSTONE series test explosions in the

from Col. L. E. Seeman, U.S. Army Corps of Engineers, subject: Critical Assemblies; memorandum dated 6 September 1946 to Eric Jette, Acting Director, LASL, from Darol Froman, subject: The Order to Test Certain Composite Bomb Cores for Safety in Assembly.

²⁰⁸Los Alamos Technical Program for 1947, p. 6. "Christy type" pits were unlevitated nearly-solid hemispherical plutonium bomb cores with a small cavity in their centers to hold the initiator.

²⁰⁹"History of Atomic Energy Commission Proving Ground Group (Task Group 7.1 of Joint Task Force Seven)," Report 1, in "Report to the U.S. Atomic Energy Commission on Operation SANDSTONE Atomic Weapon Proof Tests," Eniwetok Proving Ground 1948, Part I, Volume I, pp. I-2, I-3.

²¹⁰AF ATOMIC ENERGY PROGRAM, Vol. II, p. 678; letter dated January 13, 1947 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory to Carroll Wilson, General Manager, Atomic Energy Commission, Washington, D. C., p. 2.

spring of 1948.²¹¹ (MK III bombs were withdrawn from the national stockpile in 1950.)

Apparently the AEC was late in getting the word on composition: an undated "Long-Term Commission Agenda" evidently prepared in early 1947 recommended that in order to maximize bomb production, two types of weapons be produced, one making maximum use of available plutonium and the other making best use of "the remaining U-235."

The AEC also noted that "there is an urgent requirement for a test as soon as possible of the types of bombs which have become available since the manufacture of those which have already been exploded," and that such a test series should include two or three explosions and should be conducted before the end of 1948.²¹²

²¹¹A 1948 report, OPERATION SANDSTONE, NUCLEAR EXPLOSIONS - 1948, Scientific Director's Report - Volume I, General Report, makes a number of statements about composite cores:

Composition is feasible. A direct inference from the first shot is that the [deleted] model would be suited for combat use. This was the model of most of the bombs in the national stockpile at the time of the tests. (p. 37)

The core is made up of nearly pure Pu-239, highly enriched U-235 or a combination of these materials, whereas the tamper in current implosion models is made of normal uranium. (p. 52)

The efficiency of Pu-239 and U-235 components of the X-RAY bomb have been determined separately. (p. 56).

The X-Ray shot of the SANDSTONE test series yielded 37 kilotons; it was a MK III weapon with a composite core. The MK 4 weapon also used composite cores:

The Mark 4 was the device anticipated for the stockpile...most important from the practical point of view, this new implosion weapon would utilize the ample supply of uranium-235 being produced at Oak Ridge. (J. Carson Mark interview, LOS ALAMOS SCIENCE, Volume 4 No. 7, Winter/Spring 1983, pp. 34, 35.)

²¹²"Long-Term Commission Agenda," undated, but probably ca.

In January 1947, while levitated composite pits were being stockpiled, Los Alamos expected to be able to proof-test one before the end of the year:

It now appears that a composite type of weapon (a type now being stockpiled) will have received sufficient experimental study at Los Alamos to warrant a full scale test in about six to eight months, and other tests having application to other types of atomic explosions are under consideration.²¹³

At its meeting in February, the GAC noted that "in regard to weapons, no real exploration of new weapons has taken place in the sense of an actual test of modified weapons of the type previously used." When GAC chairman J. Robert Oppenheimer asked the committee members for comments regarding the testing of atomic weapons, Navy Admiral W. S. "Deke" Parsons replied by emphasizing the need for testing presently-stockpiled types since they had not been subject to any orthodox form of test. In addition, the behavior of both the gun-type and implosion-type weapons already used in combat had not been accurately predicted in advance.

The GAC also recommended that, following theoretical and experimental preparation, tests be conducted to establish whether or not the composite levitated fission bomb design was a satisfactory weapon; the GAC had no strong reason to doubt that this would be the case. If composite weapons were feasible, then replacement plutonium production reactors would have to be built at Hanford to bring the plutonium-to-uranium production ratio up to that required by the new bomb cores.²¹⁴ More uranium-235 than plutonium was being produced at this time, due to the shutdown of one Hanford reactor and the power reduction in the other two.

The GAC noted that "we therefore wish to couple a recommendation for a test program of new atomic weapons with a prior and stronger recommendation for a fundamental study of the theory of atomic explosives" and "that it be accepted

early 1947.

²¹³Letter dated January 13, 1947, from Norris E. Bradbury to Carroll Wilson. David Rosenberg, in his August 1983 PhD thesis, "Toward Armageddon: The Foundations of United States Nuclear Strategy," claims on p. 90 that composite cores did not begin to enter the stockpile until June 1947.

²¹⁴Draft Minutes of the Second Meeting of the General Advisory Committee, February 3-4, 1947, pp. 2, 3, 10.

as policy to test, after appropriate theoretical study, essentially new models of atomic weapons."²¹⁵

During a meeting of the Weapons Subcommittee of the General Advisory Committee at the beginning of April, the subcommittee discussed the relationships between the efficient use of active material, the amount of active material, the size of the resultant explosion, and the availability of active material. Because of safety concerns, more than lack of availability, the Subcommittee wanted to minimize the amount of fissile material for a given yield.

The Subcommittee suggested that LASL start testing with a pure uranium-235 core in a levitated configuration, then study how increases in core size affected yield. The Subcommittee noted that there was "plenty" of uranium-235 available, and recommended that efforts be made to maximize the use of uranium-235 in composite weapons; a test of uranium implosion was just as important as a test of the composite levitated core.

The Subcommittee also recommended that weapons yielding more than 20 kilotons not be designed, if they required large increases in the amounts of active material in their cores. At the same time, efficiency was not to be sacrificed in the interest of obtaining as many weapons as possible from a given amount of active material, if the yields were low. Cores interchangeable between pits were also recommended.²¹⁶

Just a few days after the GAC Weapons Subcommittee meeting, the AEC informed President Harry S Truman that "the atomic core of the most important type of bomb currently being produced has never been tested by explosion."²¹⁷

Later in April, a preliminary test series of three explosions was recommended. The General Manager of the AEC summarized the objectives of the series and described for the first time the three types of bomb models which were to be tested:

- a. A MK III bomb to test the principle of levitation and the principle of the

²¹⁵Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953.

²¹⁶"Proposed Agenda for GAC Weapons Sub-Committee," March 22, 1947; "Notes on Weapons Sub-Committee Agenda," April 4, 1947.

²¹⁷"Report to the President of the United States from the Atomic Energy Commission, January 1 - April 1, 1947," U.S. Atomic Energy Commission, Washington, D.C., April 3, 1947.

composite core. The bulk of the production program for the next few years apparently depends on proving these two principles.

- b. A levitated U-235 bomb which, if successful, will open an opportunity to use available U-235 about three times as effectively as is being done at present.
- c. A solid composite bomb (in the event that the above tests do not produce satisfactory results).²¹⁸

At this time, the three most urgent problems confronting the AEC were (1) to increase the number of weapons which could be stockpiled from the present supply of fissionable material; (2) to proof-test the new weapons already being stockpiled; and (3) to determine the validity of the implosion theory which was the basis of the design of the new weapons. The contemplated tests were mainly to determine the performance of the test devices themselves rather than to measure material damage from atomic explosions or to investigate blast, thermal, and radiation effects on plants and animals, as had been done in 1946 during Operation CROSSROADS.²¹⁹

On April 25, AEC Chairman Dr. David E. Lilienthal wrote to Lt. General Lewis H. Brereton, chairman of the Military Liaison Committee (MLC), and stated:

The urgent requirement, as it appears to us, is for a test program to examine the characteristics of levitated type bombs, both composite core and U-235 core. Both of these bombs could probably be ready by early 1948. In the meantime, serious

²¹⁸Memorandum for the Commissioners dated April 21, 1947, from Carroll L. Wilson, USAEC, Washington, D.C., Subject: Letter to the Military Liaison Committee Regarding a Bomb Test Program. NUCLEAR TEST SUMMARY, TRINITY-HARDTACK, DASA 1220, Defense Atomic Support Agency, Albuquerque, New Mexico, August 1963 states on p. 37 that "the success of (the SANDSTONE X-Ray) and the following two shots caused the cancellation of a fourth shot" which was probably a test of a solid composite pit.

²¹⁹"History of Atomic Energy Commission Proving Ground Group (Task Group 7.1 of Joint Task Force Seven)," Report 1 in REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint Task Force Seven, p. 1-4.

consideration should be given to the testing of a solid composite bomb under the simplest conditions.²²⁰

Earlier that day, the AEC commissioners had met and recommended that "further tests of atomic bombs are a technical necessity." and that only by such testing could the AEC and the armed forces obtain information essential to future planning for atomic warfare and defense against atomic weapons. These tests would be as simple as feasible and conducted as quickly as possible.²²¹

Also in April, Lilienthal wrote to Navy Fleet Admiral William D. Leahy, Chief of Staff to President Truman, concerning the proposed weapons tests:

The Atomic Energy Commission is convinced of the necessity for some further tests of atomic bombs...

The bomb types produced during the war, which are the only types exploded to date, do not make efficient use of available material. Development during the war indicated the great value of one new type which was approved for production by the Manhattan Project and is now being stock-piled. This bomb has never been tested.

Further development during and after the war has indicated an additional new principle which should give so much more efficient use of (fissionable) materials as to make its adoption mandatory if it proves successful in test.

We would like to hold the tests early in 1948 ...²²²

At the end of April, the AEC assumed that solid plutonium cores and uranium assemblies for both implosion and gun-type weapons would be produced until the beginning of January 1949. After that date, pending the results of full-scale

²²⁰Letter dated April 25, 1947 from David E. Lilienthal to Lt. General Lewis H. Brereton.

²²¹Atomic Energy Commission, Minutes of Meeting No. 40, April 25, 1947.

²²²"Draft of Memorandum from Mr. Lilienthal to Admiral Leahy," undated, but probably late April 1947.

tests, only composite and pure oralloy cores would be manufactured, and bombs in stockpile at the beginning of January 1949 would not be converted into other types before 1951.²²³ (These plans changed after SANDSTONE.)

By June, both composite cores and levitated composite cores comprised a significant fraction of the 13 implosion assemblies then in the U.S. nuclear weapons stockpile. At the same time, three new implosion weapons were under study for application of these cores: a low-yield, low-efficiency bomb that combined the use of an unlevitated oralloy core with a known implosion technique, and which could be designed and fabricated in two months; a bomb using a levitated oralloy core that could be readied in six months; and a third device that used a levitated core that included both highly-enriched and slightly lower-enriched uranium. The latter system could be prepared for test in eight months.²²⁴

In spite of the new cores, the remainder of the nuclear and ordnance components of these bombs would remain little changed from earlier designs:

For about the next 18 months, the actual weapon characteristics of any atomic bomb tested will be those of the bomb used at Nagasaki and at Bikini. Until possibly 1 January 1949, any atomic bomb will have only nuclear material changes compared to the Nagasaki and Bikini weapon and will have otherwise identical high explosive, electronics, and mechanical components.²²⁵

A month later, during the fifth meeting of the GAC at the end of July, a question arose as to the necessity for more than one weapons test, inasmuch as the expenditure of fissionable material was of great concern. Dr. Oppenheimer replied that there were two real needs: (1) to make good use of existing material ratios in the stockpiling of weapons; and (2) to assure a proven stockpile.

²²³Memorandum for the Commissioners dated 29 April 1947, from Carroll L. Wilson, Subject: Production of HE Charges, pp. 4, 5.

²²⁴Letter dated 30 June 1947 from N. E. Bradbury to Roger S. Warner, Jr.

²²⁵June 30, 1947 letter from Bradbury to Warner. Following this statement, Bradbury proposed five new levitated and unlevitated composite and pure-oralloy weapon designs.

Admiral Deke Parsons suggested tests of presently stockpiled bombs and some new type design which would be more effective in its use of U-235. The consensus was that the object of the tests should be to prove hitherto-untested stockpile weapons, and to test a design which might develop into a weapon making appropriate use of the present types and amounts of fissionable materials available. A bomb making "good use" of U-235 was a priority.²²⁶

Following this meeting, the GAC concurred with AEC plans that the forthcoming tests be mainly materials proof-tests, without elaborate instrumentation, and devoted primarily to establishing the performance of stockpile items. The GAC requested that the shots be sequenced in order of decreasing novelty, and that if results of the first two tests were satisfactory, then those designs should constitute the basis of the U.S. atomic weapons stockpile without further testing.

If, on the other hand, performance of the first two designs were unsatisfactory, then provisions could be made for further testing. The GAC also noted that it was likely that LASL would design far-more effective, possibly even "radical," weapons in the future, and that more highly-instrumented testing would be required possibly as early as 1950 to determine the efficiency and behavior of these new designs.²²⁷

Early in October, the order of weapons to be tested had still not been resolved. There had been a change in the design of initiators for implosion bombs, and it was likely that the new initiator design would be used in one or more of the tests. There was also a question as to whether a reduced-strength initiator would be used to study its effect on predetonation and to simulate use of an "aged" initiator.²²⁸

The test program was firmed up in November; among its stated goals were the following:

- (a) to improve the short range military position of the nation through testing models which may rapidly become stockpile items, and

²²⁶Draft Minutes, Fifth Meeting of the General Advisory Committee, July 28-29, 1947, Washington, D.C., pp. 7, 8, 11, 12.

²²⁷Letter dated 29 July 1947 to David E. Lilienthal, Chairman, USAEC, from J. Robert Oppenheimer, Chairman, GAC.

²²⁸Draft Minutes of the Sixth Meeting of the General Advisory Committee, October 3-5, 1947, pp. 3, 4.

(b) to improve the long range military position by obtaining such information from the behavior of particular models so that better and more efficient weapons may be designed.²²⁹

The data from the tests was expected to enable LASL to determine more efficient weapon designs for the most effective use of available fissionable materials and to advance the general theory of implosion-type weapons.²³⁰

High-priority information to be gleaned from the tests included the behavior of mixtures of oralloy and plutonium (composition); the behavior of uranium in implosion systems; levitated systems; and systems using weaker initiators than heretofore employed.

Four new types of weapons were available for test by this time; since the behavior of one of them could be reliably extrapolated from a test of another, only three shots would be required. Calculations showed that a composite weapon with a yield two-thirds that of the 20-kiloton TRINITY FAT MAN shot would make about the same effective use of U-235 and plutonium that would occur if the materials were used separately in gun and implosion type weapons, respectively.²³¹

In mid-January 1948, the AEC reported its plans for the forthcoming SANDSTONE nuclear weapons test program to President Truman. At the time, the U.S. nuclear weapons stockpile comprised only about 50 implosion cores, most of which contained untested features and materials; the President and other government officials were justifiably concerned about the expenditure of fissionable materials during the tests:

²²⁹"Test Program Firing Sequence," Rough Draft, Revision One, John H. Manley, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, 5 November 1947, p. 3; REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint Task Force Seven, p. 1-2; Project 19-2, "The AEC Sets Up a Proving Ground," undated outline, pp. 2, 3.

²³⁰REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint Task Force Seven, p. 1-3.

²³¹"Test Program Firing Sequence," 5 November 1947, pp. 4, 5, 6.

In a memorandum to you dated November 25, 1947 on the subject of the forthcoming test of atomic weapons, the Atomic Energy Commission indicated that two shots were definitely planned, with provisions made for a third in the event data obtained from the first two shots prove inadequate.

Intensive theoretical studies by the Los Alamos Scientific Laboratory have subsequently convinced the Commission that a definite program of three shots is essential to the gathering of adequate data bearing on a primary objective of the test, which is to find substantially more economical means of employing fissionable materials in bombs.

The first weapon to be detonated will be our most advanced design containing plutonium. This shot, if successful, will permit us to almost double the number of weapons which can be made from a given amount of plutonium, while at the same time prove an element of design which has been incorporated in a considerable number of bombs now in the stockpile. Author's note: a reference to levitation.)

The second and third shots are for the purpose of exploring toward the upper and lower limits of the amount of uranium-235 which can produce a satisfactory nuclear explosion in a design radically different from that of the Hiroshima bomb, the only weapon employing uranium-235 tested to date. This new design is expected to be at least three times as efficient as the Hiroshima bomb.

For your information the second bomb to be fired will contain slightly more than [deleted] the amount of uranium-235 used in the Hiroshima bomb, while the third will contain less than one-tenth the Hiroshima amount.

If the second and third shots are successful our design data will be broadened so that a greater variety of weapons can be produced which will meet more efficiently and more economically the

various military requirements for weapons.²³²

Objectives of SANDSTONE

At this time, the official objectives of Operation SANDSTONE were:

- a. Testing the principal (sic) of composition - the use of both plutonium and U-235 in the same bomb core.
- b. Testing the principal (sic) of levitation - the suspension of the bomb core within the tamper - high explosives assembly to produce better core compression.
- c. Investigate the behavior of U-235 in an implosion weapon.
- d. Investigate the behavior of models of different tamper thickness.
- e. Test an implosion weapon with an initiator of minimum acceptable stockpile strength.²³³

More specifically, there were three major objectives for SANDSTONE:

First: To prove experimental designs which, if successful, would permit an increase in our present stockpile of bombs. One of the most important theories the Los Alamos laboratory desired to prove was that U-235 would have greater efficiency if imploded either separately or in combination with plutonium than if used (alone) in the Hiroshima-type weapon. Proof of this theory, especially considering the fact that U-235 is relatively more plentiful than plutonium, would make possible a

²³²Memorandum for the President dated January 14, 1948, from David E. Lilienthal, Chairman, USAEC, Washington, D.C.

²³³"Brief History of Weapons Testing, including Site Selection, and Expected Research Projects," Remarks of Col. Preuss at First Meeting of Joint Proof-Test Committee, July 12, 1949, p. 3.

considerable increase in the stockpile of bombs.

Second: To obtain information on the basis of which bomb designs could be varied to meet more efficiently military requirements. There were various new design elements introduced in the three models to be tested. The new designs involved different geometrical arrangements of fissionable material and surrounding components as well as varying the quantities of fissionable materials. Successful tests of those arrangements would give a broader basis for varying bomb design and would permit Los Alamos to meet more efficiently military requirements for varied-size bombs or for special-purpose bombs.

Third: To verify standards of acceptance of initiators (the neutron source which insures fission). The maintenance of the initiator stockpile is a very real and costly problem for the Atomic Energy Commission since initiators deteriorate rapidly and are difficult to produce. Certain limits on acceptability have been established for stockpile. Yet in past tests, only the initiators meeting the highest limit have been used. Theoretical studies and experiments by Los Alamos indicated that the lower limit was more than adequate. However, actual proof testing in real atomic bombs was considered highly desirable.²³⁴

In general, the tests would prove stockpiled items; investigate methods of better utilizing fissionable materials; and increase LASL's knowledge of bomb design.²³⁵ On January 15, 1948 the AEC approved expenditure of stockpiled atomic weapons required for SANDSTONE.²³⁶

²³⁴Transcript of the Oral Report to the Joint Chiefs of Staff by Lt. Gen. Hull on Operation SANDSTONE, undated, pp. 3, 4.

²³⁵"Outline of Remarks for Mr. Lilienthal's Conference with the Joint Congressional Committee; Subject: Test of Atomic Weapons," U.S. Atomic Energy Commission, Washington, D.C., January 16, 1948, p. 2.

²³⁶OPERATIONAL HISTORY OF ATOMIC ENERGY COMMISSION PROVING

The 50 FAT MAN bomb cores stockpiled at this time reflected the abundance of enriched uranium and included nine pure plutonium Christy-type solid cores; 36 composite uranium-plutonium Christy type cores; and only five levitated composite uranium-plutonium cores.

Fifty Class "A" polonium-beryllium initiators, each containing more than 25 curies of polonium, were also in stockpile, with 13 Class "B" initiators containing 12 to 25 curies of polonium each.

The national nuclear weapons stockpile also contained 44 assembled high explosive implosion assemblies and 28 unassembled assemblies. There were also enough other FAT MAN bomb components stockpiled to make 104 complete assemblies.

Critical items, those whose quantities were below the levels stated in a September 1947 stockpile directive, included aluminum pushers for the FAT MAN bombs, and detonators.

There was also enough fissionable material on hand for 10 LITTLE BOY bombs, but only enough initiators to equip six of them.²³⁷

By the end of January, whether or not the U.S. could meet military requirements for both weapons and fissionable material depended upon the outcome of SANDSTONE:

The Commission believes that the long-term objectives as currently stated by the Joint Chiefs of Staff can be met from production at existing and planned facilities, and from expected (uranium) ore receipts, providing we are successful during the forthcoming tests at Eniwetok.

GROUND GROUP (Task Group 7.1), submitted 20 May 1948, USS ALBERMARLE (AV-5), Eniwetok Atoll, Marshall Islands, p. 8-2.

²³⁷"Outline for Stockpile Reports to be Received by MLC," undated, but written between September 4 and December 31, 1947. This document was declassified (with no deletions!) by the Defense Nuclear Agency in September 1991. The "pure 49 Christy" (solid plutonium) and "composite Christy" (composite solid plutonium-uranium) cores were not tested during SANDSTONE, even though the latter comprised the majority of the U.S. nuclear weapons core inventory by the end of 1947. Presumably, all the unlevitated cores were converted to levitated composite cores after SANDSTONE. The LITTLE BOY fissionable targets and projectiles had not been officially stockpiled when this document was written.

This latter point is a most important one because the number of bombs which the Joint Chiefs of Staff now state as an objective during the next few years cannot be made from the amount of fissionable material which will be available unless we can produce a design of bomb much more economical in the use of this material than any type which has been tested to date.²³⁸

Between February 18 and February 29, seven experimental weapon assemblies (without nuclear material), six nuclear charges (cores), and spare parts were delivered to San Pedro, California, where they were loaded aboard the weapon assembly ship, the USS CURTISS (AV-4), a converted U.S. Navy seaplane tender.²³⁹ Authority was granted for only three devices to be expended during the tests; the remaining cores were to be returned to the AEC after the tests ended.

The first full-scale tests of composite cores and levitated pits were conducted during Operation SANDSTONE, the first weapons proof tests since the original TRINITY shot in July 1945 (the CROSSROADS tests in the summer of 1946 were weapons effects tests, and did not involve new bomb designs). These three experiments occurred between April 14 and May 14, 1948 at Eniwetok atoll in the AEC's Pacific Proving Grounds.

Three devices were exploded at SANDSTONE; the three shots, labeled X-Ray, Yoke, and Zebra, after international phonetic representations of the letters X, Y, and Z, yielded 37, 49, and 18 kilotons, respectively.²⁴⁰ The yield of the X-Ray test

²³⁸Letter dated January 29, 1948 to Honorable Bourke B. Hickenlooper, Chairman, JCAE, from David E. Lilienthal, Chairman, USAEC.

²³⁹Project 19-2, "The AEC Sets Up a Proving Ground," undated outline, p. 2; "Operation Report of Atomic Energy Commission Proving Ground Group (Task Group 7.1 of Joint Task Force 7)," Report 3 in REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint Task Force Seven, p. 3-13. The latter document claims that eight HE units were moved to the Long Beach Naval Shipyard and loaded aboard the CURTISS.

²⁴⁰Hewlett and Duncan, ATOMIC SHIELD, Appendix 4: Announced Nuclear Tests, Operation SANDSTONE. Components for six bombs were shipped to Eniwetok; just a year earlier, this number would have comprised half the entire U.S. nuclear stockpile. (Memorandum dated January 14, 1948 to Captain James S. Russell, USN, from U.S. Atomic Energy, Subject: Experiments to be conducted at Eniwetok Atoll during 1948.)

was first believed to be significantly above its predicted value; however, recomputation of the yield brought the predicted and actual values closer together.²⁴¹

Each of the tests was conducted on an island to the south of the preceding shot, so that each successive test site would not be contaminated by fallout carried by the southerly winds from the preceding shots.²⁴²

The SANDSTONE X-Ray bomb used a levitated composite core with both plutonium and uranium-235; the Yoke and Zebra shots used levitated uranium-235 cores.²⁴³ The X-Ray and Zebra cores were so efficient and made such good use of then-plentiful uranium-235 that they were put into immediate production in

²⁴¹Project 19-2, "The AEC Sets Up a Proving Ground," undated outline, p. 2; "Report to the U.S. Atomic Energy Commission on Operation SANDSTONE Atomic Weapon Proof Tests, Eniwetok Proving Ground 1948," Part I, Volume I, p. 2.

²⁴²Map of Eniwetok atoll SANDSTONE test sites on p. 3-5 of REFLECTIONS OF A NUCLEAR WEAPONER, Frank H. Shelton, Shelton Enterprises, Inc., Colorado Springs, Colorado, 1988. While Dr. Shelton is a reasonably authoritative source -- he participated in many Pacific and Nevada test series -- information from his book was used only when it could be corroborated by other sources. The flyer promoting the Shelton book states that it "conforms to current classification and technical policy of the Department of Energy and the Department of Defense;" for this reason, the book is often incomplete and frequently suspect.

²⁴³FINAL EVALUATION REPORT, MK IV MOD 0 FM BOMB, The MK IV Evaluation Committee, Report No. SL-82, Sandia Laboratory, Albuquerque, New Mexico, September 13, 1949, p. 92; BRITISH SCIENTISTS AND THE MANHATTAN PROJECT: THE LOS ALAMOS YEARS, Ferenc Morton Szasz, St. Martin's Press, New York, New York, 1992, p. 150; letter dated February 7, 1951 to Senator Brien McMahon, Chairman, JCAE, from David E. Lilienthal, Chairman, USAEC (this letter states that Soviet agent (Klaus) "Fuchs made efficiency estimates on various implosion system designs, including levitated and composite structures -- one of them corresponding rather closely to X-Ray shot at Eniwetok"); "Notes on Weapons Sub-Committee Agenda," April 4, 1947; "Restricted Data Obtainable from a Post-Shot Examination of Eniwetok Atoll," undated, but probably written just after the SANDSTONE tests; this document states (p. 2): "It can be discovered that bomb one contained plutonium and enriched uranium while bombs two and three contained only enriched uranium." It is interesting to note that the principle of levitation was considered so important that it was incorporated in all three test devices.

conjunction with the MK 4 explosive assembly, the design of which SANDSTONE had verified.²⁴⁴

The Yoke core was not as efficient as the other two (one source claims that Yoke "was regarded as wasteful of material").²⁴⁵ After SANDSTONE, in September 1948, LASL was contemplating "a preparation of the design of the YOKE type weapon adequate for MK 4 use but not necessarily representing the most efficient design which must await the completion of (implosion hydrodynamics studies)."²⁴⁶

There was almost an overabundance of U-235 at this time. Between 1947 and 1949, uranium-235 production exceeded AEC expectations and even disrupted the balance between plutonium and uranium production; plutonium production was also improved substantially during this period.²⁴⁷

In April 1947, the AEC told President Truman that:

Production of critical parts of the bomb, including ingredients of the atomic core, has been and is badly out of balance. For instance, plutonium production is far behind uranium-35 production.²⁴⁸

This situation had been rectified by the end of 1948:

²⁴⁴Hewlett and Duncan, pp. 164, 165, 178; "American Atomic Strategy and the Hydrogen Bomb Decision," David Alan Rosenberg, THE JOURNAL OF AMERICAN HISTORY, Vol. 66 No. 1, June 1979, p. 71; Shelton, p. 3-19. The SANDSTONE bombs used MK III HE assemblies; the MK 4 HE assembly was not tested in a full-scale nuclear test until 1951.

²⁴⁵Memorandum for the Files dated November 8, 1949, from Walter A. Hamilton, subject: Inquiry into the Aspects of a Superweapon Program, p. 4.

²⁴⁶Letter dated 7 September 1948 from N. E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Office of Santa Fe Directed Operations, U.S. Atomic Energy Commission, p. 7.

²⁴⁷THE OPPENHEIMER CASE: SECURITY ON TRIAL, Philip M. Stern, with Harold P. Green, Harper & Row, New York, 1969, p. 325.

²⁴⁸"Report to the President of the United States from the Atomic Energy Commission, January 1 - April 1, 1947," U.S. Atomic Energy Commission, Washington, D.C., April 3, 1947.

In general, it was noted that various schedules under the production program were either met or exceeded during calendar year 1948, despite the occurrence of a number of serious unforeseen (AEC production reactor) incidents at Hanford and Oak Ridge. ²⁴⁹

By May 1951, plutonium production had expanded by a factor of 12 over its production rate in 1947; uranium was being produced at a rate eight times that of its production in 1947.²⁵⁰

The uranium-to-plutonium ratio by weight in the SANDSTONE X-Ray bomb pit was on the order of 2:1 or greater. Efficiency of utilization of the plutonium in this core was around 35%; uranium utilization was in excess of 25%.²⁵¹ Yield of the X-

²⁴⁹"Oral Report on Production," agenda item for 234th AEC Meeting, January 19, 1949.

²⁵⁰USAEC, Draft Foreword to the Progress Report to the Joint Committee on Atomic Energy, December 1950 through May 1951, May 29, 1951, p. 4. The Hanford site eventually comprised nine production reactors. ("Hanford ... At A Glance," DOE THIS MONTH, Vol. 17 No. 11, November 1994, p. 10.)

²⁵¹"Known Soviet Nuclear Explosions, 1949 - 1985, Preliminary List," NWD 86-3, Nuclear Weapons Databook Working Paper No. 3, Jeffrey I. Sands, Robert S. Norris and Thomas B. Cochran, Natural Resources Defense Council, Washington, D.C., February 1986, pp. 6, 7.

If 3.2 kg of Pu-239 and 6.5 kg of U-235 were fissioned at 21% efficiency (the efficiency of the Nagasaki bomb), explosive yield would be about 36.6 kilotons. Note that the plutonium utilization in the X-Ray pit was nearly twice that claimed by Szasz (on p. 117 of THE DAY THE SUN ROSE TWICE) for the TRINITY bomb. Szasz states that the TRINITY device yielded 20 KT at an 18% efficiency (which means that the device probably had 5.55 kg of plutonium in its core; other sources claim that the TRINITY device had 6.1 or 6.2 kg of Pu in it).

If the figures cited in the text for efficiency of fissionable components in the SANDSTONE X-Ray bomb pit and the ratio of one to the other are accurate, then there were about 2.38 kilograms of plutonium (Pu) and 4.77 kilograms of uranium-235 (Oy) in the bomb, based on the following calculation:

$$0.35 \times M_{\text{Pu}} \times 20 \text{ kt/kg} + 0.25 \times M_{\text{Oy}} \times 17 \text{ kt/kg} = 37 \text{ kt}$$

Ray shot was well above the lower limit predicted by LASL before the test.²⁵²

All of the test devices used modified MK III firing units, "stripped down" to only the basic firing devices required for static firing, and explosive assemblies. The MK III firing unit had never before been used in a full-scale field operation similar to SANDSTONE; the bomb casings had to be shoehorned into the shot cabs atop 200-foot high towers.²⁵³

The three bomb pits were essentially identical in weight and volume; only their internal composition and the thicknesses of the tampers around them varied.²⁵⁴ At least one of the test devices also used an "aged" polonium-beryllium initiator of less than full strength.

where M_{Oy} (mass of uranium-235) = 2 x M_{Pu} (mass of plutonium). These figures are supported by statements in late 1950 (see Volume III) that the three to five kilograms of tritium required for each hydrogen bomb, as then envisioned, were equivalent to the plutonium required for 100 to 150 fission weapons. Since one kilogram of tritium required as many neutrons (in a reactor) to produce it as did 70 kilograms of Pu, then three to five kg of tritium was equivalent to 210 to 350 kg of Pu. If 100 to 150 bombs could be produced from this much Pu, then each bomb probably contained between 2.1 and 2.3 kg of Pu, with probably twice as much uranium. The Soviets tested a composite bomb on October 18, 1951 during their third atomic test. (NUCLEAR WEAPONS DATABOOK, Volume IV, SOVIET NUCLEAR WEAPONS, Thomas B. Cochran, et. al., Ballinger Books, 1989, p. 336.)

²⁵²Project 19-2, "The AEC Sets Up a Proving Ground," undated outline, p. 2.

²⁵³REPORT TO THE U.S. ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE, "Armed Forces Special Weapons Project Activities, Operations and Training," Report 4 in ATOMIC WEAPON PROOF TESTS (NUCLEAR EXPLOSIONS), Test Director, Joint Task Force Seven, Eniwetok Proving Ground, 1948, Part 1, Volume 1, pp. 4-8, 4-9; Part 1, Volume II, p. 7-17; memorandum dated 23 November 1947 from Darol Froman to Capt. J. S. Russell, Subject: AFSWP Teams, p. 2; inter-office memorandum dated February 20, 1948 to Darol K. Froman from Capt. B. H. Leeper, Subject: Report of Activities of AFSWP Personnel assigned to J-12 for Operation SANDSTONE; memorandum dated 15 April 1948 from Capt. E. T. O'Donnell to Scientific Director, TU 7.1.1, Subject: Report of Activities of AFSWP Personnel; Shelton, p. 3-19.

²⁵⁴OPERATION SANDSTONE, Nuclear Explosions - 1948, Scientific Director's Report - Volume I, General Report, Eniwetok Proving Ground, pp. 34, 46, 51.

Of the three tests, the Zebra shot was characterized at the time by AEC Chairman David Lilienthal as the "hardest and most important" test.²⁵⁵ The tests were conducted to prove the efficiency of the weapons, and the shots proved that the status of the U.S. in the atomic weapons field had been significantly improved.²⁵⁶

Carson Mark, then head of LASL's Theoretical Division, later stated that SANDSTONE "marked the end of the day of an atomic device as a piece of complicated laboratory apparatus rather than a weapon ... the number of units which could be made from the existing stocks of plutonium and uranium could (now) be increased appreciably merely by refabricating the fissile parts of weapons by hand."²⁵⁷

After SANDSTONE ended, the MLC recommended that because the Zebra core used much less U-235 than the wartime LITTLE BOY bomb while at the same time producing a higher yield (18 KT versus 15), it appeared desirable to refabricate the U-235 now in LITTLE BOY interior targets and projectiles into SANDSTONE Zebra-type implosion weapons, in order to maximize the number of bombs available in the near future.

The MLC also suggested that the non-nuclear components for a number of LITTLE BOYS now nearing completion be held in stockpile, and that the AEC be ready to resume LITTLE BOY production, should it become necessary to do so at the expense of the greater number of SANDSTONE Zebra pits that could be made from the same amount of U-235.²⁵⁸

²⁵⁵THE JOURNALS OF DAVID E. LILIENTHAL, Volume II, THE ATOMIC ENERGY YEARS, 1945-1950, David E. Lilienthal, Harper & Row, New York, 1964 (hereafter referred to as Lilienthal II), p. 339.

²⁵⁶Notes from an undated multi-volume Air Force Weapons Laboratory report entitled "SANDSTONE." In the short term, SANDSTONE indirectly risked the national security of the United States: all of the AEC's bomb assembly crews were in the Pacific for the tests, and none were available in Europe, had the U.S. invoked military action against the Soviets during the Berlin Blockade that began in the spring of 1948. (Rhodes, DARK SUN, p. 320)

²⁵⁷Rhodes, DARK SUN, p. 321.

²⁵⁸"Requirements for the Stockpile of Atomic Weapons," October 1, 1948, enclosure to Joint Chiefs of Staff, Decision on J.C.S. 1940/1, A Report by the Joint Strategic Survey Committee on Annual Report of Military Liaison Committee, Atomic Energy Commission, 9 December 1948.

Some specific results of SANDSTONE included the findings that "use of composite cores is feasible;" "the principle of levitation is sound;" implosion U-235 weapons are more efficient than gun-type weapons;" and "current implosion theory is sound."²⁵⁹

A general description of the results of SANDSTONE, issued soon after the tests concluded, stated in part:

The primary objective of Operation SANDSTONE, in which three atomic weapons were exploded at Eniwetok in the spring of 1948, was to test out new designs of atomic weapons and to make measurements which would lead to improvements and increased efficiency in future models. ...

Work in the Los Alamos Scientific Laboratory during the last few years led to new designs of atomic weapons which were expected to be appreciably more efficient than the wartime models and which would utilize available fissionable materials much more nearly in proportion to their supply and rates of production.

Probably the two most important results of Operation SANDSTONE are first, the newly designed nuclear weapons behaved essentially as predicted by the Laboratory, and second, a significant amount of information about the processes involved in nuclear explosions was obtained. As to which of these results is of the greater importance depends to a considerable extent upon the international situation.

If the nation should become involved in an atomic war in the near future, the first result is of great importance since it is resulting in a greatly increased stockpile of atomic weapons containing mostly weapons of improved efficiency and performance and which have been adequately tested in Operation SANDSTONE. If the nation does not become involved in an atomic conflict within the next three or four years, the second result becomes highly important

²⁵⁹ORIENTATION GUIDE, J-3 Division, Joint Task Force Seven, 1 June 1957, p. 13.

since, with the aid of the fundamental information obtained in the 1948 tests, the Los Alamos Scientific Laboratory will undoubtedly be able to further improve the design of atomic weapons so as to make much more efficient use of available nuclear-energy materials.

....

The (test) results indicate that the (atomic weapon) models tested at Eniwetok are indeed representative of their types and that we can expect stockpile weapons based on these models to perform as did the prototypes which were tested.²⁶⁰

Scientific results of SANDSTONE, and work that remained to be done, were also summarized by LASL:

The measured transit times and the values of alpha for initial neutron multiplication rates were in good agreement with those predicted. Thus the weapon performance was representative for each type and an integral check on the theory of implosion and neutron multiplication was provided.

However, the prediction of energy release differed considerably from that actually obtained in shot X-Ray (the Laboratory has since found an error in the computation for X-Ray, and a recomputation has brought the predicted (yield) into much closer agreement with the measured total energy release), and, further, the predictions for the other two shots were made only within rather wide limits.

It can be said, then, that the Laboratory's empirical work and theory on the initiation and early phases of the nuclear explosion are good and point to good weapon design. But the theory which carries the explosion on through to its total energy release, an important

²⁶⁰"Operation SANDSTONE," undated chapter or speech draft, pp. 1, 2, 6; teletype dated April 24, 1948 to USAEC, Washington, D.C. from F. Reines, Los Alamos.

consideration in studying the use of the weapon, is not too well developed.

Therefore, an important result of the Eniwetok tests was to provide considerable experimental data from which the Laboratory may advance the theoretical treatment and prediction of total energy release.²⁶¹

The most immediate military effect of SANDSTONE was to make possible within the near future a 63% increase in the total number of bombs in the stockpile and a 75% increase in the total yield of these bombs. The increase in the number of bombs resulted from the improved usage of or alloy, and the still greater increase in total yield followed from this factor and improved weapon design and technology, including application of levitation and composite cores.²⁶² (By 1950, even in the absence of proof tests after SANDSTONE, further improvements in bomb design made possible the production of four times the total stockpile yield with the same amount of fissionable material as would have been possible if early design and manufacturing methods and principles had continued without change.)²⁶³

Several other military aspects of SANDSTONE were also considered:

Various criteria may be applied to study the relative merit of the several weapons which were tested during the proofing operation and those weapons which might be designed from the data obtained. Some considerations which might apply in planning weapon programs are:

- (1) Number of weapons of acceptable types which can be made from active material on hand and being produced.
- (2) Rate of production from

²⁶¹REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint task Force Seven, p. 2.

²⁶²AF ATOMIC ENERGY PROGRAM, Vol. II, p. 694.

²⁶³Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 20.

current and projected
(production) facilities.

- (3) Damage area obtained per kilogram of active material.
- (4) Cost to natural resources in ore.
- (5) Cost in dollars, man-hours, and facilities.
- (6) Value of by-products which appear in process of manufacture of active material.
- (7) Energy release per unit weight of assembled bomb; i.e., the military load which must be carried to the target.
- (8) Versatility of employment of weapon.
- (9) Various additional military characteristics, such as: (ease) of delivery to target; accuracy of delivery to target; drag occasioned by (bomb) shape in airplane fuselage or rocket body; and many others.²⁶⁴

Three tests were conducted and all three were successful. The results obtained briefly are as follows:

a. The Atomic Energy Commission has proven the implosion of U-235, both in combination with plutonium and separately. The energy releases per kilogram indicate that efficient bombs can be made using much less U-235 than in the gun-type weapon. Roughly speaking, our stockpile potential has been increased about 2.2 times.

²⁶⁴REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint task Force Seven, pp. 7, 8.

b. Five different geometrical arrangements of fissionable material have now been tested and an additional one proven by inference. Before SANDSTONE, there were only two. These designs, with varying amounts of fissionable material, have given energy yields from 15,000 to 45,000 tons of equivalent high explosive.

The Los Alamos laboratory, after all data have been analyzed, will have added considerably to its fund of design information and will be better prepared to design bombs of varying performance and purpose as desired by the (Armed) Services. (sic)

It now appears that the Joint Chiefs of Staff should re-study their war plans, and, based on a study of means of delivery of an atomic attack, tell the Atomic Energy Commission what military characteristics are desired.

c. One initiator of strength equal to the lower acceptance limit has been demonstrated as satisfactory. This may encourage further reduced limits of acceptability thereby greatly easing the initiator stockpile problem.

I will endeavor to demonstrate on this blackboard the different arrangements utilized in the three bombs tested and give you results obtained in TNT explosive equivalent.²⁶⁵

SANDSTONE also proved conclusively that implosion of U-235 was far more efficient than assembling it in a gun-type weapon and demonstrated that current implosion theory was sound. Data were also gathered which led to more effective use of fissionable material in composite cores. New weapon designs performed as expected, and efficient bombs of the same size, weight, and shape with widely-different yields could be

²⁶⁵Transcript of the Oral Report to the Joint Chiefs of Staff by Lt. Gen. Hull on Operation SANDSTONE, undated, pp. 12, 13. In the typed draft of the transcript, the handwritten notation "2.2" replaces the typed word "three" at the end of the first paragraph quoted.

produced.²⁶⁶ The results of SANDSTONE were characterized as "radical" and representing "substantial" improvement in the military position of the U.S.²⁶⁷

The military benefits that accrued from SANDSTONE were summed up soon after the tests ended in a report that specifically addressed President Truman's concern with the expenditure of the nation's severely-limited nuclear muscle in the weapon-development test program:

The short-range military position of the country has been improved markedly as a result of Operation Sandstone. The position with respect to the national stockpile of atomic weapons is being improved rather quickly by simply engineering and beginning fabrication of actual weapons having the same parameters as the experimental models tested.

In addition, rather superficial examination of the results obtained in the experiments at Eniwetok gives rise to minor changes for the stockpile models which will improve them over the experimental models tested. It is clear that the expenditure of the fissionable material in the Sandstone tests has been repaid many fold as measured by either the number of atomic weapons or the total potential damage area in the national stockpile.

With the present state of knowledge and progress in weapon development, no hesitation, based on stockpile considerations, should be felt in expending several times as much fissionable material in future scientific tests of nuclear explosions if the proposed tests show any reasonable probability of significant advances in the understanding of the processes involved in the explosions.²⁶⁸

²⁶⁶"Summary of Test Operations at Pacific Proving Grounds," Headquarters, Joint Task Force Seven, undated (but probably 1954).

²⁶⁷Lilienthal II, pp. 342, 464.

²⁶⁸SANDSTONE Scientific Director's Report - Volume I, pp. 50-51.

As regarded the value of SANDSTONE versus its cost in both dollars and fissionable materials:

.. he stated that the cost of the operation of the Los Alamos Laboratory since AEC took it over plus the proof test operation were only 1/6 of the cost of the fissionable material which would have been needed to achieve the same weapon potential if the national stockpile had continued to be comprised (sic) of weapons of the pre-SANDSTONE proven variety, namely, of the Hiroshima and Nagasaki types.²⁶⁹

Publicly, the AEC said very little about the tests:

The tests of Operation SANDSTONE were literally and truly field laboratory tests, designed to determine how the bombs now under development by the United States would work and to determine their efficiency. We did just that. We got our answers. We liked the answers. These answers belong to the United States Government (sic) and, naturally, are not for publication. It can be said, however, that the bombs worked. We have proved the weapon development work done by the Los Alamos Scientific Laboratory during the past two years.²⁷⁰

One of the most gratifying results of the entire operation has been the confirmation of the large body of ideas, theories, and methods which have grown out of the theoretical and experimental work done since the war at the Los Alamos Scientific Laboratory.²⁷¹

²⁶⁹Memorandum for Record, June 6, 1949, by Capt. James S. Russell, USN.

²⁷⁰Statement of Lt. General John B. Hull, Commander, Joint Task Force Seven, at press conference, HQ, USARPAC, Fort Shafter, Territory of Hawaii, 18 May 1948.

²⁷¹Statement of Dr. Darol K. Froman, Scientific Director, Joint Task Force Seven, at press conference, Headquarters, USARPAC, Fort Shafter, Territory of Hawaii, 18 May 1948.

The AEC also noted that SANDSTONE results were significant:

Great importance is attached to analyzing the results of the SANDSTONE tests for both the short and the long range benefits to the weapons program which are expected to result. An early study, with such recommendations as the (Los Alamos) Laboratory may desire to make, is needed to provide a basis for decisions as to the optimum composition of the stockpile of nuclear parts to be fabricated during the next year or two, considering such factors as numbers of weapons and damage areas in relation to the amount of fissionable material (expended).

It is desirable that the study cover weapons of a wide range of expected explosive yield, including such estimates as may be made on supercritical assemblies in the LITTLE BOY type.²⁷²

The cores proof-fired during SANDSTONE figured prominently in AEC production plans for 1949:

The current Atomic Energy Commission bomb production program is, in general terms, as follows:

- a. The entire available supply of fissionable materials is being fabricated into bombs, except for allocations for research and development ...
- b. Bomb fabrication is presently confined to two improved models of bombs. These improved models were tested at Eniwetok. One of these bombs is made up of a composite of uranium enriched in uranium-235 and plutonium. The other bomb contains uranium enriched in uranium-235 alone.
- c. The fissionable material used in obsolescent bombs is being fabricated into the new improved models. Such obsolescent bombs are models similar to those detonated

²⁷²Draft Memorandum from the General Manager to the Manager, Santa Fe Directed Operations, p. 3, Appendix "A" to AEC 99, Atomic Energy Commission, Weapons Program of the Los Alamos Laboratory, May 14, 1948.

at Alamogordo, Hiroshima, Nagasaki, and Bikini.

By January 1, 1950 the above bomb production program will reduce to a minimum working level the amount of fissionable material in excess of the requirements for research and development ...²⁷³

Perhaps the most significant of these developments from the standpoint of overall plans has been the preliminary conclusions drawn from the technical findings of the weapon tests held at Eniwetok in April and May.

As a result of these determinations, as well as other recent laboratory developments, we are now able to advance by about six months the scheduled production of new bomb models. This is likely to require a request of the Congress for supplemental funds in fiscal year 1949.

The new weapons have potentially larger area of damage and can be produced in greater number from the scheduled supply of fissionable material.

Important also is the more refined knowledge we now have of the comparative value of plutonium and uranium as bomb materials. This knowledge confirms the desirability of our previously established objectives to maintain, and possibly increase, uranium production at Oak Ridge, as well as to double plutonium production at Hanford.

Since completion of the test program, the engineering of the nuclear elements of one new model of weapon for stockpiling has been completed. The engineering of the nuclear elements of another new model has progressed to the point where completion is expected in August.

²⁷³Draft letter to the President from Chairman, AEC and Secretary of Defense, January 11, 1949, signed by David E. Lilienthal, Chairman, USAEC, and James Forrestal, Secretary of Defense.

Production of the first model is already under way, and production of the second will be started as soon as engineering design is completed.²⁷⁴

This tale of industrial accomplishment which helped extend American military pre-eminence goes back to the Eniwetok weapon tests held in the spring of 1948. Analysis of these important test results touched off a period of feverish activity within the atomic energy project which rivaled the wartime development days, and culminated in successful completion of the task only a few minutes before the deadline.

The top priority job involved the design, fabrication, and operation of plant equipment and facilities for production of critical parts which were to comprise the very heart of the improved weapons tested and accepted at Eniwetok.

In order to meet the Commission's new weapons program schedule, a deadline of January 1, 1949 was set for volume production of bomb parts and responsibility for various phases of the new project were apportioned, with one plant at Oak Ridge, Tennessee being given the actual job of production.²⁷⁵

In the fall of 1948, the AEC reported to the JCAE that great progress had been made in weaponry during the preceding year:

A year ago, the wartime types of atomic weapons were being produced pending the completion and testing of new types then being developed. Now, by virtue of three principal accomplishments of the past year, it is possible to produce more efficient weapons in greater numbers, which are

²⁷⁴U.S. Atomic Energy Commission, Quarterly Progress Report April - June 1948, Report to the Joint Committee on Atomic Energy, August 20, 1948, pp. iii, 1.

²⁷⁵"One Chapter of the Weapons Production Story," Dr. Clarence Clarkson and G. R. Molesworth, Oak Ridge, Tennessee, July 11, 1949.

better suited to field use, and to increase the production of non-nuclear assemblies.

These accomplishments were: (a) the demonstration in the SANDSTONE tests at Eniwetok of the effectiveness of nuclear components based on new principles of design; (b) the completion of the MK IV weapon assembly, and (c) the sharply-increased production of high-explosive components, which was beyond expectations. To take full advantage of these achievements has required an earlier expansion of the weapons production program than had been contemplated.²⁷⁶

By October 1948, the AEC reported to the JCS that by January 1951, the U.S. would have stockpiled 400 atomic bombs.²⁷⁷ (This goal was achieved.)

By mid-1950, theories proved in SANDSTONE were being reflected in stockpile modifications.²⁷⁸ The LITTLE BOY gun projectiles and targets and the solid plutonium "Christy gadgets" were being replaced by new SANDSTONE Zebra and X-Ray cores.²⁷⁹

Uranium Hydride Implosion

To further increase supplies and sources of fissionable weapons-grade materials, research on a uranium hydride implosion bomb was resurrected at Los Alamos in 1948 after

²⁷⁶U.S. Atomic Energy Commission, Quarterly Progress Report July - September 1948, Report to the Joint Committee on Atomic Energy, November 19, 1948, p. 1. The "booster" referenced was a tritium-deuterium gas-boosted implosion weapon test device then under design at LASL.

²⁷⁷Rhodes, DARK SUN, p. 361.

²⁷⁸Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1950 to January 1954, p. 4.

²⁷⁹Memorandum for the Files dated November 8, 1949 from William L. Borden, re: Super Weapon Program, p. 2. In November 1949, some work was still being done on solid composite cores at LASL. (Memorandum for the Files dated November 10, 1949 from Walter Hamilton, subject: Inspection Trip, October 24 - November 4, Borden and Hamilton, pp. 2, 3.)

the war ended; one objective of the Los Alamos technical research program for 1949 was:

2. A program of development designed to lead to an implosion weapon of the hydride type. Such a weapon is known to have a low efficiency but the extent of this effect has not been given serious consideration since the early history of Los Alamos.

Such a weapon would probably involve amounts of uranium of the order of [deleted] ... Its specific and probable virtues lie in the fact that the [deleted] thus potentially leading to considerably smaller weapons than (the MK 5) ... The development program (for the "Booster") if successful, may then ultimately be employed in conjunction with the hydride weapon to return the efficiencies to values now attainable with weapons of presently conventional dimensions. ... This development program will clearly involve ...

- (a) A program of theoretical investigations covering the probable behavior of uranium hydride under various conditions of initiation.
- (b) A chemical-metallurgical program devoted to the preparation of uranium hydride in a physical form adequate to meet the theoretical and practical requirements of the weapon.
- (c) A critical mass determination program.

It is not anticipated that the probable specifications for this weapon will be available before the end of calendar year 1949.²⁸⁰

The Department of Defense was somewhat less than enthusiastic about this proposal:

²⁸⁰Los Alamos Scientific Laboratory Program for Calendar Year 1949, p. 4.

It is considered that work toward a hydride implosion bomb should be given a priority well below work on smaller conventional implosion weapons and below work toward the booster. The reason for this low priority is the expectation that although use of hydride might permit an even smaller weapon, it would do so at the risk of an unacceptably great loss in nuclear efficiency and weapon effectiveness.²⁸¹

The Los Alamos Technical Program for 1950, submitted in December 1949, proposed "to continue the development along the lines indicated in the 1949 program of an implosion device of the hydride type for test in 1951."²⁸²

The Committee for Weapon Development met at LASL on February 2, 1950; at this meeting, the committee recommended deleting a proposed uranium hydride weapon from the test series then scheduled for 1951 and ceasing further work on it. This recommendation was made because the weapon's efficiency and yield were extremely low; its hydrodynamic behavior was very poor; and it was not small enough to fit into rockets likely to become operational in the near future, so its diminished physical size offered no particular advantage over standard implosion bombs.

In addition, even a hollow-core fusion-boosted version offered no appreciable yield increase, and the design was extremely wasteful of scarce fissionable material, even though four times as many hydride weapons as MK 8 ELSIE gun-type penetrating bombs could be made from a given quantity of fissionable material.²⁸³ In March 1950, when LASL submitted a revised technical research program for 1950, this project was abandoned, although results of experimentation were thoroughly

²⁸¹Memorandum dated December 2, 1948, from William Webster, Chairman, Military Liaison Committee, to the AEC, Subject: Los Alamos Laboratory Program.

²⁸²Letter dated December 9, 1949 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, p. 1.

²⁸³The Committee for Weapon Development, Minutes of Meeting dated February 2, 1950, transcribed by Darol Froman; memorandum dated February 6, 1950 from Darol Froman to Members of Technical Board, Subject: Cessation of Work on Hydride Weapon.

documented and some work on it was brought to completion, in case the uranium hydride bomb program was resumed in the future.

The University of California Radiation Laboratory (later the Lawrence Livermore Laboratory, and now known as the Lawrence Livermore National Laboratory) experimented with uranium hydride in its first two fission devices tested at the Nevada Test Site -- Operation UPSHOT-KNOTHOLE shots Ruth and Ray -- in the spring of 1953.

These devices, named HYDRIDE I and HYDRIDE II, were to have been primaries for a new family of thermonuclear bombs. Both shots failed, yielding only 0.2 KT each.²⁸⁴ The two devices contained uranium deuteride, UD_3 , pits fabricated from differing mixtures. Nonetheless, UCRL continued its hydride research at least into 1954.²⁸⁵

Use of Uranium-233

In a further attempt to alleviate the severe shortage of fissionable material in the U.S. at the end of World War II, Los Alamos considered the use of uranium-233 as a bomb fuel. Of the isotopes U-235, Pu-239, and U-233, U-233 has the highest cross section for fission by slow neutrons.

Uranium-233 could be "bred" from thorium in much the same manner as plutonium was fabricated from uranium-238, by

²⁸⁴"Known U.S. Nuclear Tests, July 1945 to 31 December 1988," Robert S. Norris, Thomas B. Cochran, and William M. Arkin, Nuclear Weapons Databook Working Paper No. 2, NWD 86-2 (Rev. 2C), Natural Resources Defense Council, Washington, D.C., January 1989, p. 20; "Operation REDWING, Technical Summary of Military Effects, Programs 1 - 9 (U)," WT-1344 (EX), Office of the Deputy Chief of Staff for Weapons Effects Tests, Headquarters Field Command, Defense Atomic Support Agency, Sandia Base, Albuquerque, New Mexico, April 25, 1961, p. 40. These two tests are described in more detail in Chapter 3, which discusses UPSHOT-KNOTHOLE in the context of thermonuclear weapons development. "Preparing for the 21st Century: 40 Years of Excellence" (UCRL-AR-108618, University of California/U.S. Department of Energy, 1992) states on p. 22 that the first UCRL test was "a risky gamble on an idea once rejected at Los Alamos." Dr. Herbert York refuses to characterize these two tests as failures; he says they had lower yields than expected, but only by about a factor of two or three. (Interview of Herbert York at La Jolla, California, September 29, 1993.)

²⁸⁵Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, p. 35.

neutron absorption and radioactive decay via the reaction $n + \text{Th-232} \rightarrow \text{Th-233} \rightarrow \text{Pa-233} \rightarrow \text{U-233}$.

As early as May 1943, LASL recommended continued study of U-233 as a possible explosive material; it might be easier to chemically separate uranium-233 from thorium than plutonium from U-238. However, the Hanford piles were designed for plutonium production, rather than thorium conversion. A graphite carbon-moderated pile that produced plutonium from uranium would produce only 5% as much thorium as plutonium; a heavy-water (deuterium oxide) moderated pile would produce 20% as much thorium as plutonium.²⁸⁶

In April 1944, it appeared theoretically possible that a thorium reactor, once started, might produce enough U-233 to maintain a self-sustaining reaction without the addition of anything but more thorium. Thus thorium, estimated to be about 10 times more plentiful than uranium, suddenly assumed great strategic importance.²⁸⁷

In September 1944, measurements of the fission cross section of U-233 showed that it was a good potential bomb material. At high energies, the U-233 fission cross section was about twice that of U-235, placing it between U-235 and plutonium in both cross section and the average number of neutrons released per fission.²⁸⁸

By May 1945, it appeared likely that uranium-233 might be useful in high-yield fission weapons:

As a technical background for the discussions, Dr. A. H. Compton explained the various stages of development. The first stage involved the separation of uranium 235. The second stage involved the use of "breeder" piles to produce enriched materials from which plutonium or new types of uranium could be obtained.

The first stage was being used to produce material for the present bomb while the

²⁸⁶Notes on Conference at Los Alamos Scientific Laboratory, 15-24 April 1943, LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 11.

²⁸⁷Hewlett and Anderson, pp. 286, 287.

²⁸⁸HISTORY OF MODERN PHYSICS, Vol. II, Project Y: The Los Alamos Story, pp. 21, 23, 178, 179, 183; memorandum dated October 11, 1944 to J. R. Oppenheimer from Edward Teller, Subject: Contributions of Argonne Laboratory.

second stage would produce atomic bombs with a tremendous increase in explosive power over those now in production. ... While bombs produced from the products of the second stage had not yet been proven in actual operation, such bombs were considered a scientific certainty. It was estimated that from January 1946 it would take one and one-half years to prove this second stage in view of certain technical and metallurgical difficulties ...

Dr. Oppenheimer reviewed the scale of explosive force involved in these several stages. One bomb produced in the first stage was estimated to have the explosive force of 2,000 - 20,000 tons of TNT. ... In the second stage the explosive force was estimated to be equal to 50,000 - 100,000 tons of TNT. ...²⁸⁹

At the end of the war, research continued at Oak Ridge to devise means to convert thorium to U-233 and to separate this fissionable isotope.²⁹⁰

After the war, the thorium-to-U-233 conversion program moved slowly. The value of U-233 as a fissionable material in weapons had not been experimentally established, and the feasibility of economically breeding the isotope, in terms of U-233 production versus the amount of U-235 consumed in the process, had not been ascertained.

At its first meeting at the beginning of January 1947, the GAC called for more effort on studies relating to use of thorium as fissionable material.²⁹¹ In its next meeting during the following month, the GAC noted that very little had been done to utilize thorium and that it was not even clear whether or not thorium could really be emphasized as a raw material since so little was known about the reactor time or radiation environment required to convert thorium to U-233.²⁹²

²⁸⁹Notes of the Interim Committee Meeting, Thursday, 31 May 1945, reproduced in THE MANHATTAN PROJECT: A DOCUMENTARY INTRODUCTION TO THE ATOMIC AGE, edited by Michael B. Stoff, et. al., Temple University Press, Philadelphia, Pennsylvania, 1991, pp. 107-109. Emphasis added.

²⁹⁰Hewlett and Anderson, p. 627.

²⁹¹Minutes of the First Meeting of the General Advisory Committee to the AEC, January 3-4, 1947.

Late in November 1947, the GAC noted that thorium might turn out to be an important source material and that more incentives should be provided to explore and develop potential mining sites.²⁹³

In October 1949, the GAC recommended the production of freely-absorbable neutrons, in reactors, for production of U-233. The GAC was interested in the U-233 program for both military and civil purposes.²⁹⁴

By December 1950, approximately 300 grams of U-233 had been produced at Hanford and separated at Oak Ridge. Information to date indicated that plutonium was more useful in weapons than U-233 might be, so there was little pressure to produce large quantities of U-233. Information regarding the weapon utility of U-233 was still incomplete and was under study at LASL.²⁹⁵ One possible use for it might be in compact gun-assembly weapons.²⁹⁶

The Los Alamos technical program for 1951 stated that although production of U-233 as a weapon core material was not currently planned, research on the nuclear properties of the isotope would be carried out as part of the general laboratory program of weapon research and development.²⁹⁷

One of the first problems to be solved in determining the usefulness of U-233 was a practical means to separate it from thorium. In 1951, Oak Ridge devised a small-scale pilot extraction plant to remove U-233 from irradiated thorium, and

²⁹²Draft Minutes of the Second Meeting of the General Advisory Committee, February 3-4, 1947, pp. 2, 7.

²⁹³Draft Minutes of the Seventh Meeting of the General Advisory Committee, November 21-23, 1947, p. 10.

²⁹⁴Attachment to letter dated October 30, 1949 to AEC Chairman David Lilienthal from J. R. Oppenheimer, Chairman, General Advisory Committee.

²⁹⁵Letter dated December 11, 1950 to Senator Brien McMahon, Chairman, JCAE, from Gordon Dean, Chairman, USAEC.

²⁹⁶"Military Objectives in the Use of Atomic Energy," Department of Defense Research and Development Board Committee on Atomic Energy, 29 December 1950, p. 14.

²⁹⁷Letter dated November 17, 1950 from N. E. Bradbury, Director, LASL, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1951 and Fiscal Year 1952.

a year later, 350 grams, about 12 ounces, of U-233 were sent to Los Alamos for further measurements of its nuclear properties.

By the fall of 1951, a few process tubes in one reactor at Hanford had been loaded with thorium for U-233 production; these tubes had previously been used for tritium production so there was no loss of plutonium. Hanford planned to consume 20 tons of thorium metal per year for uranium-233 generation.²⁹⁸

In December, North American Aviation had completed a study of an appropriate type of reactor for U-233 production. There was an emergency aspect to the study, in that at this time, there was a military requirement for U-233.²⁹⁹

By the spring of 1952, on the basis of a requirement for U-233, the AEC's Reactor Division was exploring the design of a converter reactor rated at several hundred megawatts. The cost per gram of U-233 was expected to be about \$40 more than for an equal amount of plutonium, the extra cost being mainly due to additional chemical processing required for thorium. The cost of a gram of U-233 was estimated to range from \$355 to \$625, depending upon the production method.

The GAC, during its 30th meeting in late April, stated that "it is clearly desirable to assemble enough uranium-233 to permit detailed critical assembly studies and a more complete evaluation of the weapons properties and material."³⁰⁰

At the end of October 1952, just before the Operation IVY thermonuclear Mike shot, the first Savannah River reactors were about to be charged and the question of whether to load them for tritium or U-233 production arose. The value of uranium-233 versus tritium production had changed considerably since construction of the Savannah River plant. If the Mike device were successful in generating its own tritium "in situ," then the major future requirement for tritium would be for its use in boosting fission weapons. On

²⁹⁸Memorandum to File dated September 4, 1951 from Walter A. Hamilton, subject: Conference with Walter Williams.

²⁹⁹Minutes of the Twenty-Eighth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, December 12, 13, and 14, 1951, p. 19.

³⁰⁰Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 34.

the other hand, the utility of U-233 weapons was still an unknown factor.³⁰¹

By year's end, production of small quantities of U-233 was continuing at Oak Ridge and Hanford. In view of the limited quantity available, LASL believed that more could be learned from detailed and heavily-instrumented non-destructive experiments than from a full-scale nuclear test shot.³⁰²

In January 1953, Oak Ridge sent another allotment of U-233 to LASL, and this second shipment gave Los Alamos scientists sufficient material to complete a preliminary evaluation. A month later, the AEC sent to the MLC a paper summarizing the U-233 program, including the supply of the isotope, its usefulness in fission weapons, its reactor properties, and possibilities for its production.³⁰³

By March, results were less encouraging than had been anticipated, but it still seemed probable that U-233 could be used for military purposes. At the end of May, production of small quantities of U-233 at Hanford and Oak Ridge was continuing to provide enough for research purposes. Nondestructive tests and studies of the nuclear and weapons properties of U-233 had by now provided sufficient information to allow its evaluation as a weapon material.³⁰⁴

In the summer of 1953, the Army, Navy, and Marine Corps gave their full support to the program, but the Air Force objected on the grounds that U-233 production would detract from U-235 generation.³⁰⁵

By April 1954, the AEC believed that the cost of U-233 would be comparable to high-quality plutonium; plans were made to produce U-233 by loading an enriched-uranium Savannah River plant reactor with thorium during 1955. There was some indication that the available supply of thorium metal might

³⁰¹Minutes of Seventy-Third AEC-MLC Conference, October 30, 1952.

³⁰²Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952.

³⁰³Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 32.

³⁰⁴Atomic Energy Commission, Progress Report to the Joint Committee, December 1952 through May 1953, Part III - Weapons, May 29, 1953, p. 8.

³⁰⁵AF ATOMIC ENERGY PROGRAM, Vol. IV. pp. 23-25.

be a bottleneck. To separate U-233 from irradiated thorium, a Savannah River "Purex" plutonium-uranium extraction plant would probably be converted to a "Thorex" thorium extraction process.³⁰⁶

The GAC had stated that it was desirable to proceed with U-233 production, and production plans and reactor and weapon plans were being drawn up, but a full-scale core test during Operation TEAPOT in early 1955 was required before a final decision could be made.³⁰⁷

In July, the AEC's Division of Military Application recommended large-scale production of U-233 through 1961. If this suggestion were approved, the immediate dollar costs for expanded production capability would not be prohibitive; the DuPont Company would be responsible for the program.

The program would be somewhat costly in terms of reactor capacity dedicated to U-233 production instead of plutonium generation. However, it would not interfere seriously with materials required for the production of thermonuclear weapons.

To eliminate the possibility that U-233 might be disadvantageous in weapons use, as opposed to use of alloy or plutonium, a full-scale nuclear test was desirable before rushing into large-scale production. There might be enough U-233 on hand to allow a test during the forthcoming Operation TEAPOT planned for Nevada in the spring of 1955.³⁰⁸

If the production program were started immediately, one Savannah River reactor would be converted to U-233 generation in January 1955, another in October 1955, and a third soon afterwards. There was sufficient thorium on hand for the next three years, a more favorable situation than the present state of uranium ores. The use of thorium would also lessen U.S. dependence upon foreign supplies of raw materials.³⁰⁹

³⁰⁶Minutes of the 39th Meeting of the General Advisory Committee to the USAEC, March 31, April 1 and 2, 1954, p. 19.

³⁰⁷"Recommendations, Suggestions, Expressed Interests of GAC - Period, August 1953 - December 1954," December 31, 1954, p. 2.

³⁰⁸"Recommendations, Suggestions, Expressed Interests of GAC - Period, August 1953 - December 1954," December 31, 1954, p. 3.

³⁰⁹Letter dated July 16, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

U-233 apparently offered an increased degree of flexibility in weapon design, although at a slight cost of a higher reflected critical mass for a given yield. U-233 was also less toxic than U-235 or plutonium, and it offered related technological and fabrication advantages. In addition, changeover to thorium in reactors did not affect the availability of tritium. There were enough reactors now in operation to allow both programs, and current tritium requirements had dropped significantly following Operation CASTLE earlier in the year.³¹⁰

Early in August, the AEC recommended that a U-233-fueled device be tested during the forthcoming TEAPOT series in Nevada, noting that "in view of (a) the limited financial or material commitments for the U-233 production program that have to be made before TEAPOT, (b) the fact that an earlier test would involve additional funds and interfere with planning for TEAPOT, and (c) the question as to whether sufficient U-233 can be made available before TEAPOT for test purposes, it is recommended that planning proceed on the basis of this test being conducted during the TEAPOT series."³¹¹

The AEC recommendation was bolstered by earlier documents that surveyed possible methods and ramifications of a test of U-233 in a weapon:

1. Amount of U-233 Needed for Test

A comparison of weapon performance of U-233 and plutonium can be made [deleted] kilograms of U-233 are needed for a weapon test, including the excess required in fabrication of the core component.

2. Availability of U-233

[deleted] kilograms of U-233 has been produced and processed at Oak Ridge National Laboratory. Most of this is in the form a spherical shell specially fabricated for use at Los Alamos in determining the nuclear properties of U-233.

The Division of Research has stated that [deleted] additional kilograms of U-233 can probably be processed in their program at

³¹⁰Minutes of the Forty-First Meeting of the General Advisory Committee to the USAEC, July 12-15, 1954, pp. 41-47, 50-52.

³¹¹Atomic Energy Commission, Weapon Test of U-233, Memorandum by the General Manager, AEC 707/4, August 11, 1954.

Oak Ridge in time for use in Operation TEAPOT at the Nevada Proving Grounds next spring. The Thorex pilot plant, which will extract U-233 from irradiated thorium and recover thorium at the same time, is nearing completion. Cold runs are scheduled to start in August and hot runs in October of this year. Sufficient thorium, irradiated at Hanford, is available to give [deleted] kilograms of U-233 early next year.

3. Use of U-233 in Weapons Effects Test

The Department of Defense is considering a weapons effects test to have a yield of [deleted] and to be fired on a tower during Operation TEAPOT. It may be feasible to provide the yield desired [deleted] This would enable a test of U-233 to be made without increasing the total number of shots. (Author's note: this is in fact what was finally done; the TEAPOT U-233 test was conducted a joint effects-proof test.)

4. Test Before TEAPOT

In order to have a weapon test of U-233 before TEAPOT, it would be necessary to use either the material specially fabricated at Los Alamos for laboratory determination of the nuclear properties of U-233 or to set up special facilities at Oak Ridge to process additional U-233 on a "crash" basis. The use of the Los Alamos material would delay important measurements which will contribute to our knowledge of U-233.

A crash program at Oak Ridge to process additional material would interfere with the startup of the Thorex pilot plant, which is needed to provide operating experience with production processes and will also serve as an interim production facility. Even so, it is doubtful whether enough material could be processed at Oak Ridge in time for a weapon test this fall.

A weapon test of U-233 before TEAPOT would require a special operation at the Nevada Proving Ground. This would be expensive, perhaps \$1 - \$1.5 million, and would interfere with present plans for TEAPOT.

It is our understanding that implementation of the U-233 production program approved by the Commission on July 21 will permit Production Division to delay construction of major facilities until about March 1955.³¹²

The major production facility to be built for the U-233 production program and on which construction commitments are required in Fiscal Year 1955 is the thorium metal production plant. On the basis of the existing schedule for conversion of Savannah River reactors to the production of U-233, the recently programmed [deleted] exposure level for U-233 production, and the anticipated [deleted] capacity of the FMPC thorium semiworks plant, deliveries of thorium metal from the new facility must begin no later than January 1957.

Conversion of the second Purex plant at Savannah River to the Thorex process is to be accomplished according to present schedules by July 1956. ...

Relative to the availability of separated U-233 for the immediate future, there is ... a schedule showing the presently planned program for operation of the Thorex Pilot Plant at Oak Ridge.

This schedule would provide by sometime in January 1955 [deleted] indicated by LASL as the desired quantity for a spring test. It is reasonably certain that such a quantity could be delivered in January even if the Thorex plant schedule is delayed by unanticipated difficulties, since the old "Interim 23" facility could be reactivated and operated to the extent necessary to compensate for the Thorex delay.

It is (our) understanding ... that any U-233 to be used in a fall (1954) test would probably have to be delivered to

³¹²Memorandum dated August 3, 1954 to K. D. Nichols, General Manager, USAEC, from Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC.

LASL by the end of October. The only possibility of delivering U-233 from Oak Ridge on such a schedule would require immediate authorization for reactivation of the "Interim 23" plant. Under these conditions, we believe that [deleted] would be the maximum quantity of U-233 which could be delivered by the end of October.

We strongly recommend against such a crash program, unless it is unavoidable, principally because of the diversion of effort from the Thorex program which would result. The timing of the Purex-Thorex conversion at Savannah River Plant is such that continued concentrated effort leading to the earliest feasible operation of the Thorex Pilot Plant is essential.³¹³

Soon afterwards, plans for a test of a U-233 during TEAPOT took shape:

Consideration is being given to the test of a weapon containing U-233 in Operation TEAPOT. The main purpose of such a test would be to check the estimates of the relative masses of U-233 and plutonium required in a weapon for a given yield. [deleted] The results of the test may effect the amounts of U-233 that would be produced and the corresponding losses in plutonium production.

....

... The Los Alamos Scientific Laboratory states that the yield will be calculated and results made available by November 1, 1954, and believes that the yield can be predicted at that time to somewhat better than 25 percent.

After the weapon core containing U-233 is fabricated and criticality measurements are made on it, Los Alamos believes that the prediction of yield can be made within ± 10

³¹³Memorandum dated July 30, 1954 to Brig. Gen. K. E. Fields, USA, Director of Military Application, USAEC, from E. J. Bloch, Director, Division of Production, USAEC. The number "23" refers to ${}_{92}\text{U}^{233}$.

percent. However, because all of the U-233 for the core is not expected to be extracted from irradiated thorium until about February 1, 1955, the criticality measurements can probably not be finished until shortly before the date of the shot and the refined yield estimate will not be available until then.³¹⁴

In May 1955, before the TEAPOT test of a U-233 core, the AEC decided to slow down U-233 production for at least a year. Due to the generation of tantalum-208 during U-233 irradiation, the resultant U-233 radiated intensive gamma rays, creating many processing and handling difficulties, comparable or worse even than those caused by the separation of plutonium from its intensively-radioactive associated fission products.

Even recovered thorium and U-233 would be unavoidably contaminated, so handling problems persisted for even purified material. DOD had not expressed much interest to date in U-233 weapons, so these separation and handling problems were not likely to be overcome anytime soon. Nonetheless, the AEC planned to accumulate 20 to 40 kilograms (44 to 88 lbs.) of U-233 for experimental purposes.³¹⁵

Application of U-233 to Weapons

Application of U-233 to weapons also proceeded slowly. As a bomb component, U-233 had a significant drawback: it was more radioactive than either plutonium and uranium-235, thus complicating design and degrading the reliability of other weapon parts.³¹⁶

In April 1952, U-233 was briefly considered as fuel for an 8" diameter gun-assembly howitzer shell; however, in late February 1953, results of tests at LASL indicated that U-233 could not be produced economically, returning only a 10% gain as opposed to a 60% gain for uranium-to-plutonium conversion, and the Army changed its requirements to use uranium-235 as the nuclear fuel for the 8-inch shell.³¹⁷

³¹⁴Memorandum for the Chairman, Military Liaison Committee, dated August 17, 1954 from K. D. Nichols, General Manager, USAEC.

³¹⁵Minutes of the Forty-fifth Meeting of the General Advisory Committee to the USAEC, May 2-4, 1955, pp. 8, 9, 20.

³¹⁶NUCLEAR WEAPONS DATABOOK, Vol. I, p. 23.

³¹⁷U.S. Atomic Energy Commission, Weapon Test of U-233, AEC

The lab concluded that although U-233 had fission characteristics which made it appear to be useful in certain types of small-caliber artillery shells, its strong spontaneous gamma ray emission introduced serious fabrication and handling problems.³¹⁸

U-233 was considered as an implosion weapon fuel as early as 1951. At the end of that year, U-233 was expected to have unique advantages in boosting the energy release from very-high yield weapons as well as reducing the size of gun-type weapons. The current U-233 production schedule called for the manufacture at Hanford, separation at Oak Ridge, and fabrication at Los Alamos of several kilograms of U-233 by July 1952 for a full-scale test during Operation UPSHOT-KNOTHOLE, then slated for the fall of 1952 in Nevada.³¹⁹

In addition, U-233 could be produced to some extent in the new Savannah River reactors in South Carolina, although its manufacture would reduce tritium output. Consideration was also being given to construction of a "converter" pile to change U-235 into U-233.³²⁰

The chief use of U-233 at this time seemed to be for high-yield implosion weapons.³²¹ However, the GAC was not unanimous in its opinion as to the applicability of U-233 to weapons. At the end of April 1952, reporting on the deliberations of the GAC during its 30th meeting in Washington, D.C., Chairman J. Robert Oppenheimer reviewed the status and prospects of U-233 production and the likely use of the fissionable material in weapons:

We have examined the staff papers on the production of uranium-233. We believe that the statements of weapons usefulness contained in these papers are, in some

707/4, 11 August 1954; AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 129, 130.

³¹⁸Glasstone and Redman, p. 7.

³¹⁹Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 6.

³²⁰Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, pp. 12, 13, 22.

³²¹Minutes of the Thirtieth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, April 27, 28, and 29, 1952, p. 8.

respects, misleading. There is no reason to believe that uranium-233 would be advantageous in producing implosion weapons of large yield in the megaton range.

It is true that uranium-233 offers the possibility of making gun assemblies of eight to ten inches in diameter and that this probably represents the minimum diameter of any now-contemplated atomic weapons. It seems doubtful to us, however, that this reduction in diameter will prove important in military applications, and correspondingly doubtful whether a production program for uranium-233 can be justified in view of the high cost of material.

It is clearly desirable to assemble enough uranium-233 to permit detailed critical assembly studies and a more complete evaluation of the weapons properties and material.³²²

By the fall of 1952, LASL had received several grams of U-233 and was using it to determine its nuclear properties. Even more U-233 for a possible weapon test had been produced at Hanford and processed at Oak Ridge; the remainder was scheduled for processing during the next few months. In view of the limited amount of U-233 available, LASL believed that more information could be obtained by detailed non-destructive experiments than by a full-scale test shot.

At this time, LASL believed that a U-233 "breeder" reactor could begin with U-235 as fuel, which would gradually be replaced by U-233 produced from thorium in the reactor. Excess reactivity of Hanford reactors was currently being used to generate U-233. Three new reactors at the AEC's Savannah River plant would also be used to make U-233.

Thorium metal was being fabricated at a pilot plant at the Ames Laboratory at the Iowa State College in Ames, Iowa; in April 1952, the plant had reached a production rate of four tons of thorium metal per month. Large-scale thorium fabrication was to be done in the future at the AEC's Feed Materials Production Center in Fernald, Ohio; a new facility was under construction there to produce five tons of thorium

³²²Letter dated April 30, 1952 to AEC Chairman Gordon Dean from J. R. Oppenheimer, Chairman, GAC. Later developments would prove Oppenheimer wrong in his opinion of the military application of small-diameter weapons.

metal slugs per month, which would then be taken to the Savannah River plant for irradiation and conversion to U-233. If the Fernald facility were expanded, it could raise its output to 25 tons per month, sufficient for the Savannah River plant and an additional special production reactor dedicated to U-233 only.

Oak Ridge had developed a thorium extraction process and expected to test it in a pilot plant starting in January 1953.³²³

At the end of 1952, LASL estimated that the value of one gram of U-233 in standard implosion weapons of less than 100 KT yield was about the same as that of 0.7 grams of plutonium. One U-233 nucleus produced about 2.35 neutrons per fission, as opposed to 2.11 and 2.0 average neutrons per fission of one nucleus of U-235 and Pu-239, respectively.³²⁴

In weapons yielding 200 KT or more, U-233 could be substituted for the special grade ("supergrade") of plutonium currently used in them. Estimates of the cost of U-233 production indicated that this substitution could be made only at an appreciable increase in the dollar cost per unit of explosive force delivered to a target.

Production of sufficient quantities of U-233 to lower its cost and make a substitution like this more economical would probably require the construction of a converter-type reactor, and a special separation plant which would remove U-233 from thorium by burn-up of U-235 obtained from gaseous diffusion plant operation.³²⁵

Further studies of the nuclear properties and weapons applications of U-233 continued after 1952. In January 1953, at a joint AEC-Military Liaison Committee meeting, a Joint Chiefs of Staff requirement for U-233 was reviewed, and there was a discussion of the need for prototype weapons to determine the utility of U-233 as a bomb material.³²⁶ By late

³²³Letter dated October 6, 1952 to Honorable Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

³²⁴These figures differ slightly from those shown in Table 1-1 in "Weapons Physics" in Volume I.

³²⁵Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952.

³²⁶Minutes of Seventy-Seventh Conference of the Atomic Energy Commission - Military Liaison Committee, January 22, 1953.

March, a study was underway at LASL on a one-kiloton U-233 test device.³²⁷

Plans for U-233 production continued in 1954. Late in March, the cost of U-233 was expected to be comparable to that of high-quality plutonium. Some U-233 production was to start in 1955 when an enriched Savannah River plant reactor was loaded with thorium. Since there was an expectation that the supply of thorium metal might be a production bottleneck, a Savannah River plutonium-uranium extraction plant (Purex) probably would be converted to a thorium extraction (Thorex) process.³²⁸

By the end of June, the AEC's Division of Military Application had recommended that the General Manager approve, on a planning basis, the large-scale production of U-233. If approval were granted, the DuPont Company would be instructed to prepare plans for production plants.

It was still possible that U-233 might offer some disadvantages, so a field test was deemed necessary before full-scale production of U-233 started. If large-scale production were approved, as many as three Savannah River reactors could be brought on-line by early 1956. If U-233 production were started, then ended, a Thorex plant could be converted back to the Purex process. A test shot could be made as soon as enough U-233 was available.

There was enough thorium ore on hand to permit U-233 production for the next three years. This was a more favorable situation than presently existed for uranium ores. The use of thorium in lieu of uranium might lessen the U.S. dependence upon foreign supplies of scarce raw materials.

The chief advantage of U-233 was that it offered increased flexibility in weapon design, despite its larger critical mass requirements. Changeover of excess reactor capability to thorium irradiation would also not interfere with current tritium production requirements.³²⁹

³²⁷Minutes of the Thirty-Fourth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 23-24, 1953, Washington, D.C., pp. 10, 11.

³²⁸Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, p. 19.

³²⁹U.S. Atomic Energy Commission, July 12-15, 1954, pp. 41-47, 50-52.

Among ideas being considered during the summer of 1954 was the use of U-233 to produce high yields in gun-assembly weapons of 8" and 175MM calibers. Composite designs seemed best for these weapons, using tungsten carbide tampers.³³⁰

By mid-September, U-233 appeared to be a potentially good substitute for plutonium, and could serve well in thermonuclear initiators and might open up a large new low-yield weapons family.³³¹

At the end of 1954, the AEC noted that U-233 had not been produced in sufficient quantity to permit its use in the weapons program. There were certain advantages to be gained from a weapons design viewpoint in using U-233 rather than "clean" plutonium, i.e., plutonium with a low-neutron background and containing only microscopic amounts of the Pu-240 isotope, in thermonuclear weapons and in other weapons then under development.

The AEC had approved an interim U-233 production program during 1955 for essential technological and research uses and for a test of a weapon using a U-233 core during Operation TEAPOT. In the meantime, planning was underway for future production of U-233 for thermonuclear weapons.³³²

Although it was first scheduled for test during Operation UPSHOT-KNOTHOLE in 1953, a full-scale U-233 nuclear explosive experiment was not conducted until the MET shot of Operation TEAPOT in Nevada in mid-April 1955.³³³ The test device used a

³³⁰Letter dated July 16, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC; Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, pp. 25, 47.

³³¹Memorandum to Corbin Allardice dated September 14, 1954 from Walter A. Hamilton, subject: Project Proposal.

³³²Atomic Energy Commission, Program Status Report, December 31, 1954, Part III - Weapons Program, p. 9.

³³³AEC 384/8, Weapon Development Effort, November 20, 1951; Atomic Energy Commission, Nevada Proving Ground Test Activities for Calendar Year 1955, Note by the Secretary, August 9, 1954, pp. 2, 4, 7. The MET (Military Effects Test) device may have used a pit called the "Viper": according to McPhee (p. 167), the "Viper," designed by Ted Taylor, was tested in a 20 KT shot that involved "spheres of steel covered with graphite, the size of big pumpkins, dangled from wires 30 feet" from air zero. McPhee claims this shot occurred at Eniwetok; no date is given, however, Taylor left LASL in 1956, after TEAPOT. MET was a 22 KT explosion; to

composite Pu/U-233 core in a MK 7 HE assembly,³³⁴ and one of its purposes was to obtain data on the substitution ratio of U-233 for "supergrade" plutonium.³³⁵

The shot had dual purposes: as its name implies, in addition to being a test of a U-233 fueled core, MET was also a military effects test. Early in September 1954, DOD accepted substitution of a U-233-fueled device for the previously-tested weapon originally scheduled for firing during the MET shot:

The DOD has expressed an interest in the development of the U-233 device. The proposed substitution of the U-233 test device for the originally-selected weapon for the DOD MET shot is believed compatible both with the accomplishment of the objectives of the DOD effects programs and the Atomic Energy Commission development program.

The only proviso attached to this change was that LASL guarantee actual device yield to within $\pm 10\%$ of its

determine the effects on missiles of thermal radiation from a high altitude burst, steel and aluminum spheres and cylinders were exposed to a nuclear fireball at 78-foot intervals from air zero out to 390 feet (AF ATOMIC ENERGY Program, Vol. IV, p. 301). The MET shot included Project 5.4, Evaluation of Fireball Lethality Using Basic Missile Structures, which was to determine the effects of thermal radiation on spheres and cylinders and ceramic materials. Spherical and cylindrical test specimens were positioned on five towers located at distances between 19 and 92 meters (62.33 and 301.8 feet) from MET air zero. (OPERATION TEAPOT 1955, DNA 6009F, Jean Ponton, et. al., JRB Associates, McLean, Virginia, 23 November 1981, p. 105.)

³³⁴Letter dated December 22, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC. The core-HE assembly was essentially identical to that used during the Easy shot of Operation BUSTER-JANGLE on November 5, 1951. (Memorandum dated 10 August 1954 to Commanding Officer, Chemical Corps Chemical & Radiological Laboratories, Army Chemical Center, Maryland, from E. H. Enquist and E. H. Bouton, subject: Report of Conference at the Armed Forces Special Weapons Project, Washington, D.C., on 4-5 August 1954 attended by Mr. Elmer H. Enquist and Mr. E. H. Bouton, p. 2.) The BUSTER-JANGLE Easy shot was a test of a TX-7E weapon with a composite core.

³³⁵Operation TEAPOT, Report of the Test Manager, Joint Test Organization, Nevada Test Site, Spring 1955, p. 78.

predicted value, and that the predicted yield value be ascertained by September 15, 1954.³³⁶

By mid-1955, plans for large-scale U-233 production for weapons had been deferred pending the outcome of further development work relating to both its production and its possible weapons applications. Its principal use was expected to be as a substitute for plutonium in thermonuclear weapon sparkplugs. Plutonium, with its higher-yield-to-weight ratio, was more desirable for sparkplugs than U-235; however, plutonium also had a much higher spontaneous neutron emission rate due to the presence of the Pu-240 isotope.

The low rate of U-233 spontaneous neutron emission would minimize the possibility of loss of yield due to nuclear degradation of the lithium deuteride surrounding the sparkplug. Cost of U-233 production was also believed to be comparable to the cost of production of low-neutron, "clean" plutonium whose Pu-240 content was very low.

Also, results from TEAPOT indicated that U-233 was less valuable as a substitute for plutonium than had been previously believed. In addition, certain weapon development efforts, such as D-T gas boosting, were directed toward improved designs whose safety and efficiency did not depend upon the use of fissionable materials with low spontaneous neutron emission rates.³³⁷

Uranium-233 was also used in one or more of the devices fired during Operation REDWING in the Pacific in the spring and summer of 1956.³³⁸

With the vast expansion of uranium enrichment and plutonium production reactors in the U.S. in the early and mid-1950s, the requirement for U-233 in weapons was sharply curtailed.

During Operation HARDTACK Phase I in 1958, a new principle of weapon design was demonstrated which caused moderately-efficient fission in U-233. Critical mass experiments on 98.1%-enriched U-233 assemblies as both bare unreflected and

³³⁶Memorandum for Chairman, Military Liaison Committee to the Atomic Energy Commission dated 1 September 1954 from Maj. Gen. A. R. Luedecke, USAF, Chief, AFSWP, subject: U-233 Test in TEAPOT.

³³⁷Atomic Energy Commission, Program Status Report, Part III - Weapons Program, July 27, 1955, pp. 10, 11.

³³⁸Letter dated April 9, 1956 to RAdm. B. Hall Hanlon, USN, Commander, JTF 7, from K. E. Fields, General Manager, USAEC.

natural-uranium reflected spheres continued at LASL until the early 1960s.³³⁹

Testing of devices containing U-233 continued during Operations NOUGAT, DOMINIC, STORAX, NIBLICK, FLINTLOCK, LATCHKEY, and CROSSTIE between in 1961 and 1968.³⁴⁰

³³⁹"Criticality: The Fine Line of Control," Hugh C. Paxton, LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 157; "A History of Critical Experiments at Pajarito Site," LA-9685-H, H. C. Paxton, Los Alamos National Laboratory, New Mexico, March 1983, pp. 17, 32, 33.

³⁴⁰Letter to the President dated September 15, 1961 from Glenn T. Seaborg, Chairman, USAEC; teletype dated September 24, 1961 to M. K. Rex, USAEC, LVAO, Las Vegas, from A. W. Betts, USAEC; proposed draft letter to the President re NOUGAT II dated December 22, 1961; memorandum for Chairman Seaborg dated January 2, 1962 from Brig. Gen. A. W. Betts, Director of Military Application, USAEC, subject: Operation NOUGAT II; memorandum for Chairman Seaborg dated January 20, 1962 from Brig. Gen. A. W. Betts, Director of Military Application, USAEC, subject: Revision to NOUGAT II; memorandum for Chairman Seaborg dated March 17, 1962 from Brig. Gen. A. W. Betts, Director of Military Application, USAEC, subject: Underground Tests -- April 1 to June 30, 1962; Atomic Energy Commission, Status Report on Test Program, Note by the Secretary dated March 30, 1962; memorandum for the Chairman, Atomic Energy Commission dated April 11, 1962 from McGeorge Bundy; memorandum for the Chairman, Atomic Energy Commission dated April 20, 1962 from President Kennedy; memorandum for The Chairman, Atomic Energy Commission dated June 9, 1962, from President John Kennedy; letter to the President dated July 3, 1962, from Glenn Seaborg, Chairman, USAEC; memorandum to Chairman, Atomic Energy Commission dated July 12, 1962 from McGeorge Bundy; National Security Action Memorandum No. 190 dated October 1, 1962 to Dr. Glenn T. Seaborg, Chairman, U.S. Atomic Energy Commission from President Kennedy; memorandum for The Chairman, Atomic Energy Commission dated October 22, 1962, from President John Kennedy, subject; Authorization for [deleted] in DOMINIC Series; memorandum for Chairman Seaborg, et. al. dated December 17, 1962 from Brig. Gen. A. W. Betts, Director of Military Application, USAEC, subject: STORAX III; Atomic Energy Commission, STORAX III, Note by the Secretary dated January 8, 1963; memorandum for Chairman, Atomic Energy Commission dated January 11, 1963, from President John Kennedy; memorandum for The Chairman, Atomic Energy Commission dated March 26, 1963 from President John Kennedy; undated draft endorsements in form of memoranda for Chairman, AEC, from The President authorizing expenditures of special nuclear materials during the first and fourth quarter FY 1964 underground test series (NIBLICK I

Hollow Cores

Implicit in the concept of composition is the concept of hollow cores. This was the result of the employment of relatively thin, lightweight bomb pits, if anything weighing 11 ounces per cubic inch can be called light.

Studies and experiments at Los Alamos during World War II had shown that thin shells of plutonium were compressed more easily and more densely than solid spheres. However, the design of pits, tampers, and complex high explosive system geometries required to minimize hydrodynamic instabilities, which caused asymmetric core compression, were postponed until after the end of the war, mainly because of limited wartime computing resources at Los Alamos.³⁴¹

In addition, detonation simultaneity requirements were not quite as stringent with solid hemispheres as they were with

and IV, respectively); memorandum for Chairman, Atomic Energy Commission dated July 9, 1963 from President Kennedy; Atomic Energy Commission, Test Program, Note by the Secretary dated July 18, 1963; letter dated December 21, 1963 to McGeorge Bundy, Special Assistant to the President for National Security Affairs, from Glenn Seaborg, Chairman, USAEC; Memorandum for the Chairman, Atomic Energy Commission dated January 16, 1964 from McGeorge Bundy, subject: Underground Nuclear Testing Program - Third Quarter FY 1964, NIBLICK III; memorandum for the Chairman, USAEC, dated July 1, 1965 from McGeorge Bundy, subject: FY 1966 Underground Nuclear Test Program; memorandum for Chairman Seaborg, Commissioner Palfrey, Commissioner Ramey, and Commissioner Tape dated December 8, 1965, 1963, from Delmar L. Crowson, Brigadier General, USAF, Director of Military Application, USAEC, Subject: FLINTLOCK III; memorandum for The Chairman, Atomic Energy Commission, dated January 3, 1966, from McGeorge Bundy; letter to The President dated December 23, 1966 from Glenn T. Seaborg, Chairman, USAEC; memorandum for the Chairman, U.S. Atomic Energy Commission dated January 11, 1967, subject: Third Quarter FY 1967 Underground Nuclear Test Program (LATCHKEY III); memorandum for the Chairman, U.S. Atomic Energy Commission dated April 3, 1967, subject: Fourth Quarter FY 1967 Underground Nuclear Test Program (LATCHKEY IV); letter to The President dated September 14, 1967 from Glenn T. Seaborg, Chairman, USAEC; letter to The President dated March 18, 1968 from Glenn T. Seaborg, Chairman, USAEC.

³⁴¹Rhodes, TMAB, pp. 466, 467, 480; Rhodes, DARK SUN, pp. 116, 154; "Implosion Becomes the Key to the 'Gadget'," Newsbulletin, Los Alamos National Laboratory, September 3, 1993, p. 9; Hoddeson, et. al., pp. 86-88, 133; Serber, THE LOS ALAMOS PRIMER (1992 edition), pp. xvi, xxxii.

hollow shells.³⁴² By December 1944, hollow cores had been abandoned in favor of the more robust and simpler so-called "Christy gadget," a nearly-solid plutonium core.³⁴³ The design of this core also reduced significantly the problems of jetting and spalling which arose frequently in the hollow shell core designs.³⁴⁴

However, calculations indicated that the "Christy gadget" would be far less efficient than a hollow spherical core, and that it would also require the use of a modulated initiator to begin the fission process at the optimum time.³⁴⁵ If such a modulated initiator were used, the efficiency of the "Christy pit" would be improved.³⁴⁶

Hollow cores also served to facilitate both the technique of in-flight insertion and extraction (an arming/safing technique discussed in Volume VIII), and the employment of fusionable materials to deliver a "boost" to the fission efficiency.

In September 1950, the General Advisory Committee recommended that LASL "pursue without delay" the development of "hollow implosion devices;" in conjunction with other fission weapon developments, these new designs promised to essentially double stockpile yield. More specifically, the new developments promised:

- a. to make very substantial increases in the stockpile of atomic weapons from a given supply of fissionable material;
- b. to provide fission weapons with destructive effects more than tenfold greater than any so far developed;
- c. to provide very much greater flexibility than now exists as to the

³⁴²"A Few Words About This Picture," Stanley Goldberg, INVENTION & TECHNOLOGY, Fall 1991, p. 51.

³⁴³Hoddeson, et. al., pp. 161, 307, 413, 450; Rhodes, DARK SUN, p. 119.

³⁴⁴Hoddeson, et. al., p. 270.

³⁴⁵Hoddeson, et. al., p. 293.

³⁴⁶HISTORY OF MODERN PHYSICS, Vol. II, Project Y: The Los Alamos Story, p. 168.

carriers (delivery vehicles) which can be used for delivery of atomic weapons.³⁴⁷

Hollow implosion designs were discussed at length during the 23rd meeting of the GAC at LASL at the end of October; the concept offered an opportunity to further reduce the amount of fissionable material required by weapons, although their development might require a new and different type of HE-pit-core assembly.³⁴⁸

A report on long- and short-range military objectives in the use of atomic energy, prepared at the end of 1950 by a distinguished panel of civilian and military scientists and advisers, concluded that if successfully developed, hollow core weapons would offer the most efficient use of fissionable material, making possible the use of extremely small amounts of plutonium and enriched uranium, and that with suitable initiation, hollow cores could be used in atomic weapons of very high yield.³⁴⁹

By March 1951, the hollow implosion idea had not been rejected by LASL, but the lab was too busy to devote a well-organized effort to it.³⁵⁰

Hollow-core fission devices were first tested during Operation UPSHOT-KNOTHOLE at the Nevada Proving Ground in the spring of 1953; results revealed that hollow-core nuclear systems could greatly increase the efficiency of implosion weapons. By the time of Operation TEAPOT in Nevada during the

³⁴⁷Letter dated September 13, 1950 from J. Robert Oppenheimer, Chairman, GAC, to Gordon Dean, Chairman, USAEC.

³⁴⁸Minutes, Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, and November 1 1950, Los Alamos New Mexico, pp. 2, 6, 10, 13, 16, 20, 22, 28.

³⁴⁹"Military Objectives in the Use of Atomic Energy," 29 December 1950, report prepared by The Panel on Military Objectives in the Field of Atomic Energy, Committee on Atomic Energy, Research and Development Board, United States Department of Defense, p. 5.

³⁵⁰Minutes of the Twenty-fifth Meeting of the General Advisory Committee to the Atomic Energy Commission, March 15, 16, and 17, 1951, p. 3. By October 1951, Enrico Fermi said that he thought that LASL was not being "particularly negligent" in considering hollow implosion. (Memorandum for the File dated October 2, 1951 from J. Kenneth Mansfield, JCAE, subject: Conversation with Dr. Enrico Fermi on Monday, September 24, 1951.)

spring of 1955, all but one of the prototype LASL air-defense warheads tested were essentially hollow-core devices.³⁵¹

Interchangeability

Another major early postwar advance in fission weapon design was the concept of interchangeable pits and cores, offering a range of yields for a given weapon using a specified HE assembly. By November 1950, interchangeable nuclear components were an important part of the national stockpile.³⁵² The Los Alamos Technical Program for Calendar Year 1951 and Fiscal Year 1952 had as one of its objectives "means whereby cores may be interchangeable to the greatest extent practicable in weapons of various external geometries."³⁵³

The Joint Committee on Atomic Energy (JCAE), during a meeting in mid-August 1951 noted:

General McCormack: This core as being manufactured will fit into any of the non-nuclear assemblies we are making. On hand today we have only the 60" non-nuclear assembly.

After the first of the year we will have the 45 (inch diameter bomb) and after that the 30 (inch diameter bomb). The nuclear cores are interchangeable in all (of them). This is a point that could result in a struggle. We made the cores interchangeable. Everybody has his own weapon. There will be an operational decision as to which type of non-nuclear assembly and, therefore, which type of carrier, which type of aircraft will be used, but the cores are good for all of them, which is a very neat technical trick. Los Alamos earned their year's pay when they developed that one.

³⁵¹Letter dated December 1, 1954 to Brig. Gen. K. E. Fields, Director, Division of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

³⁵²Minutes of Atomic Energy Commission, Meeting No. 483, October 18, 1950, minutes prepared November 7, 1950.

³⁵³Letter dated November 17, 1950 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, General Manager, Santa Fe Operations Office, USAEC, p. 8.

The Chairman: You have a uniform core?

General McCormack: All the cores will fit into all of the non-nuclear assemblies.³⁵⁴

During an AEC briefing for then-President-elect Dwight D. Eisenhower on November 19, 1952, AEC Chairman Gordon Dean stated:

First, as to the weapons picture, our research and development effort has been directed toward five things: (1) getting more and more energy release from smaller amounts of fissionable material; (2) developing weapons which are lighter and smaller and which can be carried by fighter aircraft; (3) developing weapons of high yield, including both hydrogen and fissionable weapons; (4) developing the warheads for guided missiles; and (5) the suitcase weapon.

In speaking of weapons we use the terms nuclear and non-nuclear. Nuclear means that small portion of the center of the weapon made up of plutonium, U-235, or both. All the rest of the weapon, including the high explosive, the case, the firing unit, we call non-nuclear. We have several types of nuclear cores containing small or large amounts of fissionable material. In most of our weapons we can use any type of core that we wish.³⁵⁵

The "suitcase weapon" was a reference to a nuclear device that could be delivered piecemeal and covertly onto enemy territory and then detonated remotely. The weapon was so dependent upon its fuzing system that the device had to be considered as essentially a fuzing mechanism.³⁵⁶

³⁵⁴Minutes of JCAE Meeting, August 16, 1951, pp. 1270, 1271. A list entitled "Items Proposed to be Conveyed to NATO," attached to a letter dated 3 March 1955 to The President from Lewis L. Strauss, specifically mentions "the extent to which there is interchangeability of nuclear components among the various types of weapons."

³⁵⁵Anders, FORGING THE ATOMIC SHIELD, pp. 234, 235.

Later in the postwar U.S. nuclear weapons program, more sophisticated methods of yield variability were developed with the advent of externally-initiated boosted primaries (see below).

Linear Implosion

Yet another significant advance in the postwar design of U.S. fission weapons was the development of linear implosion, which allowed abundant supplies of plutonium to be used in artillery shells.

Because of its tendency to predetonate, caused by the neutrons contained in the alpha particles emitted by its Pu-240 isotope, plutonium could not be used in conventional gun-type assembly nuclear artillery shell designs. A plutonium projectile simply could not, within the limited distance available, be accelerated quickly enough to completely preclude predetonation.³⁵⁷

However, if a slender solid cylinder of plutonium were encased within a hollow cylinder of high explosive, and the explosive detonated simultaneously along its entire length, predetonation would be obviated. This technique was also both conceptually and practically much simpler than multipoint spherical implosion of a solid plutonium core.

Experiments on cylindrical critical masses were conducted at Los Alamos during the earliest days of the Manhattan Project; once the implosion technique was conceived, the first tests used cylindrical rather than spherical core shapes.³⁵⁸ The

³⁵⁶AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 90. By the end of 1953, experiments were underway at the AFSWP on remote radio detonating devices compatible with the Project A and Project B atomic demolition munitions. (Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 July 1953 - 31 December 1953, pp. 227, 228, 263.)

³⁵⁷Predetonation of nuclear weapons was also a concern in other circumstances. The possibility of "fratricide," i.e., the destruction or predetonation of one attacking warhead by another, was of such concern that by July 1952, DOD war plans stated that implosion-type detonations of MK 5 and 6 bombs should not follow each other closer than five seconds in time and two miles in distance. Gun-type weapons should not be used at intervals of less than 10 seconds at separations of three miles, 30 seconds at two-and-a-half miles, and 90 seconds at two miles. (WT-202, p. 93.)

first early HE castings made at S, or "Sawmill," site at LASL during World War II were cylinders of Composition B, ranging in weight from 30 to 500 lbs.³⁵⁹

By October 1943, early experiments had shown that with a small charge, cylinders could be collapsed symmetrically.³⁶⁰ Cylindrical critical masses remained under both theoretical and experimental consideration at LASL during the war.³⁶¹ The biggest impediment to symmetrical collapse was interfering shock waves that intersected and liquefied jets of hot metal.³⁶²

One serious problem that became apparent in experiments conducted during the war was the phenomenon of "jetting," in which tongues or knives of molten material squirted ahead of the collapsing main shell envelope. These jets, which originated from the interaction of detonation waves and from timing imprecision in the firing of neighboring detonators, drastically upset the implosion symmetry and caused undesirable turbulence in the center of the compressed assembly.

Another serious problem was spallation, the breaking off of pieces of the imploding metal at its surface, caused by the reflection of the rapid detonation waves at the surface. These problems also occurred in spherical shells of fissile materials.³⁶³

Although LASL designed and tested cylindrical HE "lenses," they were abandoned in the winter of 1944-1945 because they did not prevent jetting. Primacord, flexible tubing filled

³⁵⁸"Implosion Becomes the Key to the 'Gadget'," Newsbulletin, Los Alamos National Laboratory, September 3, 1993, p. 9; Hoddeson, et. al., pp. 88, 89, 133, 135, 137; THE HISTORY OF MODERN PHYSICS, 1800-1950, Vol. II, p. 68. The latter is a further-declassified edition of LAMS-2532 and it contains some information deleted from the earlier edition released in 1961.

³⁵⁹LA-1052, p. 2.

³⁶⁰Minutes of the Meeting of the Governing Board, October 28, 1943, 7:45 PM.

³⁶¹Report of Hugh H. Clegg and Robert J. Lamphere covering Interviews with Klaus Fuchs in London, England, between May 20 and June 2, 1950, pp. 28, 34; Hoddeson, et. al., pp. 129, 133, 141, 143, 145, 146, 158, 159, 180, 181, 446.

³⁶²Rhodes, DARK SUN, p. 117.

³⁶³Hoddeson, et. al., pp. 140, 279.

with PETN, a relatively slow explosive, was the most-frequently used implosion HE, although cast cylinders of HE were also used; in addition, electrical detonators which could guarantee better simultaneity of detonation, were not yet available. Later experiments in cylindrical implosion used Pentolite, a 50/50 mixture of PETN and TNT, cast around metal cylinders.³⁶⁴

Typical cylinder diameters imploded at LASL during World War II ranged from three to 12 inches and in lengths up to 18 inches.³⁶⁵

The laboratory concluded that detonation synchronization problems, chemical inhomogeneities in the HE and the fissionable material, the geometric arrangement, and other intrinsic characteristics of the HE all militated against success in this field.³⁶⁶

After the war, LASL did not do any work on non-spherical implosion systems in part because the process could not be easily calculated.³⁶⁷

By the end of May 1954, the University of California Radiation Laboratory (UCRL) was conducting linear implosion experiments with non-fissionable U-238 cylinders at both LASL's "S" ("Sawmill") site and at the U.S. Navy's Naval Ordnance Test Station at Inyokern (China Lake), California.³⁶⁸

In July, linear implosion appeared to be a promising alternative to the gun-type or hollow implosion methods and was therefore, in the GAC's opinion, "interesting to explore."³⁶⁹

³⁶⁴Hoddeson, et. al., pp. 453, 454.

³⁶⁵Hoddeson, et. al., pp. 143, 145.

³⁶⁶Hoddeson, et. al., p. 448.

³⁶⁷"An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 37.

³⁶⁸Minutes of the 40th Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 27-29, 1954, p. 24; letter dated July 16, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC; "An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, pp. 37, 38.

³⁶⁹Letter dated July 16, 1954 to Lewis L. Strauss, Chairman,

In the fall, LASL director Norris Bradbury downplayed the importance of this new UCRL research effort:

The first area in which this problem may appear is typified by the UCRL linear implosion system. This system is not new -- it certainly was discussed at LASL at least as early as 1944 -- and represents an obvious interpolation of [deleted] It is presumably of interest in tactical atomic weapon applications, but its high cost in plutonium makes it of very dubious early applicability in any area where, if there is to be any real military significance, large numbers are required.³⁷⁰

By the end of 1954, UCRL was pursuing the design of linear critical mass assemblies; research had reached a point that tests of such devices were proposed for Operation TEAPOT in the spring of 1955.³⁷¹ In early December, UCRL director Dr. Herbert York proposed that among three nuclear devices to be detonated by the lab during Operation TEAPOT were to be "two .. for the purpose of investigating the principles of the linear implosion device."³⁷² The primary application of the new principle was to be small-caliber artillery shells.³⁷³

USAEC, from I. I. Rabi, Chairman, GAC.

³⁷⁰Letter dated September 22, 1954 to Brig. General K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

³⁷¹In a 1989 interview, Dr. Ted Taylor said that "Livermore's trademark in the beginning was non-spherical things ... the reason for wanting to consider something that looks more like a football rather than a basketball is because by that time we were interested in ramming things through guns." ("An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 23.)

³⁷²Letter dated December 7, 1954 to Brig. Gen. K. E. Fields, Director, Division of Military Application, USAEC, from Herbert F. York, Director, UCRL - Livermore.

³⁷³"There is, however, another type of artillery shell which does not employ gun assembly." (Glasstone and Redman, WASH-1037, p. 12.). This statement is followed by a paragraph citation in a chapter of the Glasstone-Redman work entitled "Implosion Systems." In "Small Is Beautiful: Livermore, the Army, and Atomic Artillery," a paper presented at the Annual Meeting of the Society for the History of Technology at

The General Advisory Committee, however, was not so enthusiastic about these devices, compared to other more-conventional designs; the GAC recommended that whether or not a second linear implosion test would be conducted during TEAPOT should be contingent upon the results of the first test of this principle:

The Committee's position is somewhat different in respect to the proposed detonation of two "linear implosion" test devices. This is an interesting but at present somewhat problematical line of weapon development, in which the ideas have been changing rapidly and in which one can expect considerable additional information to become available from non-nuclear experimentation. We do not disapprove of these shots but regard them as less urgent than the small implosion and (thermonuclear device) case tests (also slated for TEAPOT).

We understand that Gen. Fields and Dr. York will discuss these matters in the near future, particularly the question of the criteria for deciding whether a second "linear implosion" shot is to be fired (during) the TEAPOT operation.³⁷⁴

The AEC's Division of Military Application took a somewhat more charitable view of the Livermore tests proposed for TEAPOT:

The Division of Military Application has reviewed the program submitted by the (LASL and UCRL) and considers, with one exception, that the testing of the devices proposed by the laboratories is essential to the weapons development program to meet military requirements for air defense, tactical and thermonuclear weapons and to

Lowell, Massachusetts on October 1994, Ms. Sybil Francis, a PhD candidate at the Massachusetts Institute of Technology, explores the interaction between the Army and Livermore that resulted in early linearly-imploded small-caliber artillery shell warheads. She specifically states (on p. 16) that "Livermore planned two tests of a fission bomb designed explicitly for Army use for the spring 1955 TEAPOT test series in Nevada."

³⁷⁴Letter dated December 22, 1954 to Lewis L. Strauss, Chairman, USAEC, from I. I. Rabi, Chairman, GAC.

further weapon research in the fission and fusion fields.

The exception is that of the two linear implosion devices proposed by the University of California Radiation Laboratory. The present University of California Radiation Laboratory proposal for linear implosion devices differs from its previous proposals, which at first were for differing size devices and later for unboosted and boosted devices.

The two now proposed are similar to each other in all respects but will be so detonated as to develop the time behavior of the system. The Director of the Division of Military Application is developing with the Director, Livermore Laboratory, the need for such knowledge of the system at this time, and in particular whether it is necessary to perform two tests in order to understand the capabilities of the linear implosion system sufficiently to determine the nature and extent of further effort on it.

It is hoped that recommendations on this can be submitted to the Commission ... by January 20 (1955). However, it is recommended that in order not to delay the TEAPOT program, the Commission seek Presidential authorization at this time for the full program, thus permitting latitude to include two such devices if it is determined (to be) essential.

...

RECOMMENDATION

The General Manager recommends that the Atomic Energy Commission:

- a. Approve the Los Alamos Scientific Laboratory and the University of California Radiation Laboratory weapons development programs for TEAPOT as circulated ... subject to further recommendations with respect to the linear implosion devices.³⁷⁵

³⁷⁵Atomic Energy Commission, Proposed Program for Operation

The GAC's doubts notwithstanding, linear (cylindrical) critical masses were tested during UCRL shots Tesla and Post during Operation TEAPOT, a series of nuclear test explosions conducted at the Nevada Test Site in the spring of 1955.³⁷⁶ The firing of the second test was contingent upon satisfactory results of the first.³⁷⁷

These successful tests led to development of a family of linear implosion devices, primarily plutonium-fueled artillery shells and air defense rocket warheads. The Tesla device was 10 inches in diameter.

In addition, during the suspension of nuclear tests between 1958 and 1961, weapon design efforts were redirected to the task of simulating chemical implosions for new, non-spherical geometries.³⁷⁸

Advances in Tamperers and Reflectors

To reduce the critical mass of a specified shape and composition of fissionable material, it is usually surrounded by a material which scatters some of the escaping neutrons back into the fissile core. By reducing the fraction of neutrons which escape completely, a smaller size or mass of fissile material can become critical. This scattering device, on account of its function, is sometimes referred to as a neutron reflector or a neutron tamper.

In nuclear weapons, the fissile material is surrounded by a tamper or, more specifically, an inertial tamper, the mass of which delays expansion of the exploding material and permits

TEAPOT, Report to the General Manager by the Director of Military Application, December 30, 1954.

³⁷⁶"The effects of weapons design, whether a given device was of the spherical implosion or linear implosion type..." (Operation TEAPOT, Project 2.2, Neutron Flux Measurements, February - May 1955, Nevada Test Site, WT-1116 (EX), p. 390; "3.2.3 Linear Implosion Systems (Shots 3 and 11, Tesla and Post). Four gamma-dosimeter lines were instrumented for shots 3 and 11 to determine the asymmetry of linear-implosion devices." (Operation TEAPOT, Technical Summary of Military Effects, Programs 1 - 9, February - May 1955, Nevada Test Site, WT-1153 (EX), p. 91). The TEAPOT Post and Tesla shots yielded 2 and 7 KT, respectively.

³⁷⁷Minutes of AEC Meeting No. 1051, January 4, 1955.

³⁷⁸"Nuclear Tests Mean New Weapons," Josephine Anne Stein, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 42 No. 9, November 1986, p. 9.

a higher energy yield to be obtained from the system undergoing fission.

This inertial tamper also serves as a neutron reflector or neutronic tamper. In some cases, however, the neutronic aspect is more important than the inertial character of the tamper.

Up to a point, increasing the thickness of the tamper reduces the number of neutrons which escape and thus makes possible a smaller critical mass and higher yield. However, a thicker tamper also carries a significant weight penalty.

The earliest U.S. implosion weapons used natural uranium tampers. Contemporary implosion weapons use beryllium.³⁷⁹

Tampers and reflectors, which frequently are the same physical entity, could be improved in ways affecting both the weight and the efficiency of the nuclear core. The heavier the tamper, the greater its inertia, and the longer it can resist the expansive forces of the exploding core to allow time for more fission to occur before disintegration, thus increasing the efficiency of the explosion.

On the other hand, the heavier the tamper, the greater the overall size and weight of the warhead pit and the larger the amount of HE required to compress it. A reflector needs to be dense and resistant to neutron penetration, in order to reflect escaping neutrons back into the core to cause more fission, but it also must be as light as possible to reduce weight and HE volume.

Some of the materials first considered at Los Alamos for use as tampers included gold, tungsten, rhenium, iridium, lead, platinum, iron, beryllium, tungsten carbide, and uranium.³⁸⁰ The FAT MAN implosion weapon of World War II finally used a thick and massive natural uranium (mostly uranium-238) tamper surrounding a small plutonium core, for both high yield and efficient use of weight and space.

Another reason for the use of a uranium tamper was to reduce the likelihood of Taylor instabilities in the implosive shock wave, by using a tamper whose density was very close to that of the material in the fissile core, either plutonium or uranium.³⁸¹

³⁷⁹Glasstone and Redman, WASH-1037 (Revised), p. 10.

³⁸⁰Hoddeson, et. al., pp. 72, 73, 187, 337; Serber, LOS ALAMOS PRIMER (1992 edition), p. 29.

³⁸¹Hoddeson, et. al., p. 161.

In boosted fission devices, a natural uranium tamper also serves as an additional fissile fuel after being "ignited" by the high-energy neutrons radiating from the fusion reactions in the center of the exploding bomb core.

The designers of the FAT MAN left nothing to chance: the outer surface of the natural uranium tamper was coated with boron-10, to absorb any stray neutrons that might enter the bomb casing from an external source and cause predetonation.³⁸²

Although the FAT MAN U-238 tamper was about 2 3/4" thick, later uranium tampers on other U.S. nuclear weapon pits varied in thickness between 1/2" and 2 1/2."³⁸³

A major breakthrough in tamper and reflector technology after World War II was the selection of beryllium (chemical symbol Be) for reflectors in small low-yield weapons.³⁸⁴ This was the result of a search for the lightest, most efficient material suitable for use as a reflector in a fission weapon.

³⁸²Rhodes, DARK SUN, pp. 194-195; Document #13, Moskovskaya Pravda, "Hunters of Nuclear Secrets," Mikhail Makhlin, June 1991, with additions from Visgin, V. P., ed. 1992. At the Source of the Soviet Atomic Project: The Role of Espionage, 1941 - 1946. Problems in the History of Science and Technology 3:97, pp. 126-129. This is an English translation of an October 1945 Soviet document which contains "information on the design of the atomic bomb compiled on the basis of agent materials received from the People's Commissariat of State Security of the USSR;" the Russian-language version of this paper was circulated widely in the former Soviet Union in 1993 and 1994 and has presumably been thoroughly compromised by now. Klaus Fuchs was presumably a major source of this information passed to the Soviets. This technique of protecting the core with boron also provided protection against "fratricide," the unintended premature detonation of a nuclear weapon caused by neutrons emanating from a nearby nuclear burst, although this was not the main purpose of coating the FAT MAN tamper.

³⁸³"Capsule Storage and Density-Analog Techniques," LA-5930-MS/UC-46, H. C. Paxton, issued May 1975, p. 1.

³⁸⁴McPhee, pp. 91-94; "The H-Bomb Secret," Howard Morland, THE PROGRESSIVE, Vol. 43 No. 11, November 1979; "Controlling Weapons-Grade Fissile Material," J. Rotblat, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 33 No. 6, p. 39.

In addition, the replacement of heavy, nearly-solid plutonium cores by thin shells of plutonium and uranium allowed lighter tampers to be used, without inducing Taylor instabilities.

The use of lightweight beryllium tampers was also a result of success in the fusion boosting of fissile cores. (See "Advances in Boosted Fission Weapons" below.) To attain high efficiencies, i.e., to consume as much fissionable material as possible in a given explosion, expansion of the reacting core had to be delayed. This important function was originally fulfilled in pure fission weapons by the use of a heavy inertial tamper, such as uranium.

In boosted devices, however, the introduction of neutrons from an extraneous (D-T) source results in a considerable increase in the fission rate before the reacting system expands significantly. The inertial tamper was thus less important than in an unboosted system, and lighter tamper materials could be profitably employed.³⁸⁵

Beryllium tampers also minimized loss of x-rays from fission reactions, making its use mandatory in primaries of thermonuclear weapons, which required high x-ray fluxes to compress their fusion stages.³⁸⁶

Beryllium is the fourth element of the periodic table, with a density of about 115 lbs. per cubic foot. It has a low thermal neutron absorption cross-section (a measure of the probability of a beryllium atom absorbing a neutron with a specific kinetic energy), and a high neutron scattering cross-section (because Be has more atoms per cubic centimeter than any other element).³⁸⁷

³⁸⁵Glasstone and Redman, WASH-1037 (Revised), p. 29.

³⁸⁶"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 8.

³⁸⁷McPhee, p. 215. MCPhee (pp. 91-94) seems to credit Ted Taylor with the idea of using beryllium as a bomb pit reflector; however, this was yet another concept that considerably predated Taylor's employment at Los Alamos. In his letter of 7 March 1986 to the author, Taylor stated:

I certainly did not invent that basic concept, but did make some contributions to ways that it could be used effectively, stimulated by the well-known and widely-published fact that beryllium is an extraordinarily

In addition, when struck by high energy neutrons from fusion reactions, beryllium serves as a neutron source, enhancing those reactions (see note 4 under "Fusion Physics" in Volume I). This neutron multiplication is useful in boosted fission devices.

Spherical uranium metal -- enriched to 93.5% uranium-235 -- has the lowest critical mass when encased by any thickness of beryllium, as opposed to its critical mass when encased by identical thicknesses of other materials such as tungsten carbide, natural uranium, copper, graphite, thorium or aluminum.³⁸⁸

Within a 4" thick Be reflector, the critical mass of uncompressed uranium enriched to 93.5% in the U-235 isotope is as low as 14.1 kilograms, compared to the critical mass of a "bare" unreflected mass of the same material, which is 52.5 kilograms.³⁸⁹

Comparable uncompressed critical masses for 100% Pu-239 and 100% U-233 in the same thickness of Be are 4.4 and 5.8 kilograms, respectively.³⁹⁰

To maximize the rapid production of fast neutrons in an exploding assembly, elements of low mass number must usually be eliminated from the cores and reflectors of fission weapons. Light elements slow down, or moderate, neutrons much more effectively than heavier ones. Beryllium, however, has neutronic properties which overcame these concerns.³⁹¹

Metallic beryllium had been examined as a possible tamper material as early in the U.S. nuclear weapons program as October 1943; while it was an excellent neutron reflector, its use in large quantities at that time would have exhausted

good neutron reflector.

³⁸⁸CRITICAL DIMENSIONS OF SYSTEMS CONTAINING U²³⁵, Pu²³⁹, AND U²³³, H. C. Paxton, J. T. Thomas, Dixon Callihan, E. B. Johnson, TID-7028, Los Alamos Scientific Laboratory and Oak Ridge National Laboratory, Los Alamos, New Mexico, and Oak Ridge, Tennessee, June 1964, p. 56.

³⁸⁹Glasstone and Redman, WASH-1037 (Revised), p. 12.

³⁹⁰Paxton, et. al., TID-7028, pp. 56, 59; "Controlling Weapons-Grade Fissile Material," J. Rotblat, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 33 No. 6, p. 39.

³⁹¹Glasstone and Redman, WASH-1037 (Revised), p. 26.

the nation's entire supply (beryllium metal did not become readily available to American industry until 1957).³⁹²

Enough beryllium had been accumulated at Los Alamos by May 1946 to allow critical mass experiments. Louis Slotin, then leader of the M-2 division at LASL, died in May 1946 following a criticality accident involving "a combat-type plutonium core and a beryllium tamper;" this tamper was a "hollow hemisphere of beryllium."³⁹³

Fission bomb tampers made of beryllium or tungsten carbide (chemical formula WC) were considered at Los Alamos as early as 1947.³⁹⁴ Significant fabrication of tungsten carbide began at LASL in May 1944; by the end of the war, pieces weighing up to 300 pounds had been manufactured. On August 21, 1945, another LASL physicist, Harry Daghlion, had been killed in a similar criticality accident involving a spherical 6.2 kg core of delta-phase Pu-239 reflected by tungsten carbide.³⁹⁵

Tungsten carbide, a very hard, very heavy compound, is now alloyed to power saw blades and electric drill bits and is also used for the armor-piercing cores of contemporary anti-

³⁹²MANHATTAN DISTRICT HISTORY, PROJECT Y, THE LOS ALAMOS PROJECT, Vol. I, pp. 103-104, 146-147. The Beryllium Brush Company was one of the first suppliers of large quantities of beryllium to LASL. ("An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 28.)

³⁹³THE HISTORY OF MODERN PHYSICS, Vol. II, p. 326; "A Summary of Accidents and Incidents Involving Radiation in Atomic Energy Activities, June 1945 through December 1955," TID-5360, Daniel F. Hayes, Safety and Fire Protection Branch, Division of Organization and Personnel, USAEC, Washington, D.C., August 1956, p. 4; U.S. Department of Energy Historic Incident Database Entry #446000001, The Pajarito Accident; Hoddeson, et. al., pp. 341-342; OPERATIONAL ACCIDENTS AND RADIATION EXPOSURE EXPERIENCE WITHIN THE UNITED STATES ATOMIC ENERGY COMMISSION, WASH 1192 UC-41, Division of Operational Safety, USAEC, Washington, D.C., Fall 1971, pp. 28, 35.

³⁹⁴"Statistical Methods in Neutron Diffusion," R. D. Richtmyer and John von Neumann, LAMS-551, Los Alamos Scientific Laboratory, 9 April 1947, p. 3.

³⁹⁵THE HISTORY OF MODERN PHYSICS, Vol. II, pp. 107, 224, 300, 301; OPERATIONAL ACCIDENTS AND RADIATION EXPOSURE EXPERIENCE WITHIN THE UNITED STATES ATOMIC ENERGY COMMISSION, WASH 1192 UC-41, Division of Operational Safety, USAEC, Washington, D.C., Fall 1971, pp. 28, 35.

tank ammunition; in the U.S. and the Soviet Union, WC anti-tank ammunition cores are being supplanted by depleted uranium, tuballoy, cores.³⁹⁶

By early 1951, some thought was being given at LASL to using tamper materials other than tuballoy. One suggestion was to test a number of bombs, possibly even without a tamper, and then work up to the most optimum tamper.³⁹⁷ At the end of the year, LASL planned to continue the "investigation of medium and low density tamper materials."³⁹⁸

The 12 KT TUMBLER-SNAPPER How shot at the Nevada Test Site on June 5, 1952 was the first full-scale nuclear test using beryllium as a tamper material; the test device was called the SCORPION.³⁹⁹ Beryllium is now used as the reflector material in most contemporary American nuclear weapons and thermonuclear warhead primaries.

³⁹⁶Tungsten was discovered in 1781 by a Swedish mariner; its name is a combination of two Swedish words meaning "heavy stone." Most tungsten comes from Argentina and the U.S.; it is used principally to harden steel and for electric light bulb filaments. Depleted uranium is also being used as a tank armor. See, for example, "Depleted Uranium Armor for M-1," DEFENSE WEEK, Volume 9 No. 11, March 14, 1988, pp. 1, 2.

³⁹⁷"Discussion with Darol Froman," source unidentified, dated 10 April 1951.

³⁹⁸Letter dated 26 December 1951 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Norris Bradbury, Director, LASL, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 3.

³⁹⁹"Known U.S. Nuclear Tests, July 1945 to 31 December 1985," Robert S. Norris, Thomas B. Cochran, and William M. Arkin, Nuclear Weapons Databook Working Paper No. 2, NWD 86-2, Natural Resources Defense Council, Washington, D. C., February 1986, p. 14; Borden-Walker H-Bomb Chronology, p. 82. McPhee (pp. 93-94) claims the "Scorpion" was fired on a 300-foot tower and was delayed for a day to remove a rat (!) from a conduit; the TUMBLER-SNAPPER How shot was fired on a 300-foot tower and was delayed by adverse weather. (SHOTS EASY, FOX, GEORGE AND HOW, The Final Tests of the TUMBLER-SNAPPER Series, 7 May - 5 June 1952, DNA 6021F, Jean Ponton & Carl Maag, JRB Associates, McLean Virginia, 16 June 1982, p. 118.) The results of the test showed that some "competing effects cancel each other." (Letter dated July 9, 1952 to Senator Brien McMahon, Chairman, JCAE, from Gordon Dean, Chairman, USAEC.)

Advances in Neutron Sources

The neutron source, or "urchin" or initiator, as it is also commonly known, is one of the most important elements of a fission device. It has only one purpose: to generate and release an initial burst of relatively-low energy neutrons at just the right moment to begin and maximize fission reactions in compressed plutonium and/or uranium fissile fuel(s).

The ideal time to begin the fission process is during the very few microseconds just before the nuclear system is fully compressed.⁴⁰⁰ For maximum efficiency, the explosion should occur as closely as possible to maximum supercriticality and the neutrons should be introduced into the compressed core just before optimum supercriticality.⁴⁰¹ Since a nuclear explosion is self-extinguishing due to expansion of the critical mass, the faster the neutrons are generated, the larger the number of fissions and the higher the yield for a given amount of fissionable material.

These neutrons must not appear before the fissile fuel has achieved a critical mass configuration, or maximum compression. If they appear too soon, the device will explode prematurely ("fizzle"); too late, and the resultant explosion will achieve less than maximum yield.

As explained in Volume I, the number of neutrons released by any generation of fission neutrons during a chain reaction is proportional to the number which start the first generation: each fission of a plutonium or uranium nucleus releases either two or three neutrons after being struck by one neutron. To attain the highest possible number of fissioning nuclei, it is extremely desirable to have a large number of neutrons introduced at the beginning of the chain reaction.

In the early days of the U.S. nuclear weapons program, most initiators were complex frangible hemispheres of polonium and beryllium. The first initiators, called "urchins" or "screwballs," were tiny solid spheres of beryllium coated with gold, to absorb alpha radiation, and polonium; this inner sphere was in turn encased by a shell of nickel-plated beryllium whose grooved inner surfaces were coated with more gold and polonium.

The outer hollow Be ball was about 0.4" in diameter, and the whole initiator assembly was levitated and supported by a small bracket within the center of the plutonium core.

⁴⁰⁰Hoddeson, et. al., p. 316.

⁴⁰¹Glasstone and Redman, WASH-1037 (Revised), p. 26.

The modulated initiator was triggered by "jets" of the spherical implosion shock wave.⁴⁰² To prevent premature activation of the initiator by "jets" of the inwardly-focussed implosion shock wave moving along the junction of the two halves of the nearly-solid plutonium core, a thin ring of corrugated gold was placed between the two halves of the core.⁴⁰³

Although the LASL weaponeers tried to leave nothing to chance in the design of the bomb, many early wartime and postwar cores would be slightly prematurely initiated, and studies of this particular type of preinitiation problem began during Operation SANDSTONE in 1948.

Another more-advanced initiator design considered at LASL during the war used cone-shaped holes, in lieu of grooves, to use Munroe effect shock wave focusing to break a nickel plating between the polonium and beryllium and permit more-timely mixing and neutron generation.⁴⁰⁴

"Urchins" were also used in postwar MK III and MK 4 bombs.⁴⁰⁵ Each "urchin" contained 50 curies, or just over 11 milligrams, of polonium-210.⁴⁰⁶ By June 1945, the Monsanto Corporation's polonium-producing facility near Dayton, Ohio was producing 35 curies per week of polonium-210; production had started in April 1944 at a rate of 2.5 curies per month; this was increased to six curies per month by summer 1944, and to 10 curies per month at the beginning of 1945.

In December 1944, Monsanto had received an emergency request for 20 curies a month; in February 1945, the company was asked to provide 100 curies a month by June 1945 and 500

⁴⁰²Hoddeson, et. al., pp. 280, 293, 294, 317, 407.

⁴⁰³Rhodes, TMAB, p. 579; Rhodes, DARK SUN, pp. 193-194; "Iraq's Bomb Program": A Smoking Gun Emerges." SCIENCE, Vol. 254, November 1, 1991, pp. 644, 645; Document #13, Moskovskaya Pravda, "Hunters of Nuclear Secrets," Mikhail Makhlin, June 1991, with additions from Visgin, V. P., ed. 1992. At the Source of the Soviet Atomic Project: The Role of Espionage, 1941 - 1946. Problems in the History of Science and Technology 3:97, pp. 126-129.

⁴⁰⁴Rhodes, DARK SUN, pp. 187-188.

⁴⁰⁵Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 16.

⁴⁰⁶Williams, p. 191; Rhodes, DARK SUN, p. 168.

curies a month by the end of 1945. These large quantities of polonium were required for new implosion initiator designs.⁴⁰⁷

In implosion weapons, these initiators were placed in the center of the pit; when crushed by implosion, the mechanical action of the intermixing of the two elements caused a nuclear reaction to occur. Alpha particles, helium nuclei, consisting of two protons and two neutrons, released by the polonium bombarded the beryllium, producing about 30 neutrons per million alpha particles. This reaction released some rather low-energy neutrons to start the nuclear chain reaction in the surrounding compressed fissile fuels.

These initiators had to be rugged enough to withstand unreasonably rough handling, accidental drops, airplane vibrations, and, in the case of initiators for gun-type weapons, the shock of firing.⁴⁰⁸ LASL did not finally decide to use initiators in the LITTLE BOY until mid-March 1945; lower-strength units, with less polonium, were used to reduce background radiation that might preinitiate the U-235 target assembly in the bomb, and leave plenty of polonium for implosion weapons.⁴⁰⁹

Because the polonium was radioactive with a half-life of 138.5 days and decayed rapidly, these initiators had to be replaced periodically (they were usually stored apart from the fissile cores).⁴¹⁰ In one year, the alpha activity of the polonium, and hence the rate of neutron generation of the initiator, would have decreased to just 18% of its original value; in two years, this "strength" would be down to only 2.6% of its initial value.⁴¹¹

These bimetallic initiators were an expensive and scarce inconvenience, and by the spring of 1947, in very short supply.⁴¹² Since the radioactive decay of one curie of

⁴⁰⁷Hoddeson, et. al., pp. 125, 309, 310.

⁴⁰⁸Hoddeson, et. al., p. 125.

⁴⁰⁹Hoddeson, et. al., pp. 125, 126.

⁴¹⁰In a memorandum dated 1 March 1946 from Co. K. D. Nichols, USA, to Maj. Gen. L. R. Groves, USA, subject: Items to be Discussed with General Eisenhower, Nichols states that "polonium is an essential ingredient in atomic bombs" and that "it is not possible to store polonium."

⁴¹¹Glasstone and Redman, WASH-1037 (Revised), p. 20.

⁴¹²Hewlett and Duncan, ATOMIC SHIELD, pp. 41, 42, 59-62; "Toward Armageddon: The Foundations of United States Nuclear

polonium generated heat equivalent to 0.1 watts, large initiators, such as those used during Operation GREENHOUSE in 1951, were quite warm.⁴¹³

In addition, the wartime Po-Be sources were also vulnerable to moisture. During a mishap at Los Alamos in January 1949, several initiators were accidentally immersed in water for several days. The protective nickel coatings on them were ruined, requiring the re-fabrication of each source (in the presence of water, an electrolytic reaction between nickel and beryllium quickly disintegrated the beryllium shell). Protective watertight containers were devised; these containers were also crush-resistant, for obvious reasons.⁴¹⁴

By early 1945, even before the TRINITY test, LASL foresaw a requirement for better future initiators and for full-scale atomic bomb tests to determine the behavior of initiators:

Initiators. It is particularly important to develop an initiator which acts at an optimum time during the nuclear explosion. A lower background (radiation rate) for the initiators is desirable. This may mean the use of less Po or other neutron-producing material. In order, however, that this be done, a clearer understanding of the action of the initiator, and particularly of its efficiency, is needed.

Other methods of testing initiators should be investigated. The possible firing of atomic bombs for testing initiators, and particularly bombs containing a large amount of hydrogen, should be studied. Direct tests of initiators using very fast counters and large amounts of Po or other neutron-producing material should be considered and possibly studied in some detail.⁴¹⁵

Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, p. 89.

⁴¹³WT-102, p. 21.

⁴¹⁴Memorandum dated January 18, 1949 to Distribution from Simon Shlager, H-1, Subject: Care in Using Po-Be Sources.

⁴¹⁵"Recommendations for Research in Nuclear Physics," R. F. Bacher and R. R. Wilson, Los Alamos Scientific Laboratory, no date but ca. early 1945, pp. 6, 7.

Klaus Fuchs and Rubby Sherr of LASL applied in June 1946 for a patent on a "timed neutron source" to provide, in an implosion-type atomic bomb, a neutron source which would be activated by the outgoing shock wave (as opposed to the incoming implosion shock wave) to release neutrons at optimum density of the compressed core.⁴¹⁶

In October 1946, Dr. Edward Teller proposed a thorough program of initiator research and improvement:

It is clearly of very great interest to obtain better data on the functioning of the initiator and in particular to find out whether one can construct an initiator that will work when the shock is moving outward rather than at the early time when the shock is moving inward. Initiators more complicated than used at the present time might be quite difficult to construct.

It might not be wise, therefore, to push this program very strongly unless men can be found who are very eager to work on it. Consideration should be given to the possibility of eliminating initiators altogether.

Since the present initiators cannot be stored, a large usable stockpile of bombs would necessitate a very considerable program of polonium production and initiator fabrication. Tests on [deleted] implosions or [deleted] together with appropriate calculations, may prove that initiators for one of these devices are unnecessary.⁴¹⁷

At its second meeting in early February 1947, the GAC noted "that more work is required on the initiator problem," and that current initiators were made from unsatisfactory materials, because of their cost, scarcity, and short lives, and that some possible but not particularly promising substitute materials were being considered. This was an

⁴¹⁶Attachment to letter dated June 30, 1952 to William L. Borden, Executive Director, JCAE, from Roland A. Patterson, Chief, Patent Branch, USAEC.

⁴¹⁷Memorandum from Edward Teller dated October 1, 1946.

important problem that warranted additional research not only on materials and design, but also on initiator behavior.

At this time, the AEC and LASL had begun taking "emergency corrective" steps to remedy "the current deficiency in initiators and to expedite the long-term program." Research directed toward finding better initiators was continuing.⁴¹⁸

In order to guarantee sufficient polonium, one Hanford reactor was being held in stand-by condition, primarily to guarantee that a reactor of sufficient size to produce adequate amounts of polonium for initiators would be available at all times.⁴¹⁹

Dr. Teller's idea of eliminating initiators altogether was later rejected by the Weapons Subcommittee to the General Advisory Committee, which feared that weapons without initiators would result in dubious efficiencies and unacceptable variations in yield.⁴²⁰

At its third meeting late in March, the GAC noted that the Monsanto Company, which had provided the initiators for the wartime bombs, was now working two 10-hour shifts daily to complete construction on a new plant in Miamisburg, Ohio, and that the AEC was ready to negotiate a four-year contract for continued initiator production.

The GAC also discussed the substitution of actinium for the polonium then being used in bimetallic initiators. Actinium was made by irradiation of radium and might require large amounts of radium.⁴²¹

The GAC noted:

We recommend that the directive to the Los Alamos laboratory should request the production of at least 15 initiators a

⁴¹⁸"Long-Term Commission Agenda," undated, but probably ca. early 1947.

⁴¹⁹Draft Minutes of the Second Meeting of the General Advisory Committee, February 3-4, 1947, pp. 4, 6, 9.

⁴²⁰"Proposed Agenda for GAC Weapons Sub-Committee," March 22, 1947; "Notes on Weapons Sub-Committee Agenda," April 4, 1947.

⁴²¹Draft Minutes of the Third Meeting of the General Advisory Committee, March 28-30, 1947, pp. 2, 3. By October 1947, at least one change in Urchin design had been proposed by LASL. (Atomic Energy Commission, Minutes of Meeting No. 111, October 15, 1947.)

month starting in May 1947. We understand that this is approximately the limit of production with present facilities. This production may be drastically reduced when Monsanto is able to take over.⁴²²

Following the TRINITY atomic bomb test in July 1945 in the Jornada del Muerto desert near Alamogordo, New Mexico, initiators were tested at locations on the outer periphery of the TRINITY crater. These tests were conducted in tightly-sealed 14-foot wide, 12-foot high octagonal chambers with two-foot thick concrete walls, buried beneath 30 feet of earth (a 16-foot high mound over the chamber which was buried 14 feet below ground level) in order to prevent any atmospheric venting of radioactive material. Observations of the explosions were made remotely above ground 200 yards away; access to the buried chambers was via four-and-a-half foot diameter shafts.⁴²³

Plans were made for three large-scale initiator tests using a 500-curie polonium "pit" in a 25 lb. ball of HE. Two experiments were conducted, including one at 11:25 PM on July 8, 1947,⁴²⁴ and the third chamber was never used, the test having been canceled. Due to a faulty switch, the second of the two tests that were conducted was a "dud."

⁴²²Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p.

⁴²³Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, pp. 22, 23; memorandum dated February 17, 1947 to D. P. MacMillan from G. L. Williams, Subject: Resistance of Underground Chamber to Internal Shock (this memo includes a dimensioned scale-drawing of a buried test chamber); memorandum dated 22 April 1947 to Darol Froman from D. P. MacMillan, Subject: Some Factors Concerning Safe Confinement of Active Material in M-3 Experiment; memorandum dated May 12, 1947 to Darol Froman from Louis H. Hempelmann, Subject: Comments on the M-3 Trinity Experiment; memorandum dated June 19, 1947 to Dr. Beller from George Littlejohn and Charles Blackwell, Subject: Monitoring of TRINITY and Surrounding Vicinity; undated memorandum to Darol Froman from J. C. Clark, Subject: Information from OSRD #6053 (AM 18) Bulletin.

⁴²⁴Memorandum dated July 10, 1947 to Mr. Bradbury from Dr. Beller, Subject: Results of Observations made by the Health Division during the Special Test at TRINITY; memorandum dated July 10, 1947 to Commander of Troops from Dr. Beller, Subject: Special Test at TRINITY; memorandum dated July 24, 1947 to Mr. Bradbury from Dr. Beller, Subject: Monitoring of TRINITY Site.

The unexpended test initiator remained as a "sleeping beauty" for the next 20 years until March 1967 when its bunker was excavated and the device was destroyed in place by a 100 lb. explosive charge put into the bunker; the chamber was then reburied to ensure the safety of future visitors to TRINITY site.⁴²⁵

In November 1947, Los Alamos authorized the construction of a local site (Technical Area 33, or TA-33) at the laboratory for all future initiator proof testing. (Sandia Base in Albuquerque had been considered as an alternate site.)⁴²⁶ At this time, a design for a specific type of nuclear reactor dedicated to polonium production was under active consideration by the AEC.⁴²⁷

In the spring of 1947, the AEC expressed a desire for further initiator research, including a determination of the minimum strength of "aged" FAT MAN type initiators:

The Commission desires that emphasis be placed upon the development of a long-life initiator. The enlisting of assistance from

⁴²⁵Office Memorandum dated March 6, 1952 to Paul Ager, Santa Fe Operations Office, USAEC, Albuquerque, from T. L. Shipman, M.D., LASL, Subject: Trinity Crater; Szasz, p. 165; memorandum dated June 20, 1956 to Paul A. Wilson, Manager, AEC LAAO, from N. E. Bradbury, Director, LASL, Subject: Trinity Site; Memo for Record dated 23 February 1967 from Don J. White and John P. McDougall, Subject: Trinity Site Radiation Survey; letter dated March 8, 1967 to Commanding General, White Sands Missile Range, Las Cruces, New Mexico, from L. P. Reinig, Engineering Department Head, LASL; "Sleeping Beauty Project," ENG-1-Rp-3, C. W. Trask, report attached to memorandum dated June 1, 1967 to L. F. Giso from N. E. Bradbury, Subject: Sleeping Beauty; "Sleeping Beauty Awakes" (sic), photo display at McDonald Ranch House, near Trinity site, October 1991.

⁴²⁶Atomic Energy Commission, Disposition of Alamogordo Test Area, Note by the Secretary, June 18, 1948, pp. 2, 3, 6; Office Memorandum dated March 6, 1952, from T. L. Shipman, MD, LASL, to Paul Ager, Santa Fe Operations Office, USAEC, Albuquerque, Subject: Trinity Crater; memorandum dated November 7, 1947 to Manager, Santa Fe Operations Office, USAEC, from R. W. Betts, Associate Director for Administration, LASL, Subject: Requirements for TRINITY.

⁴²⁷Draft Minutes of the Seventh Meeting of the General Advisory Committee, November 21-23, 1947, p. 6.

other suitable Commission facilities in this work is encouraged.

Some redesign of the gun-type initiator may be desirable for a subsurface (bursting) weapon. In so far as is practicable, such work should be scheduled to dovetail into the subsurface weapon program of the (Navy's) Bureau of Ordnance.

It is requested that the (Los Alamos Scientific) Laboratory recommend a value for the minimum strength for combat use of the present initiator in FAT MAN weapons.⁴²⁸

Production of initiators at Los Alamos continued at the low rate of about two per month into 1949; others were manufactured during this period at the Monsanto Corporation's Mound Laboratory fabrication facility at Miamisburg, Ohio.⁴²⁹ A longer-lived initiator using something other than polonium was proposed in the summer of 1948; since the fall of 1947, actinium -- with a half-life of 13.5 years -- had been considered as an alternate alpha particle source.⁴³⁰

An alternate means of initiation was via an external initiator. The use of external initiators would eliminate the short half-life of existing internal initiators, and provide more easy activation and better timing. On the other hand, external initiators directed a smaller fraction of the neutrons emitted to the central fissioning region and required highly-advanced timing controls.⁴³¹

⁴²⁸Draft Memorandum from the General Manager to the Manager, Santa Fe Directed Operations, p. 4, Appendix "A" to AEC 99, Atomic Energy Commission, Weapons Program of the Los Alamos Laboratory, May 14, 1948.

⁴²⁹Los Alamos Scientific Program for 1947, p. 3; Memorandum to File, September 3, 1947, from Robert F. Bacher, p. 2; Los Alamos Scientific Laboratory Technical Program for Calendar Year 1949, p. 10.

⁴³⁰Memorandum dated July 13, 1948, from Carroll L. Wilson, General Manager, USAEC, to Carroll L. Tyler, Manager, Santa Fe Directed Operations, USAEC, Subject: 1948 Program of the Los Alamos Laboratory, p. 2; Los Alamos Scientific Laboratory Technical Program for Calendar Year 1949, p. 6; Memorandum to File dated September 3, 1947, from Robert F. Bacher, Subject: Suggestions Made at Los Alamos during the Past Two Months, p. 2; Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 21.

The Los Alamos technical research program for 1949 included:

A program of study on the behavior of initiators of the present type in order that their strength specifications may be set at the lowest acceptable limit consistent with suitable probabilities of performance. Associated with this program are related programs to determine whether more effective or efficient initiator designs or designs presenting greater simplicity in fabrication exist.⁴³²

Los Alamos also began searching for replacement materials for the polonium heretofore used in initiators. Fairly specific results of this search were expected by the end of 1949.

At this time, two promising research programs were underway to provide longer-lived alternatives to the polonium then used in bomb initiators. The more desirable choice was actinium-227. The other was production of polonium-208 in cyclotrons rather than in reactors.⁴³³

In the meantime, a new "TOM" initiator, first put into production at LASL in January 1950,⁴³⁴ was incorporated into the MK 5 bomb. The new polonium-beryllium initiator, which did not have a levitated sphere in its center, had half the linear dimensions and was only 2/3 the size of the standard unit at that time.⁴³⁵ A "TOM" initiator was first tested in May 1950. The "TOM" was first proposed for the BOOSTER weapon, but it was also applicable to standard fission warheads.⁴³⁶

⁴³¹Hoddeson, et. al., draft manuscript, p. 307.

⁴³²Letter dated 7 September 1948 from N. E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Office of Santa Fe Directed Operations, U.S. Atomic Energy Commission, p. 7.

⁴³³Letter dated July 15, 1949 to David E. Lilienthal, Chairman, USAEC, from J. R. Oppenheimer, Chairman, GAC.

⁴³⁴Memorandum for the Files dated November 8, 1949, from Walter A. Hamilton, subject: Inquiry into the Aspects of a Superweapon Program, p. 4.

⁴³⁵Memorandum for the Files dated November 10, 1949 from Walter Hamilton, subject: Inspection Trip, October 24 - November 4, Borden and Hamilton, pp. 1, 2.

⁴³⁶Memorandum dated January 27, 1950 to Dr. John H. Manley,

A low production rate of two "TOM" initiators per month continued through most of 1950; large-scale production and stockpiling of the new initiator began in October.⁴³⁷ This initiator was far more efficient in terms of neutrons per gram of polonium.

Production of older "Urchin" type initiators also continued during 1950; LITTLE BOY type initiators, the ABNER, were to be produced only under emergency, i.e., wartime, conditions.⁴³⁸ By November 1950, approximately five "LC" type initiators for the MK 8 weapon were to be produced per month, and would be available for emergency use only.⁴³⁹

A "TOM" was first used in a full-scale nuclear test during the Baker-1 shot of Operation RANGER in Nevada in late January 1951.⁴⁴⁰ By the end of May 1951, all new bomb cores were being made to accommodate the new initiator, and cores already in the stockpile were being converted to accept the "TOM" at a rate such that conversion would be complete by the end of the year.⁴⁴¹

Secretary of the General Advisory Committee, USAEC, from Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC; letter dated 11 June 1953 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

⁴³⁷Letter dated March 10, 1950, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for 1950, pp. 1, 3; Los Alamos Laboratory Program for Calendar Year 1951 and Fiscal Year 1952, November 17, 1950; Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 14.

⁴³⁸Letter dated December 9, 1949 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, p. 3.

⁴³⁹Los Alamos Laboratory Program for Calendar Year 1951 and Fiscal Year 1952, November 17, 1950.

⁴⁴⁰Part III, Weapons, Progress Report to Joint Committee on Atomic Energy, December 1950 through May 1951, May 29, 1951, p. 5.

⁴⁴¹Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 5.

The Monsanto Company's Mound facility near Dayton, Ohio started work in May on the use of proto-actinium as an initiator material. Monsanto had also re-arranged its assembly line, doubling the potential output of "TOM" initiators.⁴⁴² By the fall of 1951, stockpiled initiators had a useful life of at least four years.⁴⁴³

In April 1952, Monsanto had successfully separated about 10 grams of actinium and expected to fabricate soon some special test initiators using the material.⁴⁴⁴ This research was terminated about a year later, due in part to a major increase in the estimated cost of an actinium-production pilot plant.⁴⁴⁵

Development of External Initiators

The possibility of an external neutron generator was raised at Los Alamos soon after the war ended. External neutron sources offered several advantages over core-resident "golf balls." Many more neutrons could be generated to begin the initial fission reactions, increasing total yield for a given amount of fissionable material.

The energy level of these neutrons could be controlled precisely to guarantee yield reproducibility. The timing of their release could also be controlled within a fraction of a microsecond to maximize the efficiency of nuclear reactions in the compressed pits. Bombs using polonium-beryllium "golf balls" tended to predetonate because the fissionable materials in them became critical as much as 10 microseconds, 1,000 "shakes" or fission generations, before the initiator began releasing neutrons.⁴⁴⁶

⁴⁴²Memorandum for the File dated May 7, 1951 from Wayne P. Brobeck and Walter A. Hamilton, subject: Conference with Walter J. Williams, Deputy Manager, AEC, Friday, May 4, 1951.

⁴⁴³Memorandum to File dated September 4, 1951 from Walter A. Hamilton, subject: Conference with Walter Williams.

⁴⁴⁴Memorandum for the Files dated April 4, 1952 from Wayne Brobeck and Hal Bergman, subject: Weekly Conference with Mr. Walter Williams.

⁴⁴⁵Memorandum dated April 30, 1953 to R. W. Cook, from F. K. Pittman, subject: Analysis of the Initiator Program.

⁴⁴⁶OPERATION SANDSTONE, NUCLEAR EXPLOSIONS - 1948, Scientific Director's Report - Volume I, General Report, Eniwetok Proving Ground, pp. 63-65.

Another reason for the development of external neutron sources was the advent of "boosted" fission warheads. The center of the pit was now required as a fusion chamber, to provide a location for fusion fuel. An external source of neutrons would have to be provided. This need was met by the development of compact high-voltage neutron generators, which work by accelerating small amounts of tritium or deuterium.⁴⁴⁷ External neutron generators, while expected to increase the yield of existing warheads and eliminate the problem of short initiator stockpile life, were indispensable if hollow cores were to realize their full potential.⁴⁴⁸

These external neutron generators work by accelerating tritons into a deuterium target.⁴⁴⁹ When the tritium and deuterium nuclei collide and fuse -- in this case, a difference in electrical potential is used instead of the high temperature of a fission explosion to give momentum to the nuclei -- neutrons on the order of several tens of millions are released.⁴⁵⁰

These large numbers of neutrons initiate many fission reactions at the start of the fission chain reaction explosion, thus increasing the efficiency of the pit, and permitting more of the fissionable material in it to be "burned" before it expands and disassembles.

Deuterium-tritium neutron generators are also being used in commercial applications, such as oil well prospecting; a new neutron radiography testing unit developed by the Vought Corporation employs D-T collisions to generate neutrons.⁴⁵¹

⁴⁴⁷Morland, "The H-Bomb Secret;" "Tritium: The New Genie," Howard Morland, THE PROGRESSIVE, Vol. 43 No. 2, February 1979.

⁴⁴⁸"Military Objectives in the Use of Atomic Energy," 29 December 1950, p. 5.

⁴⁴⁹Glasstone and Redman, WASH-1037 (Revised), p. 21.

⁴⁵⁰A similar technique was used at Los Alamos during World War II to ascertain the cross-section for the D-T (deuterium-tritium) reaction as an early part of research on materials for the hydrogen bomb. See Volume III.

⁴⁵¹AVIATION WEEK & SPACE TECHNOLOGY, May 16, 1983, p. 60; "Company's Shipments to Mideast In Question," column by Jack Anderson in San Francisco Chronicle, October 22, 1991, p. A31. The Anderson column cited details shipment of pulse neutron generators by a Dallas-based oil-field service company to Libya.

The advent of these small, external high-voltage neutron generators resulted in both safer weapons and more sophisticated variable-yield weapons. Yields for some U.S. fission weapons of the late 1940s and early 1950s had been varied by using interchangeable pits with varying amounts and geometries -- physical assemblies -- of nuclear materials and high explosives. Now yield could also be varied by changing the timing and strength of the charges on the external initiators.

After World War II, when polonium-beryllium initiators were still being used, weapons were frequently made "safe" from unintentional nuclear detonation by keeping the initiators physically separate from the fissile cores. Warheads could be armed and safed by techniques of manual or automatic inflight insertion and extraction (see Volume VIII). The tritium accelerators could also act as safing devices: if they did not fire, or failed to fire at the right moment, the fission reaction would not begin.

In addition, by varying the amount of tritium and deuterium used, or by changing the electrical potential between ends of the accelerator, thus altering the velocity of the deuterium or tritium nuclei and the efficiency of the neutron-generating fusion reactions, the number of neutrons released can be controlled.

This in turn affects the initial efficiency of the fission chain reaction, and the total yield: in a multistage thermonuclear weapon, the yield of each stage depends directly on the yield of the preceding stage. This method of variable yield is used in some contemporary U.S. nuclear warheads; typical yield ranges of variable-yield weapons are from a few tens of kilotons to several hundred kilotons.

External neutron generators were first proposed in December 1949 by the Los Alamos Scientific Laboratory Technical Program for 1950; specifically mentioned were initiators employing thermonuclear materials.⁴⁵² Planning started for a new division at Los Alamos which would be dedicated to the development of a high-voltage initiator.

One important consideration in the development of external fission initiators was the problem of predetonation. Perhaps the single most important factor in a nuclear reaction in a weapon is when the reaction starts. The reaction must not

⁴⁵²Letter dated December 9, 1949 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, p. 2.

start too soon, or optimum compression may not have been reached. Too late, and conditions are also not optimal; in either case, the device will probably not achieve its intended yield.

In early 1949, Norris Bradbury, then director of Los Alamos, defined this problem:

While we are considering this single picture of an atomic bomb, I wish to draw your attention to another characteristic of atomic weapons which is not widely realized. If we are in the process of assembling an atomic weapon from a slightly subcritical state to one as supercritical as possible, there is clearly a small span of time when the weapon is beyond critical but not yet as supercritical as it will get.

During this time, the weapon is essentially ready to explode, and the chance presence of a stray neutron might be adequate to initiate the chain reaction. If this were to happen, the weapon would, of course, start to expand and dissipate with the result that a much smaller fraction of active material would be consumed than had been anticipated.

This phenomenon is known as predetonation (because the weapon predetonates before it is supposed to), and while its probability may be reduced by careful design, it, nevertheless, cannot be reduced to zero. There is always a chance that any given weapon will detonate with less than its designed and characteristic yield.

The degree of deficiency is most likely to be small, but it can be fairly large. It should be clear that this is not a defect in behavior or design of any given weapon, but rather a characteristic of all atomic weapons of this type and a result of limitations imposed by nature.

Parenthetically, as far as we know, we have not yet observed a serious case of predetonation in any weapon yet exploded, so

you may assume that the probability of this occurrence is not high.⁴⁵³

In September 1950, the General Advisory Committee recommended to the AEC that LASL "pursue without delay" the development of an "external pulsed initiator."⁴⁵⁴ Soon afterwards, the General Electric company signed a contract with the AEC to furnish six betatron particle accelerators to be used in connection with the pulsed initiator program, at a cost of \$300,000.

GE hoped to gain experience so as to be in a better position to secure a future contract to provide pulsed initiators for the entire U.S. weapons stockpile.⁴⁵⁵ External initiation was considered probably feasible for cases in which timing was not a sensitive factor.⁴⁵⁶

Six months later, external initiation via betatron, an electron accelerator used to produce high-energy penetrating radiation for diagnostic purposes, was considered probably feasible for cases in which timing was not too sensitive.⁴⁵⁷

Betatrons had been used during Manhattan Project days to measure implosion efficiency. A betatron generated gamma rays, which, when made sufficiently powerful, could be used to generate neutrons. Since the average binding energy of a nucleon in a nucleus is about eight MeV, a photon of this energy level will eject a neutron from most nuclei.⁴⁵⁸

⁴⁵³"The Potentialities of the Atomic Bomb," N. E. Bradbury, 4 January 1949, pp. 6, 7. At least one of the SANDSTONE cores may have been slightly predetonated. This problem of predetonation also applied to the fissionable "sparkplugs" in the centers of thermonuclear weapon secondary and tertiary stages.

⁴⁵⁴Letter dated September 13, 1950 from J. Robert Oppenheimer, Chairman, GAC, to Gordon Dean, Chairman, USAEC.

⁴⁵⁵Memorandum to File dated November 27, 1950 from W. L. Borden, Executive Director, JCAE.

⁴⁵⁶Minutes of the Twenty-Fifth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 15, 16, and 17, 1951, p. 3.

⁴⁵⁷Minutes of the Twenty-fifth Meeting of the General Advisory Committee to the Atomic Energy Commission, March 15, 16, and 17, 1951.

⁴⁵⁸Glasstone and Redman, WASH-1037 (Revised), p. 20.

By the end of May 1951, a specially-designed betatron was in operation and was giving required neutron yields.⁴⁵⁹ Also in May, a GE betatron unit meeting weight and current specifications was completed and shipped to LASL for test. A proposed T-D initiator also looked like a promising development.⁴⁶⁰

During 1951, several devices operating on different principles were being developed for use as external initiators. At least one of them appeared rather promising; although development had not reached a point where a production or test date could be predicted, a suitable device might be available in about two years, i.e., by 1953. A laboratory model might be ready for testing in the spring of 1952 in Nevada during Operation SNAPPER.

Space limitations in existing stockpiled bombs might preclude incorporation of external initiators. These weapons had not been designed for the later incorporation of additional components.⁴⁶¹

A practical external initiator for nuclear weapons was developed by the W-5 group at the Los Alamos Scientific Laboratory between the spring of 1951 and the fall of 1954.⁴⁶² The W-5 organization was part of the "W" weapons research division at LASL that was responsible for advanced design phases of the mechanics and dynamics of methods of initiating nuclear energy releases; this group name should not be confused with the W-5 warhead.

The successful development of an external initiator made "sealed pit" weapons possible, in which short-lived internal initiators were not required, and went a long ways toward realizing the so-called "wooden bomb" concept of a long-lived, low-maintenance stockpile weapon.

⁴⁵⁹Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 18.

⁴⁶⁰Minutes of the Twenty-Sixth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 8, 9, and 10, 1951, p. 27.

⁴⁶¹"Budget Bureau Questions - 1953 Budget, Program 3000 - Weapons (and related Construction), Products, Storage and Surveillance," undated, but ca. early to mid-1951, p. 10.

⁴⁶²THIRTY YEARS AT PAJARITO CANYON SITE, Hugh C. Paxton, LA-7121-H, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, March 1978, p. 6.

By June 1951, an external initiator was expected to be ready for test by the spring of 1952;⁴⁶³ at the end of the year, LASL planned to continue studies of:

.. the general improvement of the technique of initiating fission reactions through production of special types of steady and modulated neutron sources, through increased understanding of the phenomena taking place in the activation of initiators of current designs, through external sources of neutrons or gamma rays, and through consideration of alternative materials for initiators.⁴⁶⁴

In mid-December 1951, progress was being made in the development of initiation devices, including an electronic device which could be located outside the HE sphere of implosion-type weapons. At least one type of external initiator was to be ready for full-scale testing in the spring of 1952. The chief advantage of the external initiator was expected to be its ability to start the nuclear fission reaction at a more-nearly optimum time during the implosion process.⁴⁶⁵

By February 1952, LASL had made tentative plans for a pair of initiator tests during the forthcoming TUMBLER-SNAPPER series in Nevada; the test bombs would be MK 5 models, a prototype of which had been fired earlier during Operation GREENHOUSE:

Two shots in Operation SNAPPER are strictly to determine variation of the yield of a typical implosion system with time of initiation. Theoretical calculations indicate that the yield should be increased

⁴⁶³Notes on J-7 Meeting, 9 June 1951, Discussion of Future Tests, p. 6.

⁴⁶⁴Letter dated 26 December 1951 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Norris Bradbury, Director, LASL, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, pp. 3, 4. The term "modulated" in the context of initiation refers to an initiator which would be capable of either varying the energy of its generated neutrons, or of turning the neutron stream on or off at will. (Hoddeson, et. al., p. 430.)

⁴⁶⁵Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 14.

appreciably if the injection of the neutrons which start the nuclear reaction could be delayed beyond the time at which a TOM initiator would normally perform this function.

One shot will be made with a TOM initiator, and the other will be initiated at a later time by an external device. This requires some controllable means of injecting neutrons at the desired moment.

If these tests indicate that bomb performance varies with initiation time in the manner predicted by theory, then there will be considerable interest in making practical use of increased nuclear efficiency resulting from optimum initiation time. From this may follow a definite engineering program where a [deleted] or OASD for use with atomic weapons in place of the customary initiator will depend upon a number of factors: the success of the SNAPPER tests and perhaps further experiments which demonstrate that appreciable increase in efficiency occurs as a result of delayed initiation.

The research phase on external initiation is expected to go on until January 1953.⁴⁶⁶

1. External Initiation. (Two shots)

It is planned to fire two tower shots to study the effect of changing the time of initiation. The bomb model to be used is [deleted]

The problem of delayed initiation is an old one which [deleted]

[deleted]

For any given implosion bomb design, the expected yield can be calculated as a

⁴⁶⁶Summary of Secret letter to Col. Fields, Director, Division of Military Application, USAEC, dated 9 February 1952 from Norris Bradbury, Director, LASL, in "Notes on Operation TUMBLER/SNAPPER from the Files of the Santa Fe Operations Office, U. S. Atomic Energy Commission" (undated but probably ca. summer 1952).

function of the time of initiation. The curve obtained by plotting expected yield against initiation time rises from a very low yield, at a time corresponding with the assembly becoming critical, passes through a maximum at a time slightly before maximum compression, and decreases to zero as the assembly decompresses and springs apart.

The total width of such a curve is approximately equal to the total time the assembly is supercritical [deleted]

The purposes of this (proposed SNAPPER) experiment are several. In the first place, the theoretically-predictable gains in efficient utilization of fissionable material by properly timing initiation are quite large in certain cases.

However, the development and engineering of an external initiator appropriate for use in operational, stockpile bombs is a long and expensive undertaking and the inclusion of such a device in a bomb will add to its bulk and weight and probably detract from its general reliability. It seems prudent to determine experimentally if these large gains in efficiency are attainable in practice before deciding whether or not to undertake the work necessary to introduce the external initiator into the national stockpile.

In the second place, although considerable experimental information about core compressions has been obtained in recent test operations (Author's note: especially during Operation RANGER in early 1951), essentially no experimental data exist on how the compression changes with time in any single model, especially at times near the maximum compression.

It is most desirable, for example, to find out whether or not core material continues to be compressed in accordance with theory after central (Author's note: i.e., internal) initiation time. No test has yet been made with (delayed) initiation.

In the third place, a successful demonstration of the practicability of external initiation will open up a method

for study and eventual application to radical designs of implosion systems not well suited to central initiation. Such designs may include completely hollow-implosion and multiply-levitated systems.

All of the above purposes can be accomplished by determining experimentally the yield-vs-initiation time curve and indeed the objective of the planned experiment is just to obtain points on this curve. At least three points are needed to define this curve experimentally, even in broad outline.

The curve may be thought of as defined roughly by its width in time, the height of its maximum and the time at which the maximum occurs. Thus, if these SNAPPER tests are successful, it is planned to carry out at least one more external initiator shot (during) a future test operation. It is advantageous to study the results of the first two shots rather thoroughly before planning the details of the third and, consequently, only two are planned for SNAPPER.

The reasons for the selection (of the MK 5 HE-pit assembly) merits some attention. The experiment really consists of making single observations on different bombs in order to determine the sequence of conditions (assumed) to exist in a single specimen. Consequently, the different samples should be as nearly identical as possible.

Transit-time measurements and other data [deleted] indicate that timing variations, test-to-test, would introduce considerable uncertainty in the interpretation of the results [deleted]

Thus, it was decided to use the more uniform (MK 5 HE-pit assembly) system. Care has been taken in selecting the HE charges for these tests and a group of pits has been selected which are identical within very close tolerances. Although the form of the experiment requires the different specimens to be very nearly identical, the major gains in practical application of external initiators to stockpiled bombs do not appear

to depend upon maintaining closer tolerances than those in current production models.

[deleted]

These two shots must be performed upon a tower because the external initiator equipment is in such an early stage of development that it cannot be assembled into any existing ballistic case and even for the shot with the TOM initiator, the same timing, triggering, and auxiliary equipment must be used in order tie the initiation times of the two shots together.⁴⁶⁷

A prototype initiator was tested by scientists from the LASL W-5 group during the George shot of Operation TUMBLER-SNAPPER at the Nevada Test Site in late May and early June 1952.⁴⁶⁸ Results of the test showed that external initiation appreciably increased yield, although not as much as expected or possible: the core was slightly postdetonated.

The Fox and George shots of Operation TUMBLER-SNAPPER were tests of devices named XR1 and XR2, respectively; the Fox shot used an internal TOM initiator in a MK 5 HE assembly and George used a betatron external initiator in the same HE-core geometry. Both shots were tests of initiation time versus yield; the George shot was initiated at a known time.

After TUMBLER-SNAPPER, LASL continued studying the feasibility of achieving the same effect through modification of internal initiators, as well as continuing studies of the more costly method of providing external initiators, which were not then adaptable to stockpiled weapons.⁴⁶⁹

⁴⁶⁷Letter dated February 11, 1952 to Col. Kenneth E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL, Appendix A to Atomic Energy Commission, "Proposed Experimental Devices - SNAPPER," Report by the Director of Military Application, attachment to AEC 505/13, March 10, 1952, Note by the Secretary, Proposed Experimental Devices for Operation SNAPPER.

⁴⁶⁸Memorandum dated March 17, 1952 for Mr. James S. Lay, Executive Secretary, National Security Council, from Gordon Dean, Chairman, USAEC, Subject: Request of Presidential Approval for TUMBLER-SNAPPER.

⁴⁶⁹TEST OF AN EXTERNAL INITIATOR, Lester M. Baggett, Robert S. Kike, Robert E. Dunaway, et. al., WT-557, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, December 1952;

By mid-November 1952, development by LASL of a deuterium-based external initiator had the highest priority of any development in the fission weapon field; the lab planned to test such a device in Nevada during Operation UPSHOT in the spring of 1953 or during Operation DOMINO in the fall of 1953.⁴⁷⁰

At the end of the year, both improved internal initiators and external initiators were being tested; existing external initiators would probably not be adaptable to stockpiled weapons and would require the design of new pits and cores.⁴⁷¹

One question addressed in the fall of 1952 was whether the new initiators, when built into the bomb casing and subjected to long periods of storage, changes of temperature, and high altitudes, would prove to be as reliable as they had been during laboratory use. Although there a number of good ideas about external initiator design, tests were extremely difficult to carry out.

Efficiency testing of initiator prototypes was done with a betatron. Fitting a bulky betatron unit into a bomb casing was cumbersome and difficult: the betatron, though large, was still a precisely-aligned mechanism, and not tolerant of rough handling.⁴⁷² Betatrons were used as initiators for shots Annie, Ruth, Ray, and Harry during Operation UPSHOT-KNOTHOLE in Nevada during the spring of 1953.⁴⁷³

letter dated July 9, 1952 to Honorable Brien McMahon, Chairman, Joint Committee on Atomic Energy, Congress of the United States from Gordon Dean, Chairman, USAEC; Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952; OPERATION TUMBLER-SNAPPER, APRIL-JUNE 1952, EXTERNAL NEUTRON MEASUREMENTS, PROJECTS 17.1 and 17.2, WT-555(EX), Los Alamos Scientific Laboratory, University of California, Los Alamos, New Mexico, June 1952, pp. 59, 72.

⁴⁷⁰Minutes of AEC Meeting 776, November 18, 1952.

⁴⁷¹Atomic Energy Commission, Progress Report to the Joint Committee, Part III - Weapons, June 1952 through November 1952, AEC 129/46, December 17, 1952.

⁴⁷²Minutes of Meeting No. 767, 11:00 AM, Thursday, October 16, 1952, U.S. Atomic Energy Commission, Washington, D. C., dated December 17, 1952.

⁴⁷³Minutes of Atomic Energy Commission Meeting No. 746, 2:30 PM, Thursday, September 11, 1952, Washington, D.C.

In particular, results of the Harry shot, a test of a betatron-initiated HAMLET core with a TX-13D HE-pit assembly, led to a recommendation that an external initiator be developed to be included in a future model of a 60" diameter weapon. The device selected for this was the ENS, or Experimental Neutron Source, formerly called the OSDA, One-Shot Deuteron Accelerator.⁴⁷⁴

The UPSHOT-KNOTHOLE Annie shot, the third full-scale nuclear test of external initiation, used a MK 5 HE assembly with a Type D pit; it was an experiment to normalize a yield-versus-time curve first developed after the TUMBLER-SNAPPER Fox and George shots. The test used a device named XR3, the successor to the TUMBLER-SNAPPER XR1 and XR2 devices.

The results of UPSHOT placed the future of Mound Laboratory operations into doubt. The prototype external initiators fired in 1952 and 1953 had been fabricated at LASL, where their manufacture for stockpile weapons was to continue into 1954.

Based on a five-day, one-shift operation, LASL estimated in April 1953 that two men working alone with bench-scale equipment could fabricate enough external initiators within just seven months to permit the replacement of all currently-stockpiled polonium-210 TOM initiators with external initiators.

This was nearly twice as fast as the TOM initiators would normally need to be replaced due to radioactive decay of the polonium. Implosion weapons using stockpiled TOM initiators included MK 5, 6, and 7 bombs.

If a higher rate of replacement were required, LASL believed that this could be easily attained by increasing the number of work shifts. Initiators for new weapons scheduled for stockpiling during calendar year 1954 could be fabricated by operating about eight shifts per month.

Polonium-210 would still be required for POT and CELL initiators for gun-type weapons, specifically, the MK 9 280MM artillery projectile and the MK 8 gravity bomb. However, planned production of these weapons had recently been reduced drastically, so that polonium requirements for POTs and CELLS by fiscal year 1958 would be less than 30% of those

⁴⁷⁴Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 July 1953 - 31 December 1953, pp. 245-246.

originally estimated (all polonium-based initiators had a relatively short shelf life).⁴⁷⁵

By June 1953, current plans called for the present rate of polonium initiator production at the Mound Laboratory to continue for the next year, at which time production would be switched to a new initiator. The AEC had not yet finally approved the substitution of the new initiator for presently-stockpiled initiators.⁴⁷⁶

Work was progressing at this time on the packaging of an external neutron source into a 60-inch diameter sphere configuration, the size of the stockpiled high-yield gravity bombs.⁴⁷⁷

Operational external initiators first entered the U.S. nuclear weapons stockpile in the fall of 1953,⁴⁷⁸ and production of the TOM was ended in October. All new implosion-type assemblies were being fitted with external initiators, and all TOMs in the stockpile were to be replaced by external initiators.

A possible small loss of yield in implosion weapons, amounting to a small percentage, was expected; this disadvantage was more than offset by the substantially lower cost of the new initiators and the elimination of the complex

⁴⁷⁵Memorandum dated April 30, 1953 to R. W. Cook, from F. K. Pittman, subject: Analysis of the Initiator Program.

⁴⁷⁶Memorandum to the File dated June 3, 1953 from Edward L. Heller, subject: Weekly Conference with Walter Williams, p. 3.

⁴⁷⁷Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 January 1953 - 30 June 1953, p. 206.

⁴⁷⁸"Current initiators must be checked only once each six-month period. Initiators scheduled to enter stockpile in the fall of 1953 have an indefinite life." Memo from Major General Howard G. Bunker, USAF, to Commander, Strategic Air Command, Offutt AFB, Nebraska, 2 July 1953, exhibit 23 to DEVELOPING AN ATOMIC CAPABILITY IN STRATEGIC FIGHTER WINGS, 1953, SAC Historical Study #56, Ron Caywood, Historical Division, HQ SAC, Offutt AFB, Omaha, Nebraska, 21 July 1954; "[deleted] has been developed and approved which will make it possible to end production of the polonium-beryllium initiator having a very short half-life. (A small loss in weapon yield is expected.)" ("Outline of Oral Report to NSC," September 17, 1953; this paper is dated September 15, 1953 and no author is listed.)

logistic problems growing out of the required frequent replacement of the short-lived polonium-beryllium TOM initiators.⁴⁷⁹

By the beginning of 1954, preinitiation of nuclear warheads was still a slight probability, to the point that LASL wondered whether DOD would be willing to accept any likelihood of predetonation, given that, in the worst case, a second strike might have to be made against a target where the first strike weapon had "fizzled."

LASL considered studying this possibility, and offered to conduct a nuclear test of a deliberately-preinitiated implosion bomb to gather experimental data.⁴⁸⁰

A review of all 45 U.S. nuclear explosions since July 1945 showed that there had not been any detectable predetonation (the likelihood for preinitiation was greatest for gun-type weapons because of the relatively long time required to assemble the target-projectile mass)⁴⁸¹; however, this limited number of tests was an insufficient basis for safe prediction of the future possibility of pre-ignition.

In each of these blasts, the number of background neutrons emitted per second by the plutonium and initiator, if any, had been carefully measured to estimate the probability of predetonation.⁴⁸²

Fortuitously, major advances in the design of weapons with improved characteristics led at the same time to a decreased probability of preinitiation. Specifically, the use of composite cores containing both plutonium and or alloy, with smaller amounts of plutonium than in all-plutonium cores, and the development of small hollow-core weapons both reduced the incidence of preinitiation.⁴⁸³

⁴⁷⁹Atomic Energy Commission, Progress Report to the Joint Committee, June 1953 through November 1953, Part III - Weapons, AEC 129/55, November 27, 1953, p. 9.

⁴⁸⁰Letter dated January 18, 1954 to Brig. General K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

⁴⁸¹Glasstone and Redman, WASH-1037 (revised), p. 33.

⁴⁸²Letter dated January 21, 1954 to Brig. General K. E. Fields, Director of Military Application, USAEC, from Darol Froman, LASL; Minutes of Thirty-Ninth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 31-April 2, 1954, p. 3.

Early in May 1954, about two-thirds of the production space of the Monsanto Mound, Ohio facility had been made unnecessary by a "new development in the initiator program."⁴⁸⁴

In July, external initiators worked by generating and accelerating tritium ions which collided with a titanium-deuterium target. External initiators had been significantly reduced in size, to the point that units compatible with MK 7 and TX-12 bombs were in production. By the end of June 1954, essentially all stockpiled TOMs had been replaced by external initiators.⁴⁸⁵

At this time, initiators had a 0.58% failure rate (one in 170) which could be improved by new components and assembly techniques. An interim solution was to use two units in parallel.

Current external initiators were being tested every 90 days; improvements could allow testing to be put on a semiannual basis.

The new initiators, when compared to the TOM internal initiator, had advantages of (a) optimum neutron-generation timing, (b) more reliable nuclear safing, and (c) applicability to special assemblies, such as hollow cores.⁴⁸⁶

By 1955, the Monsanto Mound facility had switched over to the production of detonators, and a new General Electric plant in Milwaukee, Wisconsin began mass production of external neutron generators (this work was transferred in 1956 to a new GE plant in Pinellas, Florida).⁴⁸⁷

⁴⁸³Glasstone and Redman, WASH-1037 (Revised), p. 32.

⁴⁸⁴Memorandum for the File dated May 10, 1954 from Edward L. Heller, p. 2.

⁴⁸⁵Atomic Energy Commission, Status Report for June 30, 1954, Part III - Weapons, July 22, 1954, p. 10; Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 9.

⁴⁸⁶Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 9.

⁴⁸⁷NUCLEAR WEAPONS DATABOOK, Volume III, U.S. NUCLEAR WARHEAD FACILITY PROFILES, Thomas B. Cochran, et. al., Natural Resources Defense Council, Washington, D.C., 1987, pp. 59, 81. According to one source, Mound continued to fabricate plutonium-beryllium initiators until November 30, 1961,

Devices using external neutron sources (ENSs) were tested during the Moth and Post shots of Operation TEAPOT at the Nevada Test Site in 1955,⁴⁸⁸ and during the REDWING series of nuclear tests at the Pacific Proving Grounds in 1956 and the PLUMBBOB test series in Nevada in 1957.

The latter series included experiments using similar devices (possibly W-28 and W-34 primaries) that were initiated both externally and internally in separate tests (the Doppler shot test device used an internal initiator); the choice of which type of initiator to use on the XW-31 warhead was to be made after PLUMBBOB ended.⁴⁸⁹

More advanced external neutron generators called "Zippers" were also tested during the Tesla and Hornet shots of Operation TEAPOT in 1955; later versions of the "Zipper" were used on 28 shots during Operation HARDTACK Phase I in the Pacific in 1958.⁴⁹⁰

In 1956, a new plant was built near Pinellas, Florida to produce miniaturized pulse-powered linear ion accelerator initiators. The new facility was operated by the General Electric Company's Neutron Devices Department. Later model initiators became even smaller and, as a result of Sandia research into tritium hydrides, their stockpile life was considerably lengthened.⁴⁹¹

presumably for stockpiled weapons that did not use external initiators. ("Active United States Nuclear Weapons Production and Naval Nuclear Propulsion Facilities as of July 17, 1995," compiled by the U.S. Nuclear Weapons Cost Study Project, p. 9.) These would have included most MK-numbered weapons below the MK 25.

⁴⁸⁸"Measure time from X-unit load ring pulse to the initiating neutron pulse to measure the quantity of neutrons radiating the capsule..." (Status Report No. 1, as of November 1, 1954, Project 14.1, Operation TEAPOT); REPORT OF THE TEST DIRECTOR, OPERATION TEAPOT, LA-1966, J. C. Clark, Los Alamos Scientific Laboratory, October 1955, pp. 94, 126, 138.

⁴⁸⁹Atomic Energy Commission, Quarterly Progress Report to the Joint Committee on Atomic Energy, April-June 1957, Part III - Weapons, August, 6, 1957, p. 3; "Shasta Fired - Doppler Detonated," PLUMBBOB Test Bulletin #37, August 19, 1957.

⁴⁹⁰OPERATION HARDTACK, WT-1682 (EX), Report to the Scientific Director, Report of the Commander, Task Group 7.1, Joint Task Force Seven, 1 August 1958, pp. 22, 55. Specific details and operating characteristics of the S-102 "Zipper" are contained in Sandia report 1411-186, OPERATING MANUAL FOR THE S-102.

By the end of February 1957, external initiators in certain weapons required frequent inspections, some as often as every nine months. This placed a heavy burden on the armed services, especially the Navy, whose ships had little room for warhead teardowns and component tests requiring as long as five hours each. The Navy sought initiators that required inspections once every 12 to 18 months.⁴⁹²

During Operation PLUMBBOB in 1957, three different types of "Zippers" were used on 18 fission shots and five safety tests. "Zippers" used included 18 "laboratory" prototypes that required special handling and precautions to ensure proper operation; four surplus models XMC-624/S5D, left over from Operation REDWING; and two MC-774 standard production units, the first to be used in a nuclear test series.⁴⁹³

As with most other U.S. nuclear weapon components, external initiators became more compact as they evolved. The first prototype external initiators tested in 1952 were very cumbersome: about two feet in diameter and nearly three feet long. Modern neutron generators, such as those used in the W-80 cruise missile warhead, are about the size of man's fist.⁴⁹⁴ Most contemporary U.S. weapons use two generators to guarantee firing of at least one.⁴⁹⁵

Advances in Boosted Fission Weapons

Fusion-boosted fission devices, which are also used as triggers in multistage thermonuclear weapons, offer improved efficiency, a better fail-safe firing mechanism, and a method of controlling explosive yield. A given amount of fissionable material produces a higher yield when boosted, so less material is required for a specific yield. In addition, yield

⁴⁹¹U.S. Department of Energy Pinellas Plant Products and Capabilities booklet, September 1980, pp. 1, 5, 6, 7.

⁴⁹²Minutes of 109th AEC-MLC Conference, February 28, 1957.

⁴⁹³TEST DIRECTOR'S REPORT ON OPERATION PLUMBBOB, UCRL-5166, Gerald W. Johnson, University of California Radiation Laboratory, Livermore, California, November 15, 1957, pp. 65, 66.

⁴⁹⁴Department of Energy Fiscal Year 1984 National Defense Programs Authorization, Hearing before the Subcommittee on Strategic and Theater Nuclear Forces of the Committee on Armed Services, United States Senate, Ninety-Eighth Congress, First Session, on S. 1106 and S. 1107, 12 April 1983, p. 89.

⁴⁹⁵AT THE HEART OF THE BOMB, Debra Rosenthal, Addison-Wesley Publishing Co., Inc., Reading, Massachusetts, 1990, p. 116.

reproducibility can be guaranteed between weapons of the same design.

Yields can be varied in a number of ways: via the timing and the electrical charge on the neutron-initiating "gun," as described earlier; by controlling the amount and relative proportions of tritium and deuterium gases injected into the bomb pit center; and, in a multistage thermonuclear device, by controlling the yield of the primary which directly affects the yields of subsequent stages, or by varying the isotope ratio of the fuel in the secondary and tertiary, including the isotopes lithium-6, lithium-7, tritium, and deuterium.⁴⁹⁶

⁴⁹⁶Cochran and Arkin claim, in NUCLEAR WEAPONS DATABOOK, Vol. I, p. 31, that tritium gas alone, or a mixture of tritium and deuterium gases may be used in boosted weapons or thermonuclear warhead triggers; they imply that tritium gas alone is used. However, Herbert York, in THE ADVISORS, states (p. 22) that "a fifty-fifty mixture of tritium and deuterium reacts about one hundred times as rapidly as does pure deuterium." In addition, an official U.S. Atomic Energy Commission history entitled ATOMIC ENERGY COMMISSION THERMONUCLEAR WEAPONS PROGRAM CHRONOLOGY (drafted between October 1953 and December 1955 by the staff of the AEC Secretariat) states (p. 6) that a 1948 thermonuclear development program included "development of a weapon called the 'Booster' utilizing a small quantity of deuterium and tritium gases at the center of an ordinary atomic weapon to increase its efficiency." Tritium was (and still is) very expensive in terms of time to produce, cost, and lost plutonium production: the same reactors are used for manufacturing both plutonium and tritium.

Because tritium is an artificial radioactive isotope of hydrogen with a half-life of 12.3 years, it loses about 5.5% of its potency each year. For these reasons, it is highly unlikely that tritium is squandered in large quantities as the sole constituent of a fusion-boosting fuel in either small fission weapons or in the triggers of large high-yield thermonuclear weapons.

It is also possible that a reflector or lattice-work of lithium-6 deuteride resides in the center of the core of the primary (as claimed by Morland). If this is the case, tritium is generated in situ by lithium-6 interacting with neutrons. Many early test devices used both tritium and deuterium gases for boosting; however, for ease of portability and servicing, it is possible that deuterium was incorporated into stockpile weapons as a solid compound.

Further evidence that tritium is used in contemporary U.S.

The implementation of boosting reduced somewhat the need for extremely high core compressions, thereby relaxing the requirements for large cores, heavy tampers, and thick HE layers. The attainment of high compression was an important objective in pure fission weapon design.

In the earliest solid-core devices, an improvement in efficiency was the most important objective. In more recent hollow-core boosted weapons, however, the principal purpose of their design has been to make them as compact as possible, with low masses of both HE and fissionable material.

Thus the alpha of these systems is relatively small before boosting. The initial efficiency of the fission reaction is also small, but the total yield is greatly increased by boosting. Alpha after boosting can be an order of magnitude or more greater than alpha before the D-T reactions begin.⁴⁹⁷

Fusion-boosted fission weapons are another idea that originated at Los Alamos during the early days of the Manhattan Project in 1944:

In this connection I should like to point out that (fission) gadgets of reasonable efficiency and suitable design can almost certainly induce significant thermonuclear reactions in deuterium even under conditions where these reactions are not self-sustaining.

... It is not at all clear whether we shall actually make this development during the present project, but it is of great importance that such ... gadgets form an experimental possible transition from a simple gadget to the super and thus open the possibility of a not purely theoretical approach to the latter.⁴⁹⁸

thermonuclear weapon triggers is given by the description of tritium contamination at the site of a B-52 crash in Greenland in January 1968. The B-52 carried four MK 28 weapons which did not undergo nuclear detonation, even though the chemical high explosives in them exploded and burned, so the tritium at the crash site was not generated as fallout from thermonuclear reactions (see "Project Crested Ice," USAF NUCLEAR SAFETY, January/February/March 1970, Vol. 65 Part 2 (Special Edition), pp. 38-41).

⁴⁹⁷Glasstone and Redman, WASH-1037 (Revised), p. 32.

⁴⁹⁸Letter dated 20 September 1944 from J. Robert Oppenheimer

As early as mid-April 1944, a patent memorandum was written which proposed an implosion-type device containing deuterium and tritium, stating that the efficiency of a neutron-induced chain reaction would be greatly increased by the generation of additional neutrons via thermonuclear reactions started by the fission reaction.⁴⁹⁹

This patent application went through several drafts and was filed at Los Alamos in November 1945; it contained the first embryonic thoughts of a possible device utilizing boosting.⁵⁰⁰ By the fall of 1946, Los Alamos was already studying "the definite possibility of a weapon many times superior to the present one but lying reasonably within the capabilities of (the) laboratory."⁵⁰¹

One of the earliest references to the "Booster" appeared in the minutes of the Fifth Meeting of the General Advisory Committee on July 28-29, 1947, in Washington, D.C., which recommended:

One of the main items to be included (in a testing program) should be the thermonuclear experiment designed to test the existence of a thermonuclear reaction by the introduction of the necessary ingredient into the interior of an otherwise standard levitated (bomb pit) model.⁵⁰²

to Richard Tolman, quoted from IN THE MATTER OF J. ROBERT OPPENHEIMER, Transcript of Hearing before Personnel Security Board and Texts of Principal Documents and Letters, U.S. Atomic Energy Commission, The MIT Press, Cambridge, Massachusetts, 1970 (reprinted from 1954 U.S. Government Printing Office volumes), p. 955. The term "super" is a reference to the thermonuclear bomb (see Volume III).

⁴⁹⁹Notes on Interrogation of Klaus Fuchs from FBI serial file 65-58805-1246, dated June 12, 1950 (provided by Richard Rhodes), p. 9.

⁵⁰⁰Most of the following history of the development of boosted fission weapons is drawn from A SHORT ACCOUNT OF LOS ALAMOS THEORETICAL WORK ON THERMONUCLEAR WEAPONS, 1945-1950, J. Carson Mark, LA-5647-MS, Los Alamos Scientific Laboratory, 1974 (based on an earlier 1954 classified history).

⁵⁰¹Letter dated November 14, 1946, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to the Atomic Energy Commission, Washington, D.C.

⁵⁰²Draft Minutes, Fifth Meeting of the General Advisory Committee, July 28-29, 1947, Washington, D.C.

Another "Booster" configuration would use uranium hydride as a fissile pit material.⁵⁰³

The term "boosting" came into widespread use after Edward Teller coined it in September 1947 to refer to experiments first suggested in January of that year. Throughout 1948, studies were underway to determine the necessary characteristics of a device in which this interaction between fission and fusion might be accomplished. However, an immediate military use of such a device was not obvious.⁵⁰⁴

At a meeting of the GAC in June 1948, Dr. J. Robert Oppenheimer proposed that LASL should accelerate its work on the "booster." The device would ignite heavy hydrogen, and thus test the materials eventually required for the "Super" (hydrogen bomb); the "booster" would also increase the efficiency of standard fission weapons.

The GAC recommended continued work on the "booster," and increased tritium production. Although a boosted fission weapon might not be ready for another two to five years, it would, when available, increase war-fighting options for the U.S. military.

If work on the "booster" were to be accelerated, then tritium production would have to be increased. GAC chairman Oppenheimer believed that a simple test of thermonuclear principles could be conducted within two years, and that it could possibly be another three years before sufficient tritium were accumulated to actually detonate a Booster-type bomb. Dr. Enrico Fermi felt that five years was perhaps too long.⁵⁰⁵

A special DOD long-range nuclear weapons planning panel made its report on the BOOSTER weapon in August 1948; the panel

⁵⁰³"On the Development of Thermonuclear Bombs," Edward Teller, LA-643, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, September 26, 1947; memorandum dated December 2, 1948, from William Webster, Chairman, Military Liaison Committee, to U.S. Atomic Energy Commission, Washington, D.C.

⁵⁰⁴Draft Minutes of the Ninth Meeting of the General Advisory Committee, April 23-25, 1948, p. 22.

⁵⁰⁵"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), pp. 281, 282; Minutes of the Tenth Meeting of the General Advisory Committee to the Atomic Energy Commission, June 4-6, 1948, pp. 20-21, 26-27, 30-31.

described the device as a levitated composite implosion weapon into which would be injected deuterium and tritium gas. The BOOSTER would provide an attractive alternate to pure fission weapons and might lead to smaller and lighter atomic warheads; the replacement of plutonium by tritium to gain an equivalent yield would lower cost.

In addition, the BOOSTER would help scientists gain experience with thermonuclear fuels and reactions, information that would be directly applicable to the development of fusion weapons.⁵⁰⁶

In the fall, a full-scale test of such a mechanism was planned for the series of Pacific Ocean test shots to be held in 1951 (Operation GREENHOUSE, as it was later designated). By that time, several promising lines of approach had been identified. One potential "booster" configuration was estimated to weigh 10,000 lbs., with a diameter between 30 and 60 inches and a length of 128 inches (very close to the size of the MK 5 weapon). It would be ready for stockpiling in 1960.⁵⁰⁷

By early September 1948, LASL had high hopes for the BOOSTER:

A program of development aimed at an implosion weapon of the "Booster" type. Such a weapon involves the addition of approximately (deleted) of tritium with some additional deuterium ... It is presently anticipated that a weapon of the Zebra type would be employed with conventional explosive system and only the necessary internal modifications so that as nearly a direct comparison as possible may be made of the ratio of efficiency of this device to that of the SANDSTONE Zebra (core). A factor of two in yield is presently considered attainable.⁵⁰⁸

⁵⁰⁶Draft of The Long Range Military Objectives in Atomic Energy, Committee on Atomic Energy, Panel on Long-Range Objectives, pp. 7, 8; attachment to letter dated 18 August 1948 from J. Robert Oppenheimer, Chairman, Panel on Long Range Objectives, to Donald F. Carpenter, Chairman, Committee on Atomic Energy Committee, Research and Development Board, Department of Defense.

⁵⁰⁷AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 186, 187. There is no explanation for the long development time estimate.

⁵⁰⁸Los Alamos Laboratory Program for Calendar Year 1949, p. 4.

In December, the DOD's Military Liaison Committee recommended that the booster principle be tested in a SANDSTONE Zebra type bomb, whether or not the overall energy release from tritium versus that which could be obtained from additional plutonium or oralloy was favorable to the booster.⁵⁰⁹

Some studies on the problems of containing hydrogen isotopes under high pressure inside hollow plutonium and uranium bomb cores were begun in December.⁵¹⁰

In January 1949, the AEC reported the status of the BOOSTER to the Joint Committee on Atomic Energy:

Research is also being conducted on two new types of nuclear components. The first will incorporate a small amount of previously untested material in an effort to explore nuclear reactions in light elements as well as to achieve greater efficiency in the use of presently employed heavy fissionable material.⁵¹¹

Work on the BOOSTER continued in 1949⁵¹² and 1950; a more thorough theoretical investigation of the BOOSTER was carried out early in 1949. In the summer, the results of this study seemed to indicate that introduction of a tritium-deuterium gas mixture into a fissioning core would add little to the effectiveness of the reaction.

⁵⁰⁹Memorandum dated December 2, 1948 from William Webster, Chairman, Military Liaison Committee, to the AEC, Subject: Los Alamos Laboratory Program.

⁵¹⁰Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report on Atomic Weapon Tests at Eniwetok 1951, Annex 1.10, Cryogenics, WT-50, J. C. Potts and the Cryogenics Staff of Los Alamos Scientific Laboratory, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, November 1951, p. 1.

⁵¹¹Joint Committee on Atomic Energy, Policy and Progress in the H-bomb Program: A Chronology of Leading Events, compiled by William L. Borden and John T. Walker, January 1, 1953 (hereafter referred to as "Borden-Walker H-bomb chronology"), p. 21.

⁵¹²See, for example, LAMS-868, Progress Report, T-Division, January 20, 1949 - February 20, 1949, p. 6, and LAMS-944, T-Division Progress Report, July 20, 1949 - August 20, 1949, p. 2, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, for references to Booster program work during the periods cited.

At the time, the BOOSTER barely overcame the deleterious effects -- primarily yield loss -- of modifying the pit to accept the gas mixture. At this rate, the BOOSTER was more important for its research knowledge value to the fledgling thermonuclear weapons program than to any potential it might have for improving fission weapons.⁵¹³

Despite these apparent shortcomings, a "booster" was still expected to be ready for test by early 1951.⁵¹⁴ LASL groups CMR-6 (Metal Fabrication Group), CMR-9 (Cryogenics Group), and W-1 (Fissionable Materials Handling Group) were the leading organizations for BOOSTER engineering.⁵¹⁵ By July 1949, the Theoretical Division at LASL had completed enough calculations to allow basic BOOSTER design requirements to be specified.⁵¹⁶

The problems encountered in 1949 were later solved by three separate laboratory programs and by October 1950 the last details of an experimental design were frozen; no further changes would be allowed following conclusion of a series of calculations in September.⁵¹⁷

By the end of October, LASL believed that boosting would allow another degree of freedom in fission weapon design, and might provide a method of obtaining a significant yield from a relatively small device.⁵¹⁸

A "boosting" experiment would present an excellent opportunity to study the behavior of fusion fuels, a vital

⁵¹³Notes of a Briefing Held at the Los Alamos Scientific Laboratory on February 23, 1950, pp. 4, 5; Rhodes, DARK SUN, p. 419.

⁵¹⁴Annual Report of the Military Liaison Committee to the Secretary of Defense for the Fiscal Year 1949, 1 August 1949, p. 8.

⁵¹⁵WT-50, p. 1.

⁵¹⁶WT-50, p. 73.

⁵¹⁷September 12, 1952 Walker memorandum, p. 13; "Notes on Minutes of Meetings of the 'Family Group'," Appendix B to September 12, 1952 Walker memorandum.

⁵¹⁸Minutes of the Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, November 1, 1950, p. 31.

prerequisite to the development of a large, pure-fusion weapon using either an unboosted or boosted fission trigger.

One of the problems addressed in 1949 was the diffusion of high-pressure deuterium and tritium gases through fissile cores, specifically, hollow composite uranium-plutonium cores. LASL finally decided to use a copper pit liner to minimize gas losses.

This liner fit inside the hollow weapon core; although its presence reduced fission efficiency slightly via mixing of the copper with the fissile materials, this loss was more than offset by the fission enhancement resulting from the presence of the D-T gas mixture.⁵¹⁹

Emphasis on a boosting experiment slowed during 1950 because of demands of Operation GREENHOUSE preparations and because primary emphasis was placed on large thermonuclear explosions rather than boosting; the latter was directed mainly toward increasing the efficiency of small explosions.⁵²⁰

In addition, at its 27th meeting on November 15, the so-called "Family Committee," a group at LASL overseeing thermonuclear weapons development (see Volume III), decided that "essentially no effort is being devoted to further exploration of boosted weapons. This has come about almost automatically (due to) the lack of (Theoretical Division) manpower."

In addition, since boosting was primarily to enhance the yield of small weapons, there was a tendency to wait until after the completion of Operation GREENHOUSE for further development.⁵²¹

The first American fusion-boosted fission bomb was detonated during the Item shot of the GREENHOUSE test series in 1951.⁵²² This device had a yield of 45.5 kilotons, which suggests that it was probably a small implosion weapon with an unboosted yield in the range of 20 to 30 kilotons.⁵²³

⁵¹⁹WT-50, pp. 74, 79; WT-102, p. 32.

⁵²⁰September 12, 1952 Walker memorandum, p. 13.

⁵²¹"Notes on Minutes of Meetings of the 'Family Group'," Appendix B to September 12, 1952 Walker memorandum.

⁵²²York, THE ADVISORS, p. 76.

⁵²³OPERATION GREENHOUSE, ANNEX 6.4, FALLOUT PHENOMENOLOGY, C. E. Adams, F. R. Holden, N. R. Wallace, U.S. Naval radiological Defense Laboratory, San Francisco, California, WT-4, August 1951.

The utility of the Item shot, and whether or not it would be fired, were in question right up to its time of detonation. In 1954, Norris Bradbury noted that LASL was unwilling to strongly recommend that the test be conducted; although it was of general scientific interest, it was not, in Bradbury's opinion, an "absolute necessity for the thermonuclear weapons program."⁵²⁴

In October 1951, LASL noted that "the efficacy of simple boosting is fairly well understood and in many cases requires only application to a specific device and comparison of the practical problems presented."⁵²⁵

At the end of 1951, LASL planned to continue its research on the hydrodynamics of thermonuclear boosting.⁵²⁶ Despite Bradbury's doubts about the usefulness of boosting to the thermonuclear program, the Item shot, and subsequent tests, would pay many dividends in coming years, mainly in the form of smaller primaries for hydrogen weapons and in a wide spectrum of "sealed pit" fission warheads for specialized applications.

On May 1, 1952, the Dog shot of Operation SNAPPER in Nevada further demonstrated the practicality of using deuterium gas to boost small-diameter (30") implosion fission bombs.⁵²⁷ This shot also showed that boosting an implosion-type weapon

⁵²⁴Letter dated October 1, 1954 to AEC Chairman Lewis Strauss from Norris Bradbury, Director, LASL.

⁵²⁵Letter dated 9 October 1951 to Col. Kenneth E. Fields, Director of Military Application, USAEC, from Norris Bradbury, Director, LASL.

⁵²⁶Letter dated 26 December 1951 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Norris Bradbury, Director, LASL, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 3.

⁵²⁷MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, Dr. Hans Bethe, May 28, 1952, p. 3; Borden-Walker H-bomb chronology, p. 64; letter dated July 9, 1952 to Honorable Brien McMahon, Chairman, Joint Committee on Atomic Energy, Congress of the United States from Gordon Dean, Chairman, USAEC; GENERAL REPORT ON WEAPONS TESTS, EXTERNAL NEUTRON MEASUREMENTS 1946 THROUGH 1956, WT-9004 (EX), Wendell A. Biggers, Freeman Waddell, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, 8 October 1957, p. 127. This test was characterized as "very crude" and "(yielding) very rough results." (September 12, 1952 Walker memorandum, p. 11.)

increased the total number of neutrons emitted from the bomb.⁵²⁸

Application of boosting to stockpile weapons was slow, due partly to a major lack of emphasis and a failure by the military to realize its potential, in terms of reduced weapon diameter and weight for a given yield. The LASL Technical Program for Calendar Year 1953 and Fiscal Year 1954, issued at the end of 1952, included plans for further boosting research:

Fission Weapon Boosting - Studies on the effective methods of boosting fission weapons from both the theoretical and practical point of view will be continued. The possibility of the use of deuterium without tritium for this purpose in implosion-type weapons will be given both theoretical and experimental investigation.⁵²⁹

Nonetheless, stockpiled weapons could not be reworked to use boosting, and new warheads that featured boosting were slow in coming from the laboratories. The device tested at Eniwetok in 1951 was not readily translatable into a useful weapon, and bore little resemblance to the boosted weapon primaries that would be tested four years later during Operation TEAPOT in Nevada in 1955.

The 1951 test device was designed only to test the principle of boosting, and required much evolutionary engineering to put the idea into practice.⁵³⁰ The complicated cryogenic pumps and plumbing that serviced the 1951-model BOOSTER would have to be simplified and reduced in size, or replaced by more practical and compact D-T gas bottles, switches, and valves.

⁵²⁸THE RELATIVE IMPORTANCE OF NUCLEAR RADIATION FROM ATOMIC WEAPONS DETONATED AT HIGH ALTITUDES, AFSWP Technical Analysis Report No. 500, Paul W. Ifland, Weapons Effects Division, Headquarters, Armed Forces Special Weapons Project, April 23, 1953, p. 12.

⁵²⁹Letter dated 31 December 1952 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from N. E. Bradbury, Director, LASL, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1953 and Fiscal Year 1954 (hereafter "LASL Technical Program for CY 1953 and FY 1954").

⁵³⁰Conversation between C. Hansen and J. Carson Mark, Los Alamos, New Mexico, October 2, 1991.

In a February 1953 GAC meeting, Dr. John von Neumann implied that tritium availability was a limiting factor in stockpiling boosted weapons. The best use of tritium would be for boosting low-efficiency weapons, and for allowing the development of smaller-diameter warheads which might have utility as antisubmarine weapons and for other tactical weapons.⁵³¹ (By the summer of 1952, the nation's entire tritium stockpile was only a few hundred grams; a whole kilogram of tritium might not be stockpiled until 1955.)⁵³²

A month later, at the next GAC meeting in Washington, D.C. in March 1953, General K. E. Fields stated that DOD hoped to accumulate field data on the logistics and handling facilities required for boosted implosion weapons.

Although there was a current DOD requirement for a limited number of boosted bombs, Fields thought that thermonuclear weapons might use up most of the projected tritium production (solid-fueled hydrogen bombs that generated their own tritium had not yet been tested).

While boosting might be able to "achieve a big bang with a relatively small weapon," thermonuclear weapons might be cheaper than fission bombs for "hard" targets requiring high overpressures or for large, widely-dispersed target complexes.⁵³³

Another deuterium-boosted device was tested during the Dixie shot of Operation UPSHOT-KNOTHOLE in Nevada in the spring of 1953, using a boosted core in a MK 5 HE assembly.⁵³⁴ The shot was predicted to yield 10±2 KT; measured yield was 11 KT. The Badger shot of UPSHOT-KNOTHOLE was also boosted.⁵³⁵

⁵³¹Minutes of the Thirty-Third Meeting of the General Advisory Committee to the USAEC, Washington, D.C., February 5-7, 1953, p. 34; September 12, 1952 Walker memorandum, p. 15.

⁵³²September 12, 1952 Walker memorandum, p. 20; NUCLEAR WEAPONS DATABOOK Vol. II, p. 180.

⁵³³Minutes of the Thirty-Fourth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 23-24, 1953, Washington, D.C., pp. 9, 10.

⁵³⁴An untitled, undated classified poem issued sometime soon after the end of UPSHOT-KNOTHOLE states that the fourth shot "was deuti-full." The device may have used a solid or crystalline lithium deuteride boosting charge.

⁵³⁵WT-9004 (EX), p. 130.

At the end of May 1953, practical problems involved in designing and fabricating gas-tight boosted pits were being investigated; design and testing was proceeding on the use of tritium-deuterium gas in the new generation of "sealed-pit" weapons on the drawing boards at LASL.⁵³⁶

The use of boosting also allowed for relaxation of the chemical purity requirements for plutonium, i.e., "dirty" plutonium with a high Pu-240 content could now be used in weapons.⁵³⁷ This "dirty" fissile material, besides being harder to handle due to its radioactivity, was also more likely to lead to a "fizzle" yield.⁵³⁸

Soon afterwards, the laboratory planned more boosting tests to determine the extent of mixing of thermonuclear and fission materials in the core, and to study the possibility of using a solid compound such as lithium deuteride instead of deuterium gas for boosting.⁵³⁹

At the end of July, the AEC reported to President Eisenhower that important weapon research then underway included the use of tritium and deuterium to boost yield.⁵⁴⁰

⁵³⁶Atomic Energy Commission, Progress Report to the Joint Committee, December 1952 through May 1953, Part III - Weapons, May 29, 1953, p. 5.

⁵³⁷Minutes of the Thirty-Fifth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 14, 15, 16, 1953, p. 21. At this time, plutonium produced by the reactors at Hanford contained only about 1.4% Pu-240 with a neutron background of about 15 neutrons/gram-second. A power reactor might produce plutonium containing 13% Pu-240 and a neutron background 140 neutrons/gram-second. (Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 24.) By comparison, the neutron background in ordinary alloy is approximately 1.6 neutrons/kg-sec, and U-238 has a neutron background of 15 neutrons/kg-sec. (Glasstone and Redman, WASH-1037 (Revised), p. 36.)

⁵³⁸The higher neutron background and higher Pu-240 content of reactor-grade plutonium results from its long exposure to neutron flux in the reactor; this long exposure is required for economical generation of electricity. ("Nuclear Weapons Supply and Demand," Michael M. May, AMERICAN SCIENTIST, Volume 82, November-December 1994, p. 530.)

⁵³⁹Memorandum to All Concerned dated July 16, 1953 from Paul C. Fine, Division of Military Application, USAEC, Subject: Trip Report, MLC Tour, July 5-11, 1953.

In December, Norris Bradbury noted that boosting appeared to be essential to small-diameter primaries for thermonuclear weapons.⁵⁴¹ By February 1954, he noted that if tritium were used for boosting, its radioactive characteristics might be a drawback to its use in stockpiled weapons:

The present military requirements for thermonuclear weapons, together with the foreseeable trend towards still smaller systems of this nature having the highest possible yield, requires that serious attention be given to the problem of the primary bomb under these circumstances. Fundamentally, the problem is this: To [deleted]

[deleted]

Boosting of such weapons is certainly technically feasible. However, present production requirements for tritium where its use is inescapable, together with the problems incident to its handling and to the fact that its half-life is relatively short, indicate this is not ordinarily an attractive solution. [deleted]⁵⁴²

Soon afterwards, several of the primaries in the first solid-fueled thermonuclear weapon prototypes fired during Operation CASTLE in the spring of 1954 were boosted.⁵⁴³

By the summer of 1954, LASL planned to test in a "post-TEAPOT" series a lithium-6 deuteride boosted device.⁵⁴⁴

⁵⁴⁰Atomic Energy Commission, Semiannual Report to the President, AEC 21/12, Report by the General Manager, July 27, 1953.

⁵⁴¹Letter dated 2 December 1953 to Honorable W. Sterling Cole, Chairman, JCAE, from Norris Bradbury, Director, LASL.

⁵⁴²Letter dated January 18, 1954 to Brig. General K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.

⁵⁴³Atomic Energy Commission, Progress Report to the Joint Committee, June 1953 through November 1953, Part III - Weapons, AEC 129/55, November 27, 1953, p. 3, Table 1 (listing CASTLE test devices): "In addition to the (fusionable) materials indicated, small quantities of tritium are used for boosting the primary [deleted] fission reactions involved in these devices."

A number of boosted primaries for small thermonuclear weapons were tested during Operation TEAPOT in Nevada in the spring of 1955, including shots Hornet and Bee.⁵⁴⁵ Results of the tests were very satisfactory:

We wish to comment specifically at this time on the favorable outcome of the experiments on gas boosting of small implosion weapons. Subject to further experiments on the range of applicability of this technique, the results are of consequence in several ways.

The energy yield of the fissionable material used in particular weapon assemblies can be greatly increased. The technique promises implosion weapons which can meet the requirements of safety against one-point detonation.⁵⁴⁶

By the end of July, smaller, lighter, and more-rugged immediately-ready "sealed pit" weapons were entering the U.S. nuclear weapons stockpile, due in part to the application of gas boosting which, for a given-sized pit, increased yield and improved economy of fissionable material.⁵⁴⁷

In addition, boosting obviated pre-initiation and resulted in fission warheads and devices with predictable and reproducible yields, the exact characteristics required for dependable thermonuclear weapon primaries.⁵⁴⁸

At the end of 1955, the AEC had decreed that in general, all future U.S. nuclear weapons would be gas-boosted, if

⁵⁴⁴Minutes of the Forty-First Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, July 12-15, 1954, p. 27.

⁵⁴⁵McPhee, p. 110; WT-9004 (EX), p. 135.

⁵⁴⁶Letter dated May 4, 1955 to Dr. Willard F. Libby, Acting Chairman, USAEC, from I. I. Rabi, Chairman, General Advisory Committee.

⁵⁴⁷Atomic Energy Commission, Program Status Report, Part III - Weapons Program, July 27, 1955, p. 4.

⁵⁴⁸"An NHP Interview with Donald R. Cotter," (NHP Oral History Transcript #2, Center for International Security Studies at Maryland School of Public Affairs, University of Maryland, 18 December 1990, p. 12.

possible, in order to provide greater yield than could otherwise be obtained from the same amount of fissionable materials if it were unboosted.

In the spring of 1956, all presently-planned applications of gas boosting involved the use of a mixture of gaseous tritium and deuterium under pressures of a few thousand pounds per square inch, stored in stainless steel reservoirs within the weapons but external to their implosion systems.⁵⁴⁹ Explosively-actuated valves would release the gas into the center of the core as part of the weapon arming sequence.⁵⁵⁰

Since the tritium decayed radioactively, the reservoirs would have to be emptied and refilled at regular intervals in order to maintain an optimum amount of fresh gas within the weapon.

To meet requirements for boosting gases, construction of a gas packaging facility at the AEC's Savannah River plant had been authorized. The new facility would provide for the bottling of tritium and deuterium gases under very high pressures in reservoirs to be incorporated into stockpile weapons.⁵⁵¹

There were a number of reasons for the relatively-slow paced development of boosting technology. Among them were substantial material fabrication problems involved in producing gas-tight weapon cores. (The Huron shot of Operation REDWING in 1956 had to be rescheduled when the first attempt to fire it failed because of a leaky pit.)

The D-T gas mixture was injected into the pit when the weapon was armed; although only a few grams of the gases were used⁵⁵², injection pressures were typically on the order of several thousand pounds per square inch.

⁵⁴⁹Administrative Plan for Johnston Island (Operation WILLOW Concept #2, Headquarters, Joint Task Force Seven, Arlington Hall Station, Arlington, Virginia, undated, but after August 18, 1956, p. C-1-D-1.

⁵⁵⁰Letter dated March 23, 1956 to Commander, Field Command, Armed Forces Special Weapons Project, from M. F. Roy, Assistant Director for Production, LASL, subject: Storage and Handling of Tritium-Boosted Atomic Weapons.

⁵⁵¹Atomic Energy Commission, Program Status Report, December 31, 1955, Part III - Weapons, February 15, 1956, p. 6.

⁵⁵²"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 7. On March

The AEC's tritium production capability was also a retarding element: in a February 1953 GAC Meeting, Dr. John von Neumann implied that tritium availability was a limiting factor in the stockpiling of boosted weapons. In addition, until extensive testing showed otherwise, there were significant concerns about the inherent one-point safety of the new generation of boosted sealed-pit warheads.

The initial operation of a reactor fully loaded with enriched material for tritium production was begun at the AEC's Savannah River facility in Aiken, South Carolina in September 1956 and continued until December, when discharge was affected. Operation was resumed when the reactor was recharged with a second loading of enriched fuel.

Large quantities of tritium were scheduled for weapons use in fiscal year 1958, at which time significant increases in tritium output were projected and fabrication facilities for new-type gas-boosted weapons would be completed.

Peak tritium production was expected during fiscal year 1959, when all available tritium would be required for new weapons and for new boosted primaries replacing unboosted primaries in certain existing weapons.

Plans to incorporate tritium into a substantial portion of the stockpile were based upon DOD guidance concerning desired stockpile composition. By March 1957, conversion of stockpiled weapons to tritium-boosted designs was expected to be completed by July 1, 1959.⁵⁵³

Another reason for slow development of boosted warheads was that the utility of boosting may not have been readily apparent to LASL weaponeers. At the end of 1950, boosting was thought to be useful only for small, relatively low-yield weapons.⁵⁵⁴

23, 1995, the DOE released a classification bulletin (WNP-93) which stated that "a reservoir typically contains less than 20 grams of tritium." (OD COMMUNIQUE, Vol. VII No. 4, U.S. Department of Energy, Office of Declassification, May 1995, p. 3.)

⁵⁵³Proposed Changes to the Program Status Report to the Joint Committee on Atomic Energy, December 31, 1956; Memorandum for Commissioner Murray dated March 8, 1957 from R. W. Cook, Acting General Manager, USAEC, subject: Revisions in the Program Status Report to the Joint Committee.

⁵⁵⁴Enclosure (Appendix B), entitled "Thermonuclear Effort," to letter dated November 27, 1950 from Dr. Frederick de Hoffmann to Gordon Dean.

In October 1954, LASL director Norris Bradbury noted that "no boosted weapon has yet entered the stockpile, and the major uses which have been made of the (GREENHOUSE) Item shot results lie primarily in the [deleted] calculations."⁵⁵⁵

By the latter part of the 1950s, boosted fission explosives were being used routinely as the triggers in small-diameter thermonuclear warheads, such as the W-28; they were also being used in some low-yield battlefield tactical weapons.⁵⁵⁶

A modern boosted thermonuclear primary requires very little high explosives and only a lightweight tamper; a typical primary might include four kilograms of plutonium in the form of a spherical shell between four and 6.3 inches in diameter, a thin beryllium reflector, and about 18 lbs. of HE in the form of a multipoint-detonated hemispherical shell. Such a device yields between a few and 15 kilotons.⁵⁵⁷

Production of tritium ended in the U.S. in 1991. The national stockpile during Fiscal Year 1984 was estimated to be about 70 ± 25 kg with one Savannah River reactor then producing about 11 kg per year; with a net loss rate of 5.5% annually, there was expected to be a net gain of about 7 kg per year at that time.⁵⁵⁸

⁵⁵⁵Letter dated October 1, 1954 to AEC Chairman Lewis Strauss from Norris Bradbury, Director, LASL.

⁵⁵⁶Diary page dated October 25, 1957, Dwight D. Eisenhower, Papers as President of the United States, 1953-1961. Ann Whitman File, ACW (Ann C. Whitman) Diary series, Box 9, Folders: ACW Diary - October 1957 (1), Dwight D. Eisenhower Library, Abilene, Kansas. This particular diary page citation was never classified, although it clearly dealt with concepts (staging and radiation implosion) that were classified before and during the PROGRESSIVE case in 1979. See also pp. 1 and 2 of HYDRONUCLEAR EXPERIMENTS, by Robert N. Thorn and Donald Westervelt (LA-10902-MS/UC-2, Los Alamos National laboratory, February 1987), where the authors discuss the advent and importance of boosted primaries for lightweight thermonuclear weapons.

⁵⁵⁷"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 8.

⁵⁵⁸NUCLEAR WEAPONS DATABOOK, Vol. II, pp. 63, 75.

Assuming an FY 1984 stockpile of 70 kg, then by the end of 1995, with production ending in 1991, the national stockpile would be around 104 kg. The U.S. tritium inventory is decaying steadily at a rate of 5.5% per year; without replacement, it will eventually all be gone. The DOE is currently contemplating ways to produce more tritium either in reactors or in particle accelerators.

All of the so-called "sealed pit" weapons produced by the U.S. since about 1957 have depended upon boosting for much of their power. All of the weapons in the so-called "enduring stockpile" -- the B61-4, B61-7, B61-10, W76, W80-0, W80-1, B83, W87, and W-88 -- use boosted primaries. Boosting is the key to small-diameter primaries.

Summary

By the end of the 1950s, radically-new nuclear weapons designs developed at the laboratories at Los Alamos and Livermore had led to dramatic improvements in the performance of fission warheads. Weapons then in the U.S. stockpile bore little resemblance, in performance and versatility of military use, to the bombs dropped on Hiroshima and Nagasaki 15 years earlier (many of these improvements are detailed in the weapons descriptions in Volumes VI and VII).

Perhaps the greatest single improvement in fission weapons was their employment as triggers in thermonuclear weapons. All other modifications paled in significance beside it: a small stockpile of fission triggers, little more than tactical weapons by themselves, could be used as primaries for an equal number of much more powerful strategic weapons.⁵⁵⁹ The story of the development of the American H-bomb follows.

⁵⁵⁹There is some evidence suggesting that this is precisely what has happened in Israel. See "Revealed: The Secrets of Israel's Nuclear Arsenal," The Sunday Times, No. 8,461, London, England, 5 October 1986, pp. 1, 4, 5. In December 1953, at the dawn of the thermonuclear age, LASL director Norris Bradbury noted that the relationship between fission weapons and thermonuclear weapons was very close, "since the primary bombs for the latter are developed in parallel (and frequently represented by almost the same device) with the former." (Letter dated 11 December 1953 to Brig. Gen. K. E. Fields, Director of Military Application, USAEC, from N. E. Bradbury, Director, LASL.)

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Table II-1: Early AEC Facility Code Names

Apple	Rocky Flats, Colorado plant, opened April 1951
Centerline	Naval Ordnance Plant, Center Line, Michigan weapons storage site
Eye	Salt Wells Pilot Plant of Naval Ordnance Test Station, Inyokern, California, operational July 1945 - 1954
Lakeside	Paducah, Kentucky (?) (inactivated in fall of 1948)
Marsh	Pocatello, Idaho (inactivated in fall of 1948)
Orange	Pantex plant, Amarillo, Texas, activated October 1951
Pepper	U.S. Army Picatinny Arsenal; AEC activity began 1948
Plum	??
Plum-A	Spoon River plant (canceled)
Roger	??
Royal	Bendix Kansas City, opened February 1949
Sugar	Burlington, Iowa ordnance plant activated late 1947
Sunflower	Possibly Sunflower Army Ammunition Plant, DeSoto, KS
Tee	U.S. Army Rock Island Arsenal, Illinois; AEC work began in 1947

Early Nuclear Weapons Storage Sites

Able	Monzano Base, Albuquerque, New Mexico
Baker	Killeen, Texas base (near Ft. Hood Army base)
Charlie	Clarksville, Tennessee site (near Ft. Campbell Army base)
Dog	Bossier, Louisiana (near Barksdale AFB)
Easy	Caribou AFS (Loring AFB, near Limestone, Maine)

Fox	Rushmore AF Station (near Ellsworth AFB, Rapid City, SD)
George	Deep Creek AF Station (near Fairchild AFB, Spokane, WA)
How	Fairfield AFS (later Travis AFB, near Fairfield-Suisun, CA)
Item	Stonybrook AF Station (near Westover AFB, Chicopee, MA)
Jig	Skiffles Creek Annex (near Yorktown, Virginia naval weapons station and mine depot)
King	Medina Base (near San Antonio, Texas)
Love	Lake Mead Base (near Nellis AFB, Las Vegas, Nevada)
Yoke	North Depot Activity, Seneca/Romulus Army Depot, NY

Table II-2: STATUS & NUMBER OF WEAPON PROGRAMS, JULY 1950 & JULY 1953

	<u>Stage 1</u>		<u>Stage 2</u>		<u>Stage 3A</u>		<u>Stage 3B</u>		<u>All Stages</u>	
	<u>7/50</u>	<u>7/53</u>	<u>7/50</u>	<u>7/53</u>	<u>7/50</u>	<u>7/53</u>	<u>7/50</u>	<u>7/53</u>	<u>7/50</u>	<u>7/53</u>
1. NEW WEAPONS										
(a) Implosion-type bombs	2	1	0	1	3	1	1	4	6	7
(b) Gun-type bombs	0	0	2	1	1	0	0	1	3	2
(c) Thermonuclear bombs	0	2	0	3	0	0	0	0	0	5
(d) Artillery shells	0	2	1	1	0	0	0	1	1	4
(e) Warhead installations	0	9	0	10	0	3	0	1	0	23
(f) Total (a + b + c + d + e)	2	14	3	16	4	4	1	7	10	41
2. MAJOR WEAPON MODIFICATIONS										
(a) Implosion type bombs	0	0	0	1	0	2	0	2	0	5
(b) Gun-type bombs	1	0	0	0	0	0	0	1	1	1
(c) Other	0	0	0	0	0	0	0	0	0	0
(d) Total (a + b + c)	1	0	0	1	0	2	0	3	0	6
3. GRAND TOTAL (1f + 2d)	3	14	3	17	4	6	1	10	11	47

NOTES: Stage 1 indicates that studies incident to formulation of of program were in progress.
Stage 2 indicates that full-scale design and development were underway.
Stage 3A indicates that development was complete or nearly complete, and that production engineering had been started or completed, but that the first production units were not yet complete.
Stage 3B indicates that production and stockpiling were underway.

SOURCE: Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 61.

Table II-3: Warhead/Missile Marriages, December 1953

<u>Warhead-Missile Combination</u>	<u>Responsible Service</u>	<u>Warhead-Missile Combination</u>	<u>Responsible Service</u>
XW-5/REGULUS	Navy	XW-7/NIKE-B	Army
XW-5/TRITON	Navy	XW-8/REGULUS	Navy
XW-5/B-61 MATADOR	Air Force	XW-12/TALOS-W	Navy
XW-5/B-63 RASCAL	Air Force	XW-12/F-99 BOMARC	Air Force
MK 7/CORPORAL	Army	XW-13/REDSTONE	Army
MK 7/HONEST JOHN	Army	XW-13/B-62 SNARK	Air Force
XW-7/BETTY	Navy	XW-13/B-64 NAVAHO	Air Force
XW-7/BOAR	Navy	XW-M/ATLAS (study)	Air Force

Source: Semiannual Historical Report, Headquarters, Field Command, Armed Services Special Weapons Command, Sandia Base, Albuquerque, New Mexico, Activities for the Period 1 July 1953 - 31 December 1953, p. 230.

Table II-4: Composition of Weapon Stockpile by Types
(By end of Fiscal Year), 1950-1956 (1)

<u>1950</u>	<u>1951</u>	<u>1952</u>	<u>1953</u>	<u>1954</u>	<u>1955</u>	<u>1956</u>	<u>Type</u>
Little Boy							Gun
MK 3							Implosion
MK 4C (2)	MK 4C						"
	MK 4D (3)	MK 4D					"
		MK 5	MK 5	MK 5	MK 5	MK 5	"
					W-5 (4)	W-5	"
		MK 6	MK 6	MK 6	MK 6	MK 6	"
		MK 7	MK 7	MK 7	MK 7	MK 7	"
					W-7 (4)	W-7	"
		MK 8	MK 8	MK 8	MK 8		Gun
		MK 9	MK 9	MK 9	MK 9		"
						MK 11 (4)	"
					MK 12	MK 12	Implosion
				TX-14			Thermonuclear
					MK 15	MK 15	"
				TX-17	MK 17	MK 17	"
				MK 18	MK 18		Implosion
						MK 19 (5)	Gun
						MK 21	Thermonuclear
						MK 23 (5)	Gun
				TX-24	MK 24	MK 24	Thermonuclear
						MK 26	"
						MK 33 (5)	Gun

Source: Atomic Energy Commission, Program Status Report, December 31, 1954, Part III - Weapons Program, p. 3.

Notes: (1) Fiscal Year N includes period of July of year N-1 to June of year N.

(2) MK 4 core with Type C pit.

(3) MK 4 core with Type D pit.

(4) No warheads were stockpiled as such in FY 1955. Capability existed through stockpiling of components to convert stockpiled gravity bombs to warheads.

(5) Production of non-nuclear assemblies of these weapons was the responsibility of the Department of Defense.

Table II-5: U.S. Nuclear Weapons Stockpile Size, 1945-1994

<u>Date</u>	<u>In USAEC Custody</u>			<u>In DOD Custody</u>			<u>Total</u>	
	<u>Nuclear Cores</u>		<u>Total</u>	<u>HE Assemblies</u>		<u>Complete Weapons</u>		
	<u>Gun Type</u>	<u>Implosion Type</u>		<u>Gun Type</u>	<u>Implosion Type</u>			
12/31/45	0	2	2	0	2	2	0	2
6/30/46	0	9	9	0	9	2	0	9
6/30/47	0	13	13	0	29	29	0	13
6/30/48	0	56	56	2	53	55	0	56
6/30/49			169	12	228	240 (est.)	0	169
6/30/50			298	28	660	688 (est.)	0	298
1951			429				9	438
1952			823				9	832
1953			1152				9	1161
1954			1463				167	1630
1955			1499				781	2280
1956			2262				1358	3620
1957			3578				2250	5828
1958			3385				4017	7402
1959			3968				8337	12305

Regarding the estimated HE assembly counts for 1949 and 1950, at least through the end of 1954, the total number of HE assemblies stockpiled was somewhat greater than the number of nuclear assemblies (which were interchangeable within HE assemblies) to enhance stockpile flexibility. The sharp increase in gun-type cores and HE assemblies in 1949 and 1950 resulted from preparation of cores and HE assemblies for the MK 8 weapon. Six implosion weapon cores were retired (i.e., recycled) in 1949. Two of the implosion cores on hand in June 1946 were later expended during Operation CROSSROADS.

In June 1994, the DOE released a table comprising the numbers of warheads stockpiled and retired between 1945 and April 1994, and the corresponding total megatonnage of the U.S. stockpile during that period. The numbers of total weapons stockpiled after 1961 were withheld, so the numbers below after 1961 are only estimates. There is no explanation currently available for the few discrepancies between the 1945-1959 numbers above released in September 1992 by DOD and those below released in June 1994 by DOE.

Build numbers shown below are total annual quantities for fully retired weapon systems, and as such, represents a subset of U.S. nuclear weapon production. Retirement numbers below reflect weapons retired for conversions, modifications, or disassemblies for disposal; the numbers depend critically on how "retirement" is defined. Disassemblies shown below reflect only the number of weapons dismantled for actual disposal, and not for repair, modification, or inspection (those disassembly numbers are much higher, since a given weapon might be disassembled more than once for more than one

purpose).

Between 1945 and 1961, the total stockpile number for any given year = the previous year's total + builds - retirements. Builds and retirements after 1961 are incomplete; the numbers of warheads still in stockpile that were built after 1962 are still classified.

It is not known at this time what the megatonnage numbers represent for the years before fixed-yield, sealed pit weapons entered the stockpile. Before then, many weapons used interchangeable cores with a variety of yields. Total megatonnage for that period may represent either the highest potential explosive power of all fissionable cores when mated with the HE assembly with which they were compatible for their highest yield, or it may represent an average of explosive powers of particular cores and pits mated with specific HE assemblies. Total megatonnage for the period following introduction of sealed pit weapons is presumably a summation of all yields for all weapons stockpiled, including weapons which had several versions, each version with a different fixed yield.

Numbers in the "Average" column are average kilotonnage per warhead; values after 1961 are estimates. This number rises sharply after the introduction of thermonuclear weapons into the stockpile in 1954-1955 and decreased gradually after 1963.

<u>Date</u>	<u>Total</u>	<u>Megatonnage</u>	<u>Avg. (KT)</u>	<u>Builds</u>	<u>Retirements</u>	<u>Disassemblies</u>
1945	2	0.04	20	2	0	
1946	9	0.18	20	7	0	
1947	13	0.26	20	4	0	
1948	50	1.25	25	43	6	
1949	170	4.19	25	123	3	
1950 ¹	299	9.53	32	264	135	
1951	438	35.35	81	284	145	
1952	841	49.95	60	644	241	
1953	1169	72.80	62	345	17	
1954	1703	339.01	200	535	1	
1955	2422	2879.99	1190	806	87	
1956	3692	9188.65	2489	1379	109	
1957	5543	17545.86	3165	2232	381	
1958	7345	17303.54	2355	2619	817	
1959	12298	19054.62	1549	7088	2135	
1960	18638	20491.17	1099	7178	838	
1961	22229	10947.71 ²	492.5	5162	1571	
1962	27100	12825.02	473	4529	766	
1963	29800	15977.17	536	3185	830	
1964	31600	16943.97	536	3493	2534	
1965	32400	15152.50	468	3519	1936	
1966	32450	14037.46	433	2429	2357	
1967	32500	12786.17	393	1693	1649	
1968	30700	11837.65	386	536	2194	

1969	28200	11714.44	415	684	3045	
1970	26600	9695.20	364	219	1936	
1971	26500	8584.40	324	1073	1347	
1972	27000	8531.51	316	1546	1541	
1973	28400	8452.00	298	1171	544	
1974	29100	8325.22	286	959	807	
1975	28100	7368.38	262	748	2240	
1976	26700	5935.51	222	427	2181	
1977	25800	5845.00	227	221	998	
1978	24600	5721.16	233	50	1148	
1979	24300	5696.34	234	170	730	
1980	24300	5618.86	231	0	904	732
1981	24400	5382.91	221	30	1887	1577
1982	23000	5358.89	233	338	1537	1535
1983	23400	5232.47	224	217	749	1120
1984	23600	5192.20	220	187	1143	994
1985	23500	5217.48	222	195	1322	1075
1986	23400	5414.54	231	140	1224	1015
1987	23700	4882.14	206	0	958	1189
1988	23400	4789.77	205	0	1023	581
1989	22500	4743.34	211	0	1794	1208
1990	21000	4518.91	215	0		1154
1991 ³	19500	3795.94	195	0		1595
1992	18200	3167.88	174	0		1856
1993	16750	2647.31	158	0		1556
1994	14900	2375.30	159	0		1376

- Notes:
1. The stockpile was almost completely rebuilt in 1950. MK III weapons were retired and replaced by MK 4 weapons.
 2. Drop in total megatonnage caused by withdrawal of approximately 950 MK 36 weapons, each with a 10 MT yield.
 3. Stockpile estimate totals subsequent to 1990 include warheads in active operational service and warheads in reserve and in retirement awaiting dismantlement.

Sources: NUCLEAR WEAPONS DATABOOK Vol. I, p. 15; "U.S. Weapons Secrets Revealed," Nuclear Notebook, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 49 No. 3, March 1993, p. 48; Nuclear Notebook, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 50 No. 6, November/December 1994, p. 59; A HISTORY OF THE NUCLEAR WEAPONS STOCKPILE, FY 1945 - FY 1985, Office of Military Application, USDOE; "Summary of Declassified Nuclear Stockpile Information, Declassified Stockpile Data, 1945 to 1994," handout at DOE press conference on June 27, 1994; Atomic Energy Commission, Program Status Report, December 31, 1954, Part III - Weapons Program, p. 2.

Dr. David A. Rosenberg of the Naval War College in Newport, Rhode Island, first disclosed the stockpile numbers between 1945 and 1948 in an article in the BULLETIN OF THE ATOMIC SCIENTISTS (Vol. 38 No. 5, May 1982, pp. 25 - 30); Dr. Rosenberg deserves most of the credit for getting this early

data declassified and making it available to the public. (These numbers were formally released by DOE on February 19, 1982 in Classification Notice WNP-46A, "Declassification of Early Stockpile Data.")

Table II-6: Characteristics of High Explosives Used in U.S. Nuclear Weapons

<u>Symbol</u>	<u>Name</u>	<u>Chemical Composition</u>	<u>Molecular Weight</u>	<u>Loading Density, lb/ft³</u>	<u>Detonation Rate, ft/s</u>	<u>Heat of Detonation, cal/gm</u>
	Baratol	24% TNT, 76% Ba(NO ₃) ₂		163	15978	740
	Boracitol	40% TNT, 60% B(OH) ₃		95	15946	400
	Composition B	60% RDX, 40% TNT	347.07	105	25723	1240
	Cyclotol 75/25	75% RDX, 25% TNT		109	27068	1570
DATB	Diaminotrinitrobenzene	C ₆ H ₃ (NO ₂) ₃ (NH) ₂	243	112	24673	980
FEFO	Fluorodinitroethyl	C ₅ H ₆ N ₄ O ₁₀ F ₂	320.1	100	24607	1280
HMX	Cyclotetramethylenetetranitramine	(CH ₂) ₄ N ₄ (NO ₂) ₄	296	119	29857	1320
	LX-04	85% HMX, 15% binder		116	27757	1310
	LX-07	90% HMX, 10% binder		117	28348	1490
	LX-09	93.3% HMX, 2.3% FEFO plasticizer, 4.4% binder		115	28906	1610
	LX-10	95% HMX, 5% binder		116	28988	1550
	LX-11	80% HMX, 20% binder		117	27298	1120
	LX-14	95.5% HMX, 4.5% binder		114	28971	1580
	LX-17	92.5% TATB, 7.5% binder		119	25034	1310

Table II-6: Characteristics of High Explosives Used in U.S. Nuclear Weapons

<u>Symbol</u>	<u>Name</u>	<u>Chemical Composition</u>	<u>Molecular Weight</u>	<u>Loading Density, lb/ft³</u>	<u>Detonation Rate, ft/s</u>	<u>Heat of Detonation, cal/gm</u>
NC	Nitrocellulose		262.6	94	23951	1160
	Octol	70% HMX, 30% TNT	388.16	112	27485	1074
	Octol 75/25	75% HMX, 25% TNT		113	27823	1570
	PBX-9010	90% RDX, 10% binder		112	27462	1470
	PBX-9011	90% HMX, 10% binder		110	27888	1530
	PBX-9404	93% HMX, 6.5% nitrocellulose, 0.5% binder	315.07	117	29132	1380
	PBX-9501	95% HMX, 5% binder		115	28971	1590
	PBX-9502	95% TATB, 5% binder		119	25296	1150
	PBX-9503	80% TATB, 15% HMX, 5% binder		117	25329	1220
PETN	Pentaerythritol-tetranitrate	$C(CH_2ONO_2)_4$	316.15	100	26576	1420
RDX	Cyclotrimethyl-enetrinitramine	$(CH_2)_3N_3(NO_2)_3$	222.13	103	27560	1320
TATB	Triaminotrinitrobenzene	$C_6H_3(NO_2)_3(NH)_3$	258	119	25920	1200
TNT	Trinitrotoluene	$C_6H_2(NO_2)_3CH_3$	227.1	97	21786	1080

Explanation of symbols:

HMX = Higher Melting (point) Explosive
LX = Livermore Explosive
PBX = Plastic-Bonded Explosive
RDX = Research Department Explosive / Rapid Detonation Explosive

Sources: PRINCIPLES OF NAVAL WEAPONS SYSTEMS (David R. Frieden, ed., Naval Institute Press, Annapolis, Maryland, 1985); THE MILITARY CRITICAL TECHNOLOGIES LIST (Office of the Undersecretary of Defense for Research and Development, Washington, D.C., 1984); "Shock Initiation of High Explosives" (ENERGY & TECHNOLOGY REVIEW, Lawrence Livermore National Laboratory, Livermore, California, UCRL-52000-83-2, February 1983); CRC HANDBOOK OF TABLES FOR APPLIED ENGINEERING SCIENCE, 2nd Edition, 1976 (CRC Press, Inc., Boca Raton, Florida); DETONATION PROPERTIES OF CONDENSED EXPLOSIVES (C. L. Mader, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, LA-2900, 1963); LLNL EXPLOSIVES HANDBOOK: PROPERTIES OF CHEMICAL EXPLOSIVES AND SIMULANTS, UCRL-52997 (B. M. Dobratz and P.C. Crawford, Lawrence Livermore National Laboratory, Livermore, California, January 31, 1985); EXPLOSIVES, PROPELLANTS, & PYROTECHNICS (A. Bailey & S. G. Murray, Brassey's Ltd., London, England, 1989); transcript of questions & answers between Representative Fazio and W. Graham Claytor, Assistant Secretary for Defense Programs, DOE, undated (received May 1991); AN INTRODUCTION TO NUCLEAR WEAPONS, Samuel Glasstone and Leslie M. Redman, WASH-1037 (revised), U.S. Atomic Energy Commission, Division of Military Application, Washington, D.C., June 1972, pp. 156, 157.

**Table II-7: U.S. Nuclear Weapons High Explosives
Employment**

<u>Explosive</u>	<u>HE Type</u>	<u>Weapons Used In</u>
Melt & Cast	Cyclotol Composition B	MK 28; MK 53 MK 28; MK 53
Conventional Plastic Bonded	PBX-9404	MK 43; W-48; W-50; W-55; W-56; MK 57 Mod 2; MK/B 61 Mods 0, 1, 2, 5; W-69
	PBX-9011	MK 57 Mods 1, 2
	PBX-9010	MK 43; W-50
	PBX-9501	W-76; W-78; W-88
	LX-04	W-62; W-70
	LX-07	W-71
	LX-09	W-68
	LX-10	W-68; W-70; W-79; W-82
	LX-11	W-71
Insensitive Plastic Bonded	PBX-9502	MK/B 61 Mods 3, 4, 6, 7, 8, 9, 10; W-61; W-80; W-85; W-90; W-91
	LX-17	B 83; W-84, W-87; W-89

Sources: Transcript of questions & answers between Representative Fazio and W. Graham Claytor, Assistant Secretary for Defense Programs, DOE, undated (received May 1991); REPORT TO CONGRESS: ASSESSMENT OF THE SAFETY OF U.S. NUCLEAR WEAPONS AND RELATED NUCLEAR TEST REQUIREMENTS, UCRL-LR-107454, R. E. Kidder, Lawrence Livermore National Laboratory, Livermore, California, July 26, 1991, pp. 3, 4; "U.S. Nuclear Weapons Safety And Control Features," Nuclear Notebook, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 47 No. 8, October 1991, p. 49.

Table II-8: U.S. Nuclear War Plans, 1945 - 1957

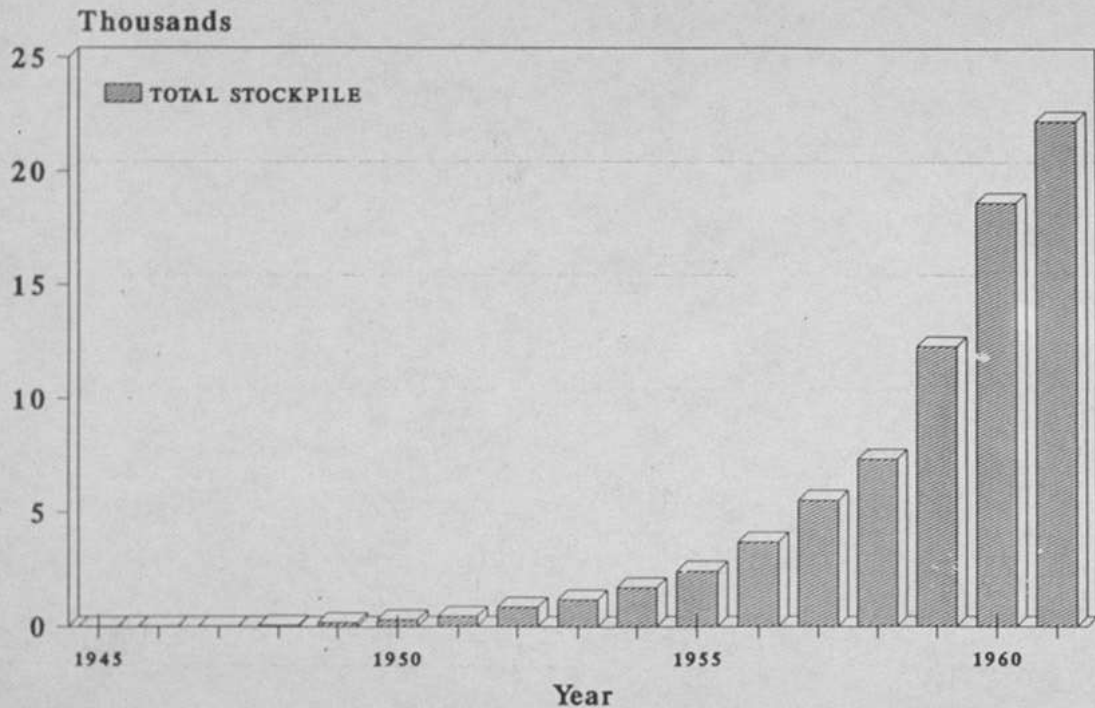
<u>Name</u>	<u>Date</u>	<u>Targets & Weapons</u>	<u>U.S. Stockpile</u>
Project 2532	Sep. 1945	204 bombs; 66 cities/urban areas	2 complete weapons
JIC 329/1	Dec. 1945	20 - 30 bombs; 20 cities	2 complete weapons
PINCHER	June 1946	50 bombs; 20 cities	9 complete weapons (3 later expended during Operation CROSSROADS)
BROILER	March 1948	34 bombs; 24 cities	45 (estimated; based on production between June 1947 and June 1948)
FROLIC (GRABBER)	May 1948	50 bombs; 20 cities	52 (estimated)
SIZZLE (FLEETWOOD, HALFMOON, DOUBLESTAR)	Dec. 1948	133 bombs; 70 cities	112 (estimated)
TROJAN	Jan. 1949	133 bombs; 70 cities	160 (estimated)
SHAKEDOWN (OFFTACKLE, CROSSPIECE)	Oct. 1949	220 bombs; 104 cities	212 (estimated)
DROPSHOT	1949	300 bombs; 200 cities	234 (estimated)
SAC Basic War Plan	1950	300 targets	298 (as of 6/50)
SAC Basic War Plan	March 1954	735 bombers carrying one or two bombs each	1,630 warheads & complete weapons (offensive & defensive)

<u>Name</u>	<u>Date</u>	<u>Targets & Weapons</u>	<u>U.S. Stockpile</u>
SAC Basic War Plan	1956	2,997 targets	3,620 warheads & complete weapons (offensive & defensive)
SAC Basic War Plan	1957	3,261 targets	5,828 warheads & complete weapons (offensive & defensive)

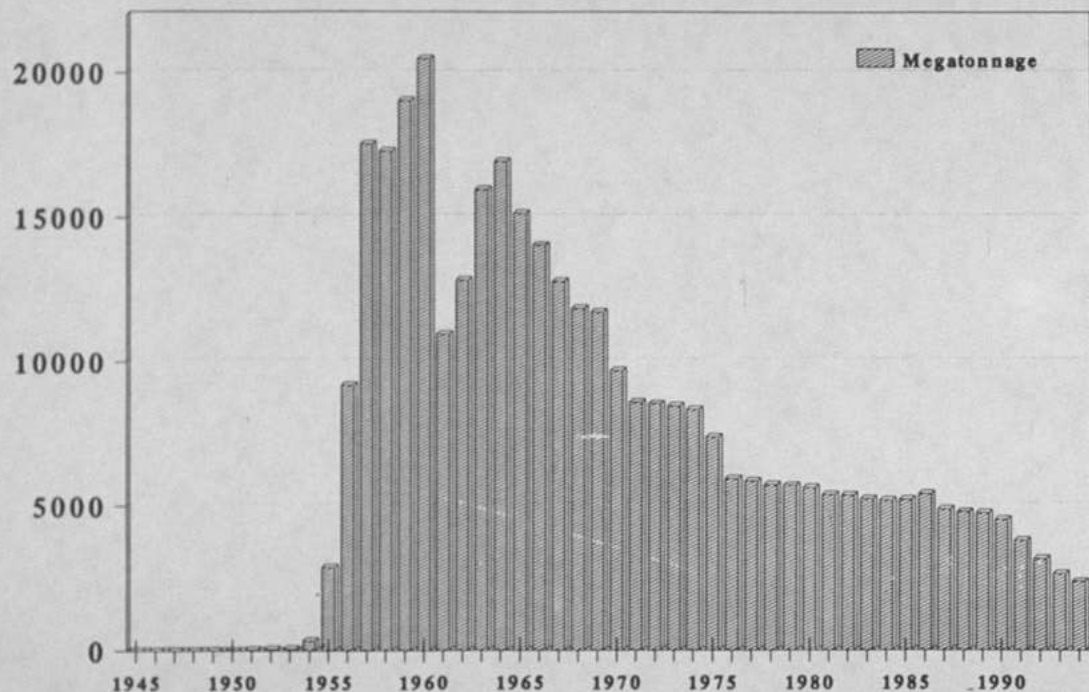
Sources: To Win A Nuclear War, Michio Kaku and Daniel Axelrod, South End Press in Boston, Mass., 1987, pp. x, xi; memorandum for Major General L. R. Groves dated 15 September 1945 from Maj. Gen. Lauris Norstad, USA, subject: Atomic Bomb Production; "Toward Armageddon: The Foundations of United States Nuclear Strategy," PhD thesis by David A. Rosenberg, University of Chicago, August 1983, pp. 32-33, 59-61, 74-77, 96-98, 107, 110, 117, 240.



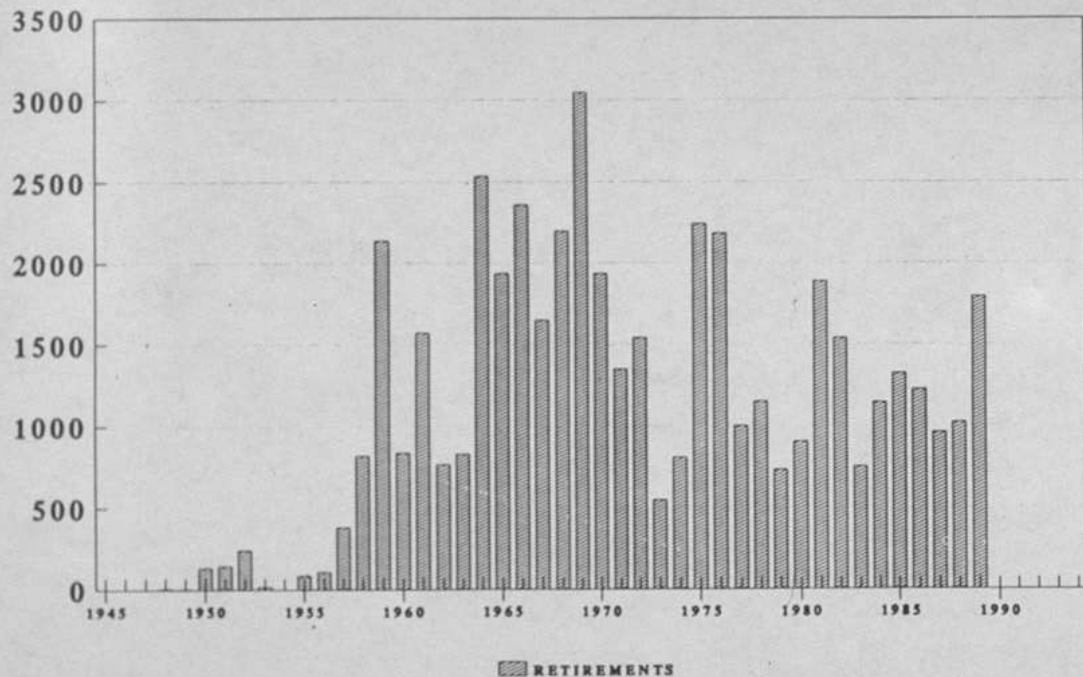
TOTAL UNITED STATES NUCLEAR STOCKPILE 1945 to 1961



TOTAL MEGATONNAGE OF THE U.S. NUCLEAR WEAPON STOCKPILE



TOTAL WEAPON RETIREMENTS BY YEAR



Note: 1984-1989 values are adjusted

BUILDS OF UNITED STATES NUCLEAR WEAPONS NOW RETIRED

